DELL Technologies

Deployment and Configuration

# Best Practices for Using Dell SRDF/Metro in a VMware vSphere Metro Storage Cluster


## Abstract

This white paper discusses best practices when configuring Dell SRDF/Metro with VMware vSphere Metro Storage Cluster (vMSC).



April 2026



Dell Engineering

# Revisions

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|---|---|
| August 2023 | New template release |
| November 2023 | Update for PowerMaxOS 10.1.0.0, vSphere 8.0U2 |
| October 2024 | Update for PowerMaxOS 10.2.0.0, vSphere 8.0U3 |
| November 2024 | Add clarification for pRDM with Microsoft failover cluster and PowerPath EOL |
| April 2025 | Remove Metro Mobility ID ALUA restriction; PP/VE info |
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# Executive Summary

The Dell PowerMax family provides disaster recovery and mobility solutions through its remote replication technology SRDF® (Symmetrix Remote Data Facility). SRDF has the capability to replicate between multiple sites, co-located or even thousands of miles apart depending on the type of replication desired.

As part of this replication family, Dell offers SRDF/Metro, an active/active version of SRDF. In a traditional SRDF device pair relationship, the secondary device, or R2, is write disabled. Only the primary device, or R1, is accessible for read/write activity. It is an active/passive solution. With SRDF/Metro, the R2 is also write enabled and accessible by the host or application. The R2 takes on the personality of the R1 including the WWN. A host, therefore, would see both the R1 and R2 as the same device.

As both devices are simultaneously accessible, the hosts in a cluster, for example, can read and write to both the R1 and R2. The SRDF/Metro technology ensures that the R1 and R2 remain current and consistent, addressing any conflicts which might arise between the pairs.

When you use SRDF/Metro in conjunction with VMware[1] vSphere across multiple hosts in a single vCenter, it forms a VMware vSphere Metro Storage Cluster (vMSC). At its core, a VMware vMSC infrastructure is a stretched cluster. The architecture extends the local network and storage, stretching it between sites. This enables these subsystems to span distance, presenting a single and common base infrastructure set of resources to the vSphere cluster at both sites.

With vMSC customers acquire the capability to migrate virtual machines between sites with VMware vSphere vMotion and vSphere Storage vMotion, enabling on-demand and nonintrusive mobility of workload.

## Audience

Dell intends this white paper for VMware administrators, server administrators, and storage administrators responsible for creating, managing, and using VMware, as well as their underlying storage devices. The paper assumes the reader is familiar with VMware infrastructure and the VMAX or PowerMax arrays and the related software.

[1] This paper will refer to VMware by Broadcom as simply VMware. It covers vSphere 8 and 9 as 7 reached End of Service on October 2, 2025.

# 1 Introduction

The purpose of this paper is to provide the best practices for running a VMware vSphere Metro Storage Cluster (vMSC) that utilizes Dell SRDF/Metro technology for stretched storage. It will include best practices for vMSC as well as SRDF/Metro and any special considerations when running the two technologies together.

This paper covers up to the PowerMaxOS 10.4.0.0 (6079) and vSphere 9.0 releases. Although some information (e.g., images) contained herein is from earlier PowerMaxOS, management, and/or vSphere versions, the best practices cover all revisions. It is always important to check product documentation for any limitations or features that are present in a particular customer's hardware/software versions.

## 1.1 NVMe over Fabrics (NVMeoF)

Beginning with PowerMaxOS 5978.444.444, the PowerMax array offers NVMeoF or NVMe over Fabrics. NVMe stands for non-volatile memory. This is the media itself such as NAND-based flash or storage class memory (SCM) which compromise the PowerMax backend. NVMe or non-volatile memory express is a set of standards which define a PCI Express (PCIe) interface used to efficiently access data storage volumes on NVM. NVMe provides concurrency, parallelism, and scalability to drive performance. It replaces the SCSI interface. NVMeoF is the specification that details how to access that NVMe storage over the network between host and storage. The network transport could be Fibre Channel, TCP, RoCE, or any other number of next generation fabrics. Dell supports Fibre Channel with NVMe on the PowerMax 2000/2500/8000/8500[2], also known as FC-NVMe, and TCP with NVMe on the PowerMax 2500/8500, also known as NVMe/TCP.

To use FC-NVMe with the PowerMax requires 32 Gb/s Fibre Channel modules (SLICs). You assign the emulation FN to ports on these modules to support FC-NVMe. In addition, in order to use FC-NVMe with a host, a supported Operating System and supported HBA card is necessary. With NVMeoF, you present targets as namespaces (equivalent to SCSI LUNs) to a host in active/active or asymmetrical access modes (ALUA).

To use NVMe/TCP with the PowerMax requires 25 Gb/s network modules (SLICs). You assign an OR emulation to ports on these modules to support NVMe/TCP. In addition, in order to use NVMe/TCP with a host, a supported Operating System and supported network card is necessary. With NVMeoF, you present targets as namespaces (equivalent to SCSI LUNs) to a host in active/active or asymmetrical access modes (ALUA).

ESXi hosts discover and use the presented namespaces and emulate the NVMeoF targets as SCSI targets internally and presents them.

### 1.1.1 Support

While SRDF supports the use of NVMeoF in active/passive configurations, it does not support active/active SRDF/Metro.

[2] FC-NVMe requires a minimum of PowerMaxOS 10.1.0.0 with PowerMax 2500/8500.

# 2 SRDF/Metro

SRDF/Metro is a feature available on the VMAX3 and VMAX All Flash arrays starting with HYPERMAX OS 5977.691.684 and all PowerMax arrays which provides active/active access to the R1 and R2 of an SRDF configuration. In traditional SRDF, R1 devices are Read/Write accessible while R2 devices are Read Only/Write Disabled. In SRDF/Metro configurations both the R1 and R2 are Read/Write accessible. It accomplishes this by having the R2 take on the personality of the R1 in terms of geometry and most importantly the WWN. By sharing a WWN, the R1 and R2 appear as a shared virtual device across the two arrays for host presentation. A host or typically multiple hosts in a cluster can read and write to both the R1 and R2. SRDF/Metro ensures that each copy remains current and consistent and addresses any write conflicts which might arise. The SRDF/Metro mode is “Active.” In VMware environments, to form a VMware vSphere Metro Storage Cluster (vMSC) you add ESXi hosts from two different data centers to the same vCenter. A simple diagram of the feature is in Figure 1.

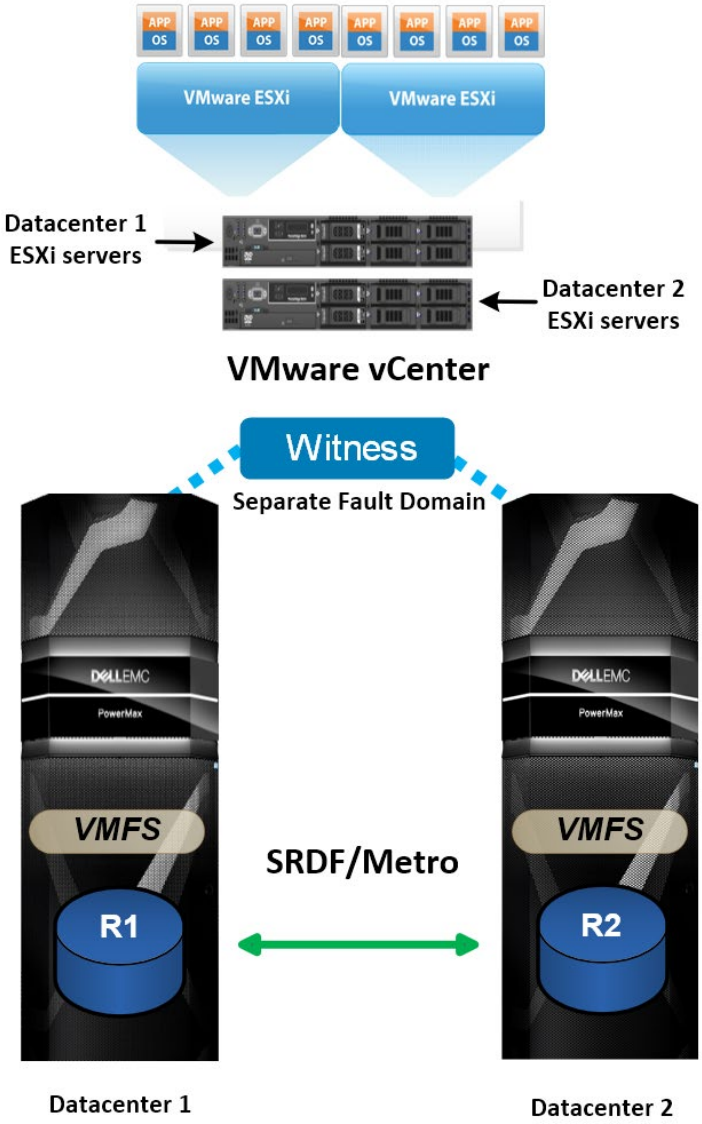



Figure 1. SRDF/Metro

This feature provides the following advantages:

- A high availability solution at Metro distances by leveraging and extending SRDF/S functionality.
- Active-Active replication capabilities on both the source and target sites.
- Witness support to enable full high availability, resiliency, and seamless failover.

## 2.1 Smart DR/MetroDR

SRDF/Metro supports 3-site configurations with a leg off either the R1, R2 or both members of a pair. This means that neither the R1 nor R2 is aware of the existence of an asynchronous or adaptive copy leg off the other since the pairs are independent. Up until PowerMaxOS 5978 Q3 2020, SRDF/Metro did not support a Star-like configuration where either the R1 or R2 could update a single, remote leg. Beginning with PowerMaxOS 5978 Q3 2020, Dell offers Smart DR (also known as MetroDR) which enhances SRDF/Metro and provides Star-like functionality. The Smart DR feature extends the high availability (HA) solution of SRDF/Metro to a third array and supports the ability to have Geo distance DR support for a device in a Metro configuration. Smart DR integrates SRDF/Metro and SRDF/A/ACP enabling HR DR by tightly coupling the SRDF/A/ACP sessions on each side of the Metro pair to replicate to a single DR device. While only one side of the SRDF/Metro pair (R1) will update the SRDF/A/ACP leg at one time, both are capable of doing so. The asynchronous replication of data from either side of the SRDF/Metro pair to a tertiary site enables Failover/Failback to the DR site while retaining the Metro environment. Smart DR requires a minimum of PowerMaxOS 5978 Q3 2020 and Solutions Enabler 9.2 running on the three arrays with Witness (array or virtual) configured. This configuration does not support bias mode alone. Dell supports both asynchronous (SRDF/A) and adaptive copy (SRDF/ACP) modes on the third leg.

Figure 2 is a schematic representation of the business continuity solution that integrates a VMware environment and Smart DR. The SRDF/Metro array pair hosts the VMware Metro Storage Cluster while the SRDF/A/ACP array becomes a true failover site. Note that the Witness in the figure should be in a separate fault domain, whether array or virtual, and the third array could be thousands of miles away.

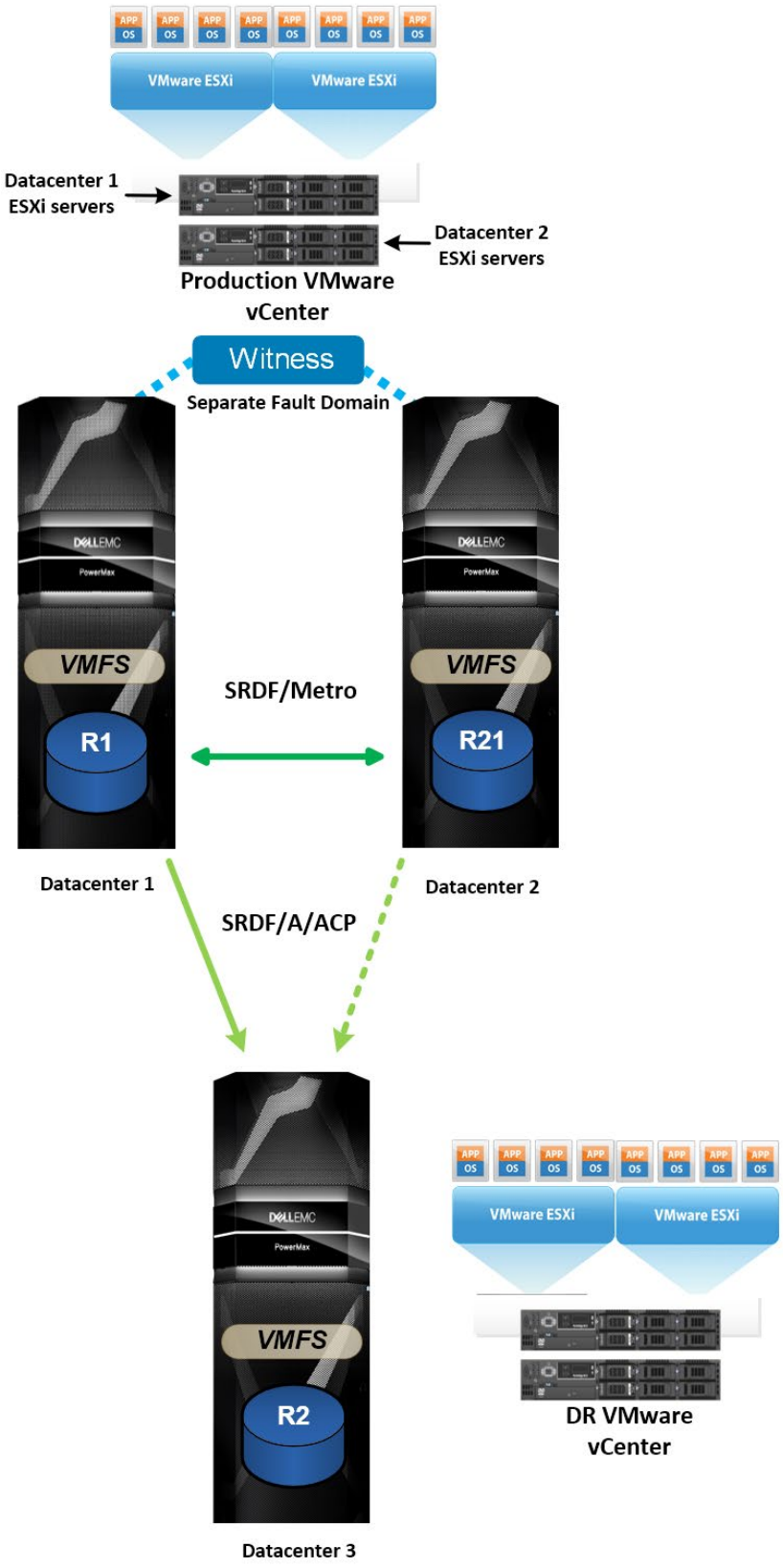



Figure 2. Smart DR

The use of Smart DR does not change the best practices for the vMSC configuration. As vMSC is an HA solution, Smart DR does, however, provide a superior DR solution over a single concurrent or cascaded leg since both the R1 and R2 can update the third site.

**Note:** You cannot replicate an SRDF/Metro pair to another SRDF/Metro pair, and thus it is not possible to have a vMSC configuration at the production and disaster recovery sites; however, if a DR event occurs that necessitates a failover, it is possible to remove the SRDF relationships between the production and disaster recovery sites and then create a new SRDF/Metro relationship and thus build a vMSC at the DR site.

## 2.2 Latency

Dell does not publish strict requirements for latency between SRDF/Metro arrays, however, keeping under 5ms round-trip time (RTT) would be an appropriate goal. VMware requires no more than 10ms RTT even if the solution could support it.

Dell recommends keeping the distance between arrays under 200 km for SRDF/Metro. What is most important, however, is what the application can sustain, rather than SRDF. SRDF/Metro can function at distances greater than 200 km; but it is unlikely applications running on the SRDF/Metro devices would tolerate the resulting response time. The key is to know the business requirements for application response time and design the solution around that. In the end, the lower the latency the better.

## 2.3 Bias and Witness

SRDF/Metro maintains consistency between the R1 and R2 during normal operation. If, however, a device or devices go not ready (NR) or you lose connectivity between the arrays, SRDF/Metro selects one side of the environment as the “winner” and makes the other side inaccessible to the host(s). There are two ways that SRDF/Metro can determine a winner: bias or SRDF/Metro Witness (array or virtual). The bias or witness prevents any data inconsistencies (e.g., split brain) which might result from the two arrays being unable to communicate.

Bias is a required component of SRDF/Metro, with or without a witness. Witness builds upon the bias functionality – bias, therefore, becomes the failsafe in case the witness is unavailable or fails. The initial `createpair` operation of SRDF/Metro will assign bias to the R1 site though it is possible to change it to the R2 after initial synchronization. Note changing the bias turns the R2 into the R1. In the event of a failure when using bias, SRDF/Metro makes the non-biased side inaccessible to the host(s) while the bias side (R1) survives. You identify bias by the state of “ActiveBias” on a device pair or SRDF group as in Figure 3. Note the SRDF Type is “R1”.

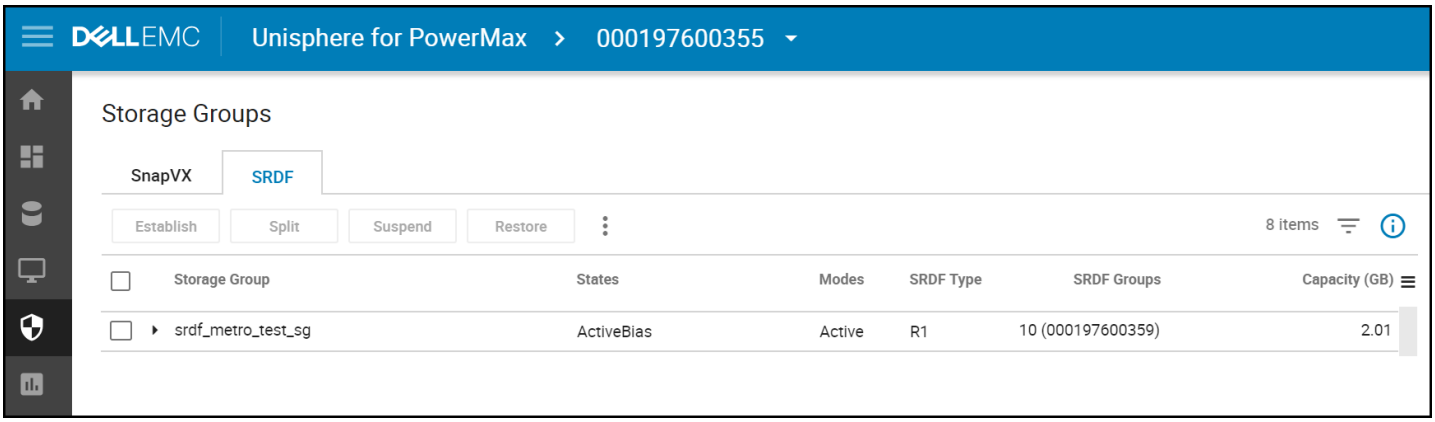



Figure 3. ActiveBias state for SRDF/Metro

If the bias side (R1) experiences the failure, then the entire SRDF/Metro cluster becomes unavailable to the hosts and will require user intervention to rectify. To avoid these types of failures, Dell offers a witness. The witness is an external arbiter that can be either array (physical) or virtual. An array witness runs on a separate VMAX/VMAX3/VMAX All Flash/PowerMax with the proper code. You create a separate SRDF group from each SRDF/Metro array (R1, R2) to the witness array and marked as witness or quorum group. In contrast, the virtual witness, or vWitness, is a VMware virtual appliance or installed as software running on a support operating system (physical or virtual). Unlike an array witness, the vWitness does not require an additional array, though it does require the eManagement Guest OS on the arrays as it communicates with a running daemon there. Both arrays in the SRDF/Metro pair must be able to communicate with the vWitness. Best practice is to place either type of witness at a third location (fault domain with separate power/network) so that it will not be subject to a failure impacting one or the other SRDF/Metro sites. Dell does not recommend using the arrays hosting the R1 or R2 of the SRDF/Metro relationship for the vWitness. It can cause undesirable failover results.

### 2.3.1 Priority

The use of the witness supersedes the bias functionality. If the SRDF witness groups are present or you configure a virtual witness before the `createpair` command, the device pair(s) will automatically enter a “Witness Protected” state upon synchronization and the state will be “ActiveActive.” In Unisphere in Figure 4 you can see the witness enabled ActiveActive state for SRDF/Metro for the same pair from Figure 3.

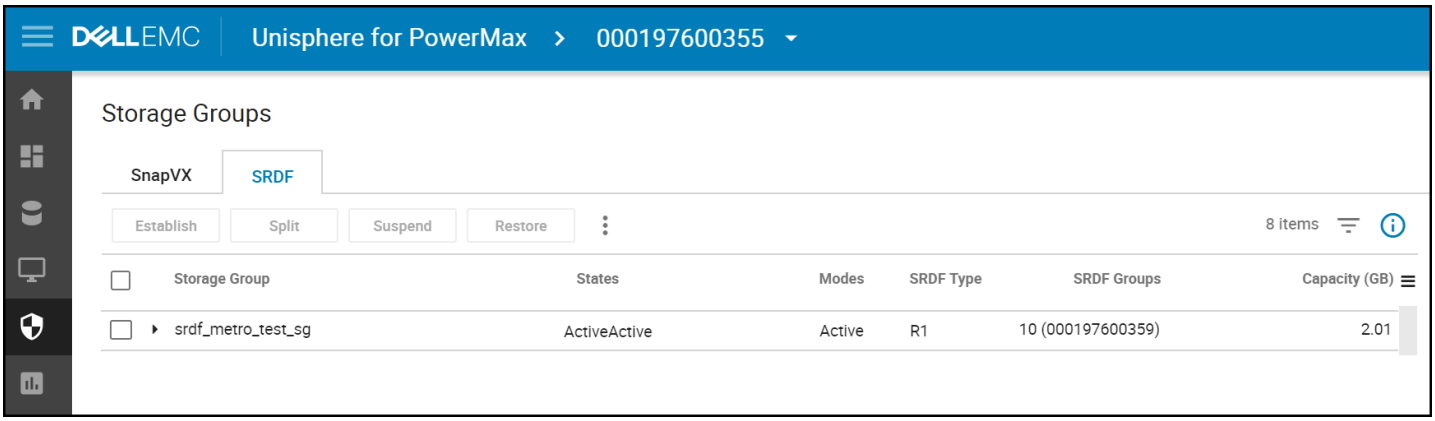



Figure 4. ActiveActive state for SRDF/Metro

Alternatively, if you add the witness after synchronization, at the next re-establish the witness will take effect. A pair, however, cannot be reconfigured to use a witness without suspending the SRDF/Metro pair first.

### 2.3.2 Witness latency

Each witness should be within forty(40) ms of the SRDF/Metro arrays. Use of a witness in a public cloud requires an RPQ.

### 2.3.3 Preferred winner

SRDF/Metro uses the concept of a preferred winner in the event of a failure. What this means is that the code grants one side the privilege of requesting the lock from the witness instance first. When either array is running HYPERMAX OS 5977, the preferred winner is always the R1. When both arrays are running PowerMaxOS 5978 and higher, there are other factors taken into consideration. These factors include whether one of the legs has an SRDF relationship to another

array, whether the device is in a masking view, and other settings. Therefore, it is possible the code will reverse the R1 and R2 during creation or re-establish. For more information, please see the Dell SRDF/Metro vWitness Configuration Guide.

**Note:** It is not possible to choose a particular witness when configuring SRDF/Metro pairs. The arrays select the witness according to a set of conditions.

#### 2.3.3.1 Cascaded and concurrent

As noted above, changes in the way the code selects a preferred winner can impact which device of the pair it assigns the R1 and thus bias. When there is a third leg off of one of the devices of an SRDF/Metro pair, the code will set that SRDF/Metro device as the preferred winner, or R1. Therefore, when using a witness, a cascaded setup is unlikely as the R2 will become the R1 and the configuration concurrent. This can be particularly problematic when using VMware SRM, and therefore the SRDF SRA does not support cascaded with SRDF/Metro. If a cascaded setup is essential, bias is the only way to guarantee it, i.e., no witness. Dell does not recommend bias but is an option for customers if their business demands it.

### 2.3.4 Multiple witnesses

Dell recommends configuring multiple array and/or virtual witnesses. Dell supports thirty-two(32) vWitnesses. In such cases SRDF/Metro manages the use of multiple witnesses so if the initial one fails, the code will pick another without user intervention. Note that array witnesses will always take precedence over virtual ones; however, SRDF/Metro is able to use either in the event of a witness failure. Figure 5, Figure 6 and Figure 7 all show the same two vWitnesses configured on three different arrays. Both witnesses are in a good **State** and **Alive**, but only one virtual witness is **In Use**, while the other is in standby mode. This means there are SRDF/Metro pairs on each of these arrays, yet they all can use the same virtual witness.

**Note:** Do not build a virtual witness on an SRDF/Metro pair that could use that very witness. The array has no way to check for this condition.

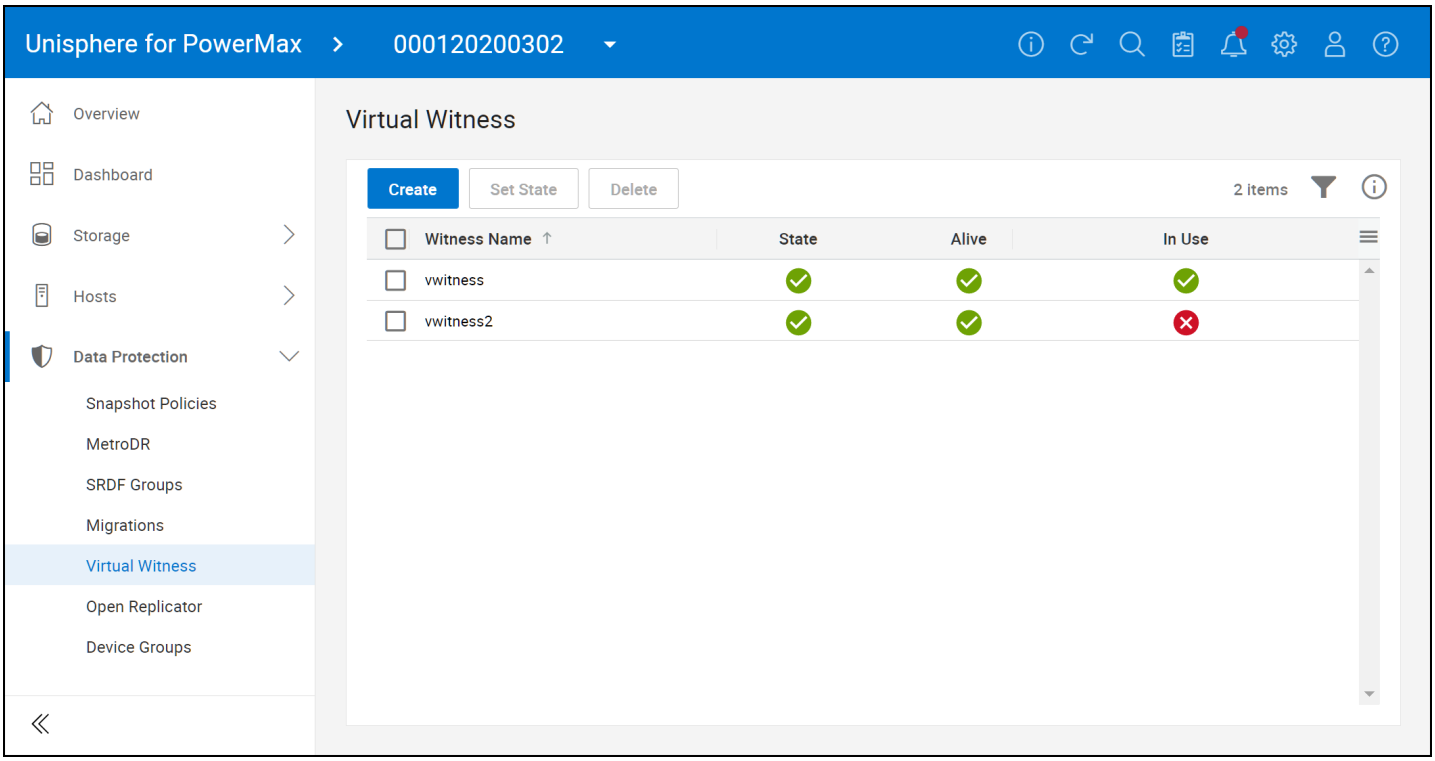



Figure 5. Virtual witnesses on array 000120200302

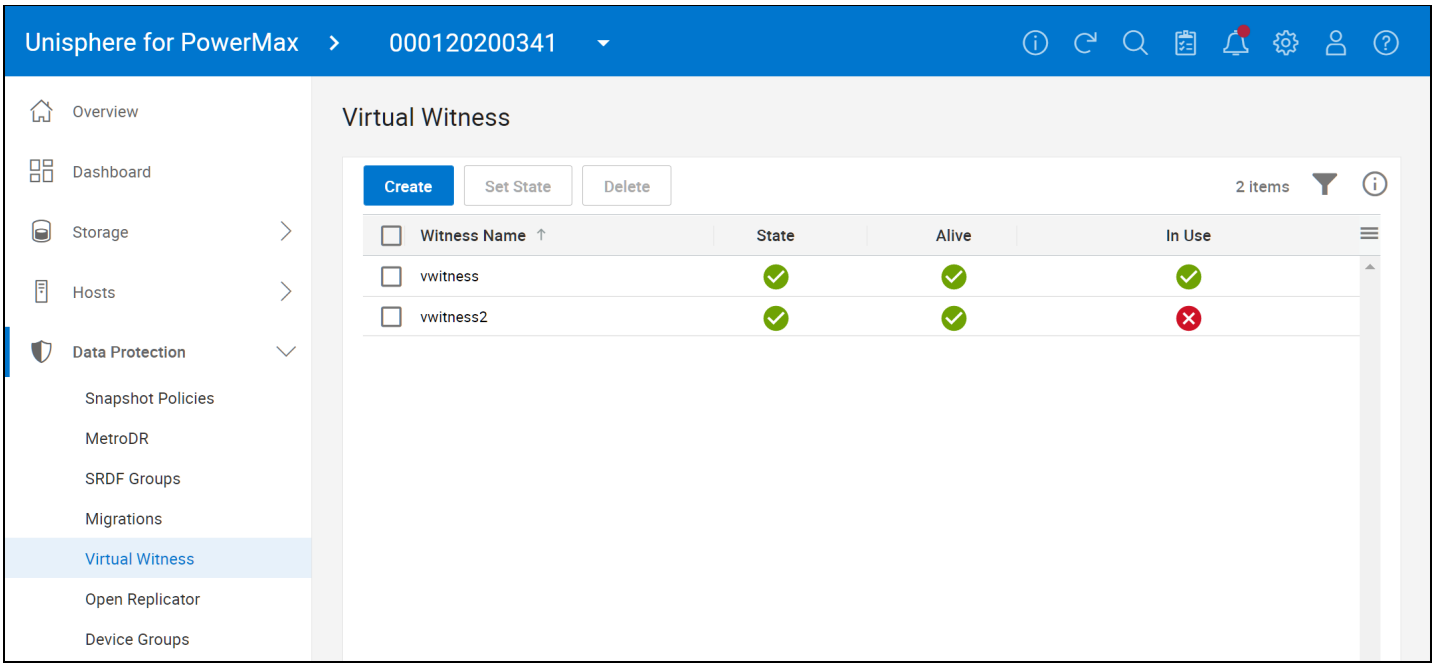



Figure 6. Virtual witnesses on array 000120200341

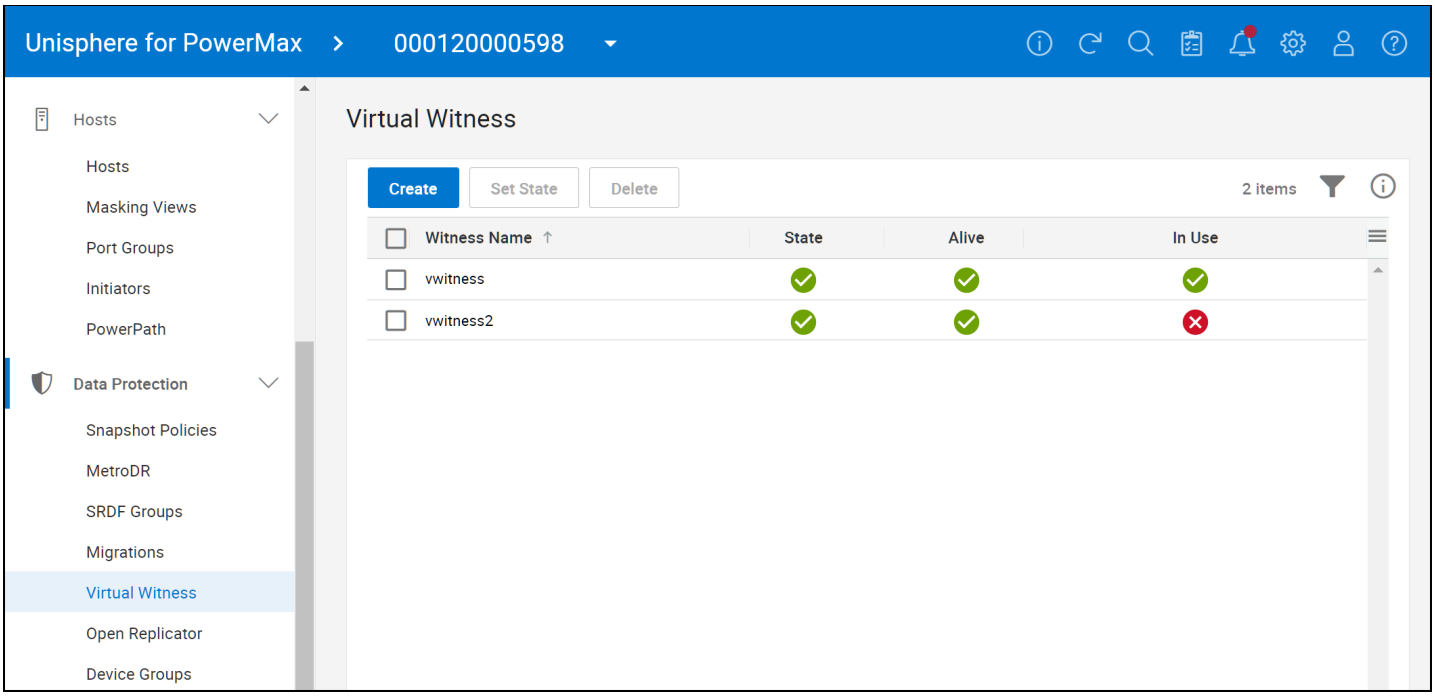



Figure 7. Virtual witnesses on array 000120000598

Note that only virtual witnesses appear in the above screen. To determine if there are array witnesses, navigate to the **SRDF Groups** screen and view existing groups for the **Type** of **Witness**. In Figure 8, you can see two array witnesses.

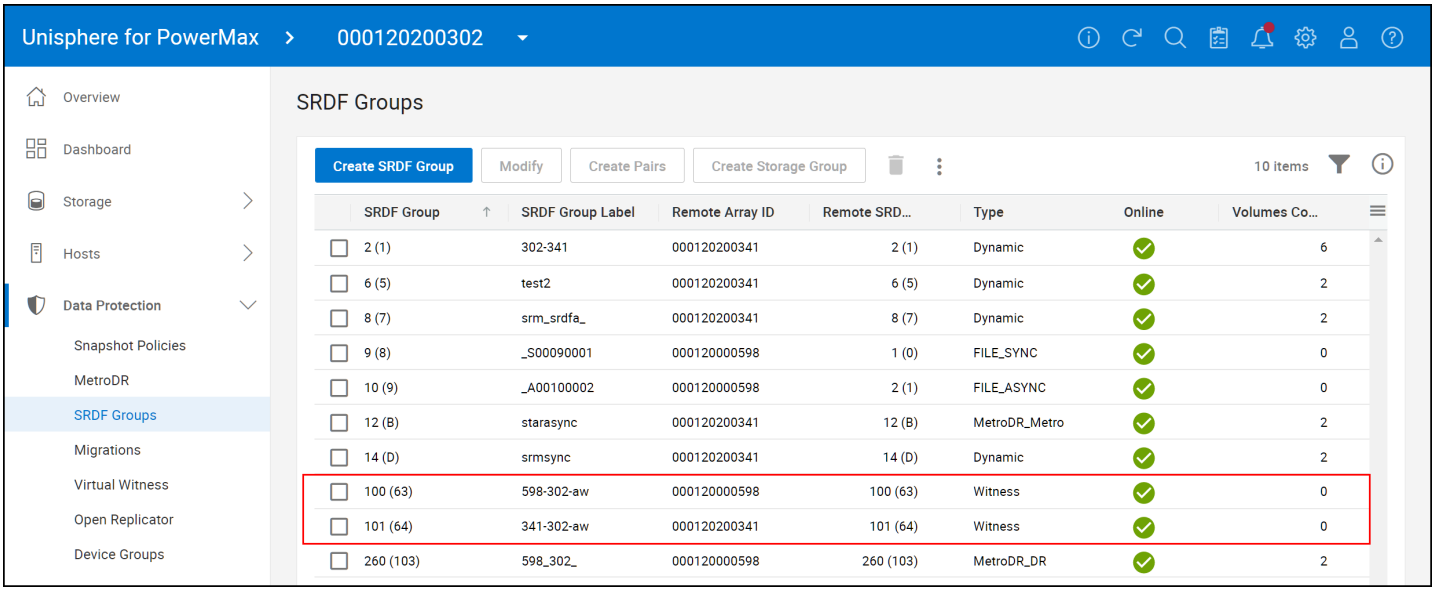



Figure 8. Array witnesses

The SRDF/Metro vWitness is available for VMAX3/VMAX All Flash storage arrays running HYPERMAX OS 5977 Q3 2016 Service Release and Solutions Enabler / Unisphere for VMAX 8.3 or later. Prior to PowerMaxOS 10, the vWitness can run as a VMware virtual appliance (vApp) or as part of Solutions Enabler on a supported OS. Dell recommends the vApp due to its lower hardware requirements. Once installed, the vWitness will then utilize the local Embedded Element Manager (EEM) installed on each pair of VMAX3/VMAX All Flash/PowerMax arrays. The Witness Lock Service daemon runs in Figure 9.

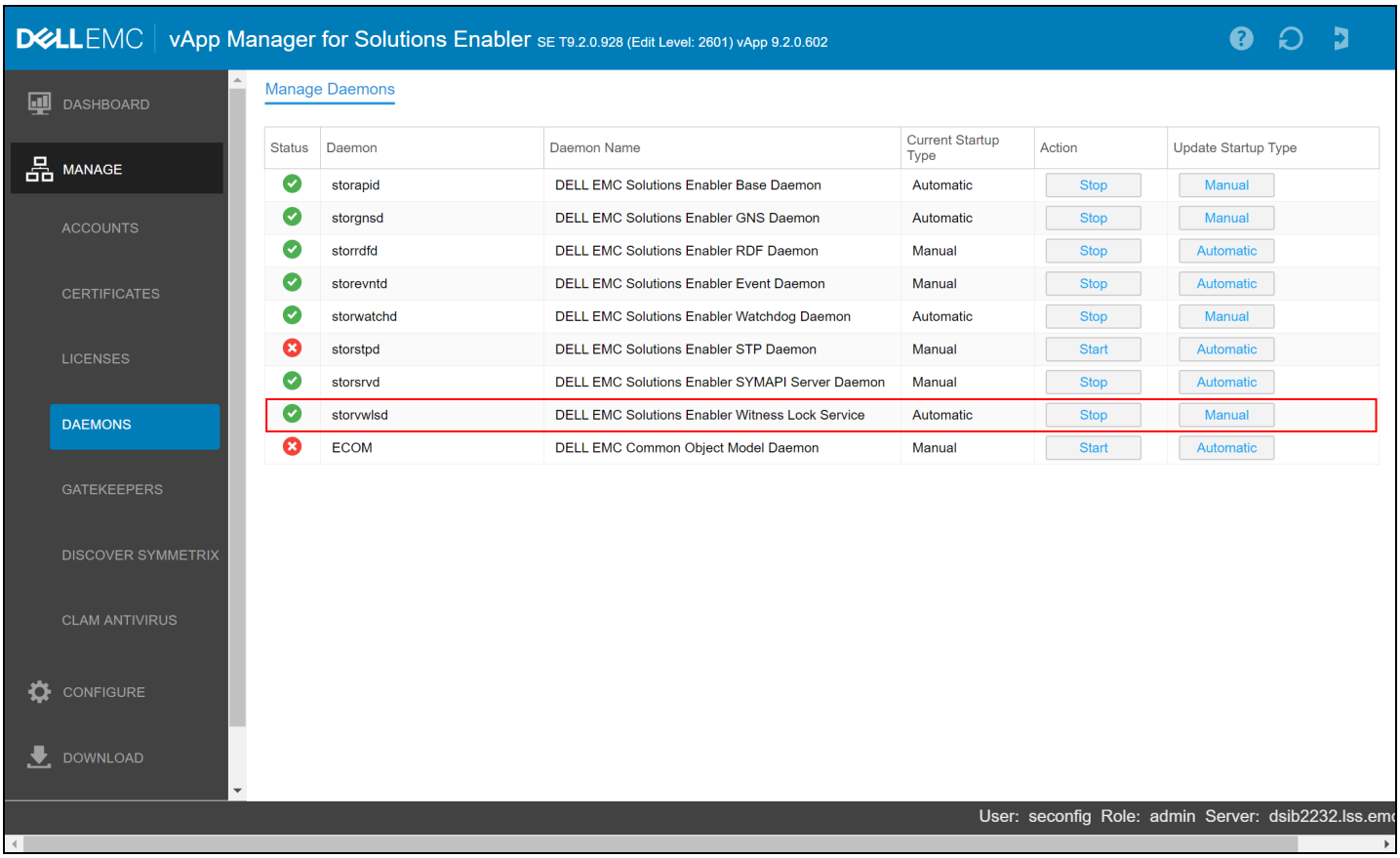



Figure 9. vWitness running on the vApp - Witness Lock Service

In PowerMaxOS 10, Dell deprecated the vApp and instead delivers the vWitness as a software package (separate from Solutions Enabler) which you install on a supported operating system. Figure 10 shows the Witness Lock Service Daemon.

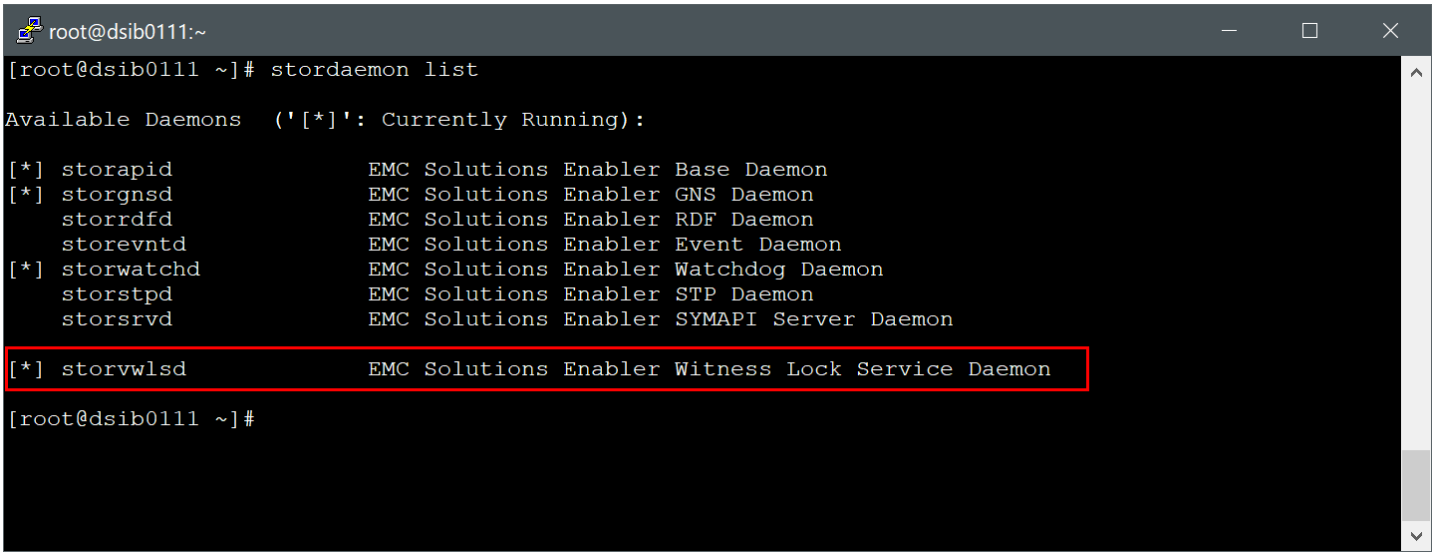



Figure 10. vWitness running on a support OS - Witness Lock Service

### 2.3.5 Witness failure

The screenshot in Figure 11 provides an example of a failure of the Witness. Note that for the SRDF group in Figure 11 the configured type(C) is Witness, but the effective type(E) is Bias. The Witness failure forces the change to the effective type.




Figure 11. An SRDF/Metro group with a failed Witness

In Figure 12 Unisphere displays the failed witness.

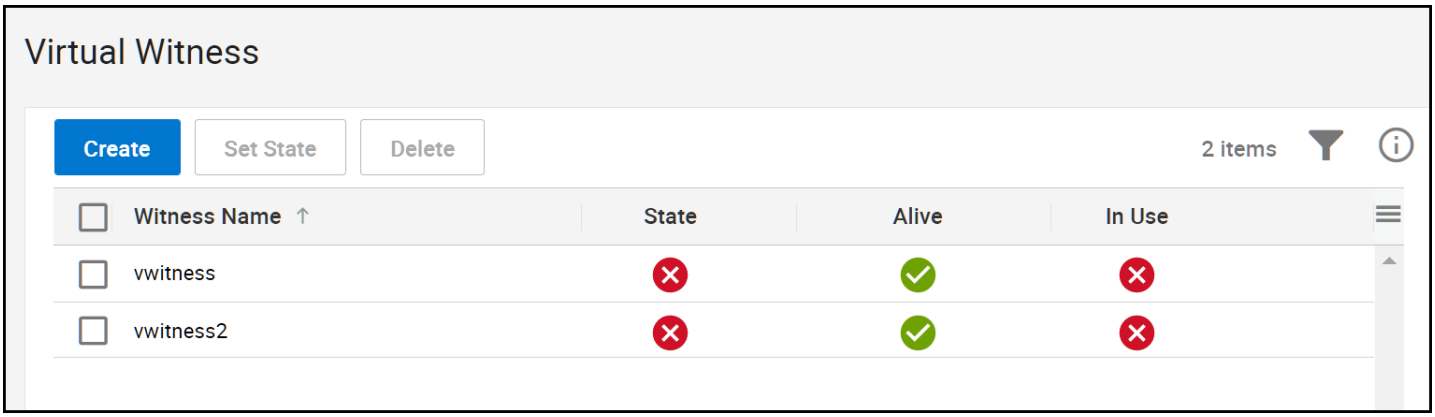



Figure 12. Failed virtual witnesses

Witnesses can experience events that impact them, yet they remain viable. In such cases there would be a “D” for the status or degraded. Repairing the Witness state and resolving the issue will return the group automatically to an effective Witness state and a Normal status as in Figure 13. You do not have to suspend the pairs.

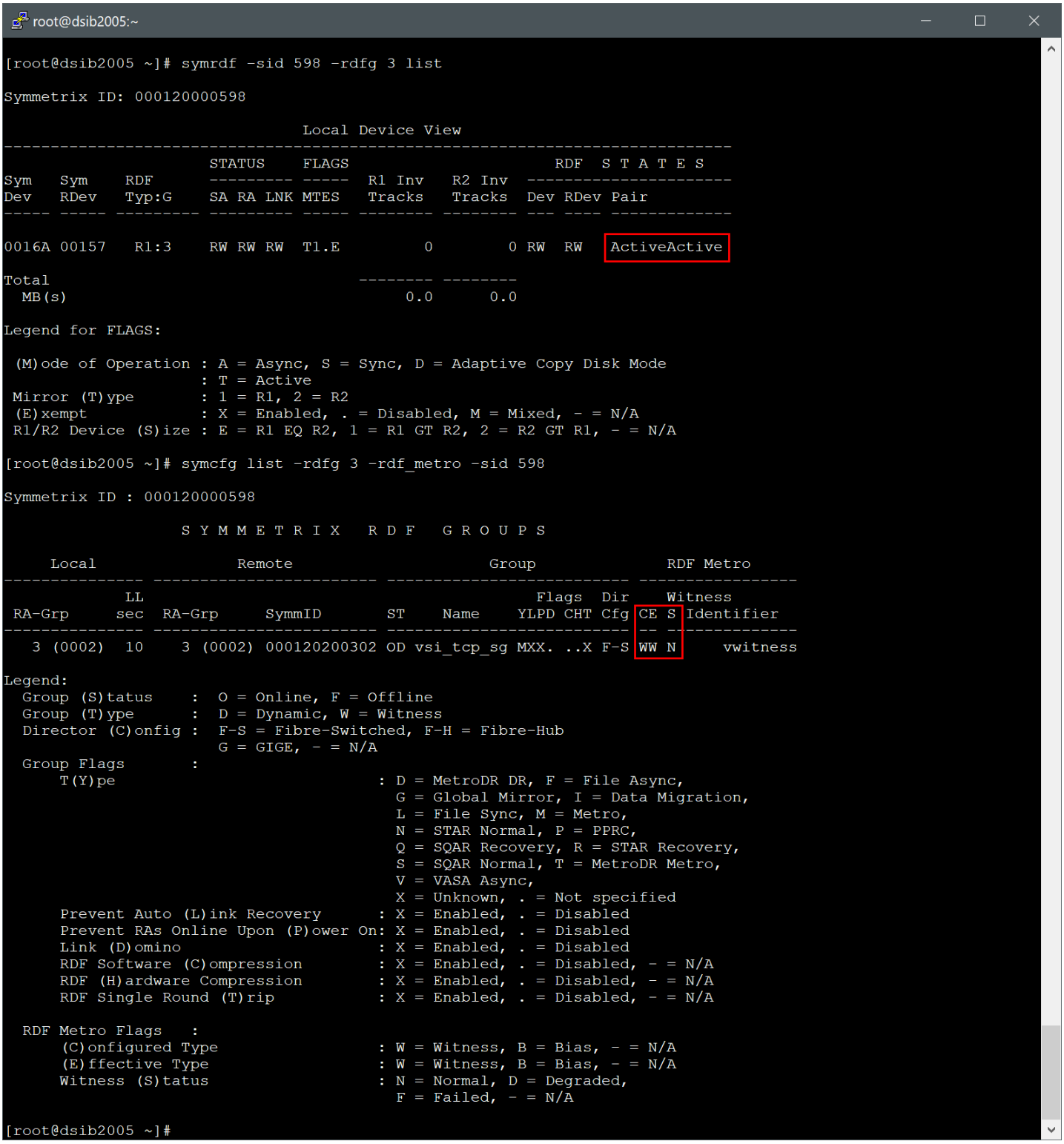



Figure 13. Restoring the Witness

Dell recommends the use of a witness instead of bias with SRDF/Metro and vMSC.

## 2.4 SRDF/Metro limitations

As SRDF/Metro is an active/active mode of SRDF implementation rather than active/passive, there are some restrictions which do not exist for other SRDF modes. We include some here for reference but for a complete list refer to *SRDF/Metro Overview and Best Practices Technical Notes* in the References section. Note that this list is based on PowerMaxOS 10 release of SRDF/Metro.

- Both the source (R1) and target (R2) arrays must be running HYPERMAX OS 5977.691.684 or higher for SRDF/Metro, or PowerMaxOS 5978 Q3 2020 or higher on all three arrays if running an MetroDR configuration.
- The R1 and R2 must be the same size.
- Devices cannot have Geometry Compatibility Mode (GCM) set prior to PowerMaxOS Q2 2018 SR.
- Devices cannot have User Geometry set.
- Concurrent and cascaded SRDF/A configurations must use HYPERMAX OS Q3 2016 SR and later.
- You cannot control devices in an SRDF group that contain a mixture of source (R1) and target (R2) devices.
- The following operations must apply to all devices in the SRDF group: `createpair –establish, establish, restore, and suspend`
- vWitness configurations require Embedded Element Management (EEM or eMgmt) on each SRDF/Metro paired array.
- SRDF/Metro does not support Star configurations.
- SRDF/Metro does not support NVMeoF.
- MetroDR does not support online device expansion. You must disable MetroDR first, and expansion done on the async R2 first, then R1, after which MetroDR can be re-enabled.
- Microsoft Cluster with SRDF/Metro requires a uniform configuration.

### 2.4.1 VAAI support

While SRDF/Metro supports all VAAI commands[3], Dell modified the implementation of Full Copy or XCOPY to accommodate the active/active nature of SRDF/Metro. These changes follow.

#### 2.4.1.1 SRDF/Metro XCOPY specifics

The following are caveats for SRDF/Metro when using Full Copy (XCOPY). In SRDF/Metro configurations the use of Full Copy entirely depends on whether the site is the one supplying the external WWN identity.

- Dell rejects Full Copy activity between a non-SRDF/Metro device and an SRDF/Metro device when the device WWN is different from the external device WWN. Typically, but not always, this means the

[3] Though technically not a VAAI command, SRDF/Metro does not support ODX.

non-biased site (recall even when using witness, there is a bias site). In such cases Dell will only use Full Copy when operations are between SRDF/Metro devices or within a single SRDF/Metro device; otherwise, it defaults to software copy.

- As Full Copy is a synchronous process on SRDF/Metro devices, which ensures consistency, performance will be closer to software copy than asynchronous copy on SRDF/A devices.
- While an SRDF/Metro pair is in a suspended state, Full Copy reverts to asynchronous copy for the target R1. It is important to remember, however, that Full Copy is still bound by the restriction that the copy must take place on the same array. For example, assume SRDF/Metro pair AA, BB is on array 001, 002. In this configuration device AA is the R1 on 001 and its WWN is the external identity for device BB, the R2 on 002. If you suspend the pair and the bias switches such that BB becomes the R1, the external identity still remains that of AA from array 001 (i.e., it appears as a device from 001). Since the device is on array 002, Dell will only use Full Copy in operations within the device itself or between devices on array 002.

# 3 VMware vSphere Metro Storage Cluster (vMSC)

A VMware vSphere Metro Storage Cluster configuration is a VMware vSphere certified solution that combines synchronous replication with array-based clustering. You deploy these solutions in metropolitan or campus environments as you must keep the arrays within limited distance. Dell SRDF/Metro represents one of those certified solutions.

A VMware vMSC requires, what is in effect, a single storage subsystem that spans both sites. In this design, a given datastore must be accessible simultaneously from both sites. Furthermore, when problems occur, the surviving ESXi hosts must be able to continue to access datastores from the surviving array transparently and do so without impact to ongoing storage operations.

The storage subsystem for vMSC reads from and writes to both locations, though, depending on pathing, it does not have to use both sites actively. For active/active solutions, all disk writes are committed synchronously at the two locations to ensure that data is always consistent regardless of the location. This storage architecture requires significant bandwidth and low latency between the sites involved in the cluster. Increased distances or latencies cause delays writing to disk, making performance suffer dramatically, and prevent vMotion activities between the cluster nodes that reside in separate locations.

**Note:** VMware (and therefore SRDF/Metro on VMware) supports ninety-six(96) hosts in a cluster in vSphere 8 and 9.

**Note:** Dell does not support Virtual Volumes (vVols) with SRDF/Metro.

## 3.1 VMware vMSC with SRDF/Metro

Dell SRDF/Metro has the ability to concurrently access the same set of devices independent of the physical location and thus enables geographically stretched clusters based on VMware vSphere. This forms the basis of a vSphere Metro Storage Cluster. This allows for transparent load sharing between multiple sites while providing the flexibility of migrating workloads between sites in anticipation of planned events such as hardware maintenance. Furthermore, in case of an unplanned event that causes disruption of services at one of the data centers, you can quickly restart the failed services the surviving site with minimal effort with the combined functionality of vSphere HA. Nevertheless, the design of the VMware environment has to account for a number of potential failure scenarios and mitigate the risk for services disruption.

### 3.1.1 Architecture

At a high conceptual level, a vMSC utilizing SRDF/Metro would appear similar to Figure 14.

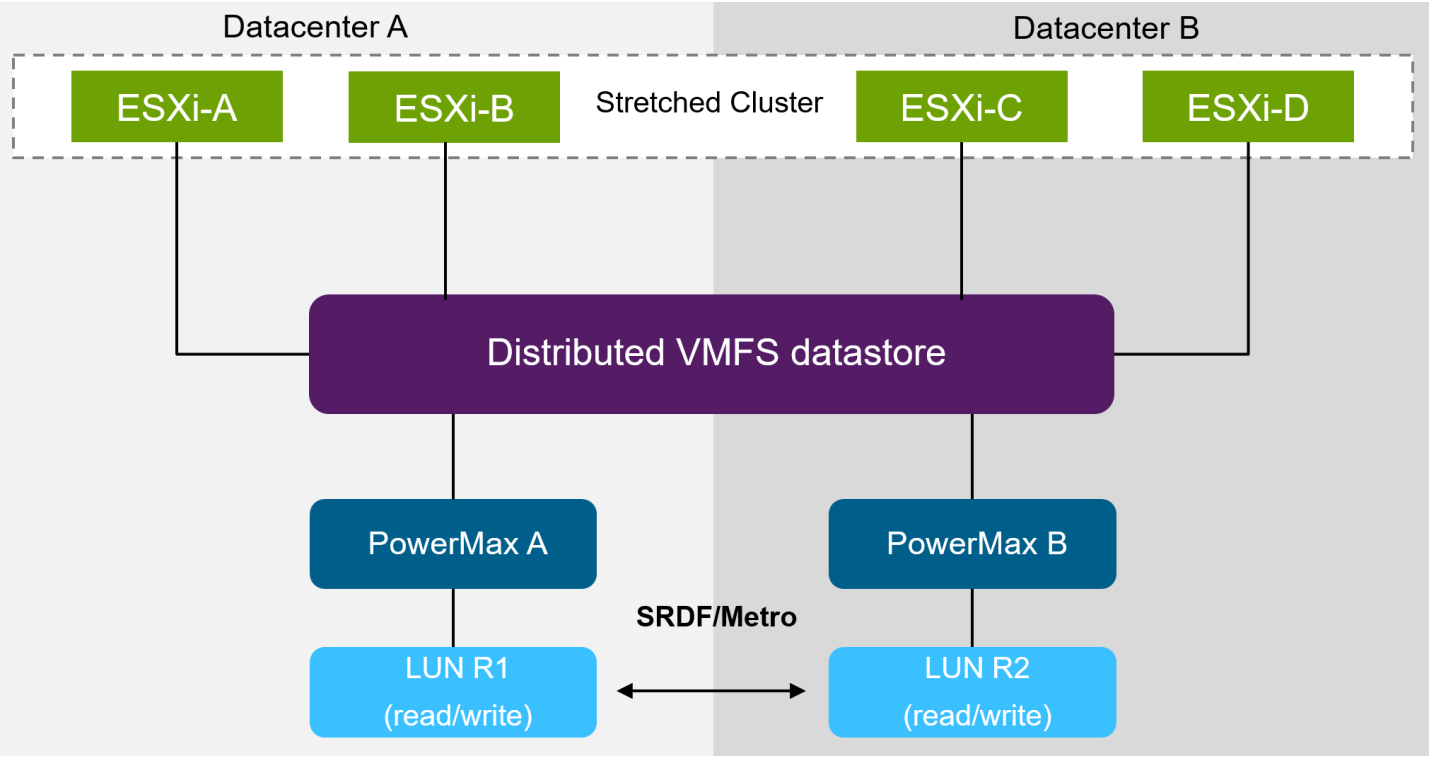



Figure 14. vMSC nonuniform configuration with SRDF/Metro

ESXi hosts A – D, despite being in separate datacenters, are in the same vCenter. Half the hosts see the R1 on PowerMax A, the other hosts see the R2 on PowerMax B; however, they all see the same VMFS datastore because from VMware's perspective, the R1 and R2 are the same device (i.e., they share an external WWN).

SRDF/Metro maintains the cache state on each array, so an ESXi host in either datacenter detects the "virtual" device as local. Even when two virtual machines reside on the same datastore but are located in different datacenters, they write locally without any performance impact on either of them. Each host accesses its own array with SRDF/Metro ensuring consistency. If a witness is not in use, there is site bias, meaning that one site will be the winner in the event of failure.

You can zone the ESXi hosts in a vMSC one of two ways. These follow.

## 3.2 Uniform and Nonuniform vMSC

There are two types of configurations available for vMSC: uniform and nonuniform. Dell supports either configuration. VMware defines them as:

- Uniform host access configuration – When ESXi hosts from both sites are all connected to a storage node in the storage cluster across all sites. Paths presented to ESXi hosts stretch across distance.
- Nonuniform host access configuration – ESXi hosts only connect to storage node(s) in the same site. Paths presented to ESXi hosts are local.

**Note:** Some non-PowerMax uniform vMSC configurations are active/passive configurations where only one device of the pair is active at one time. One array owns the device pair at any one time. This configuration is site bias (unrelated to SRDF/Metro) or LUN locality. No matter which host the IO comes from, it directs it to the array that owns the device. In these configurations it is typical to split equally the device pair ownership across the two storage environments, but of course both datacenters see both storage arrays.

### 3.2.1 General recommendation

While Dell supports both uniform and nonuniform configurations without restriction, Dell recommends nonuniform configurations. There are a number of reasons for this. Uniform or cross-connect configurations add to the complexity of the SAN design (bridging between sites) and can lead to zoning issues. Losing one of the datacenters will also generate SAN fabric events which will impact surviving hosts, and perhaps VMware HA taking place. From the hardware standpoint, and short of a complete disaster, it is also unlikely the array is the component that is going to fail in the datacenter, it will be the servers. Most arrays, but in particular the PowerMax, are incredibly resilient and in fact in the event of RAID failures in SRDF configurations, the PowerMax can even read from the remote array. Keeping the two environments isolated will reduce complexity and minimize any chance of one impacting the other. Finally, and most importantly, having servers in one datacenter accessing the array in the other datacenter will add latency to the application unless properly addressed in pathing. Pathing, therefore, is critical in a vMSC and we cover it next.

**Note:** If the vMSC environment is part of SmartDR and utilizing VMware Live Site Recovery, Dell recommends a uniform configuration, rather than nonuniform, to take advantage of the features of the SRDF SRA.

## 3.3 Pathing considerations

You can use two different technologies for path management in an SRDF/Metro vMSC: PowerPath/VE or PP/VE, and Native Multipathing or NMP. Whether both options are available for a particular customer depends on their current licensing of PowerPath. The next section addresses this.

### 3.3.1 PowerPath EOL

PowerPath went end-of-life (EOL) on July 31, 2024. Dell will not sell any new licenses. Dell will maintain standard support for PP/VE until July 31, 2027. Only customers with existing licenses may continue to use PP/VE until the end of standard support. The following KB article details PowerPath EOL: https://www.dell.com/support/kbdoc/en-us/000223522.

All customers will need to utilize NMP in their VMware environments in the future. Customers currently using PP/VE must transition to NMP at some point before July 2027. It is a disruptive process, but you can do it in a rolling manner since it is possible to have some ESXi hosts use PP/VE while others use NMP. We cover both solutions below as Dell still has many VMware customers using PP/VE.

### 3.3.2 PowerPath/VE Autostandby

Although Dell recommends a nonuniform configuration, PP/VE offers a solution for uniform or cross-connect configurations that removes the risk of latency and dead path issues. This feature is known as Autostandby or ASB. In a uniform vMSC, when ASB is enabled (default) on PP/VE, ASB is able to determine which device paths have a higher latency to storage, automatically assigning them to a standby status.

Figure 15 lists the PP/VE defaults.

**Note:** The detail supplied for PowerPath/VE below is based on version 7.2. Each version of PowerPath makes improvements to functionality and as such if using a different version than 7.2, consult the documentation. This does not change the general best practices, however.

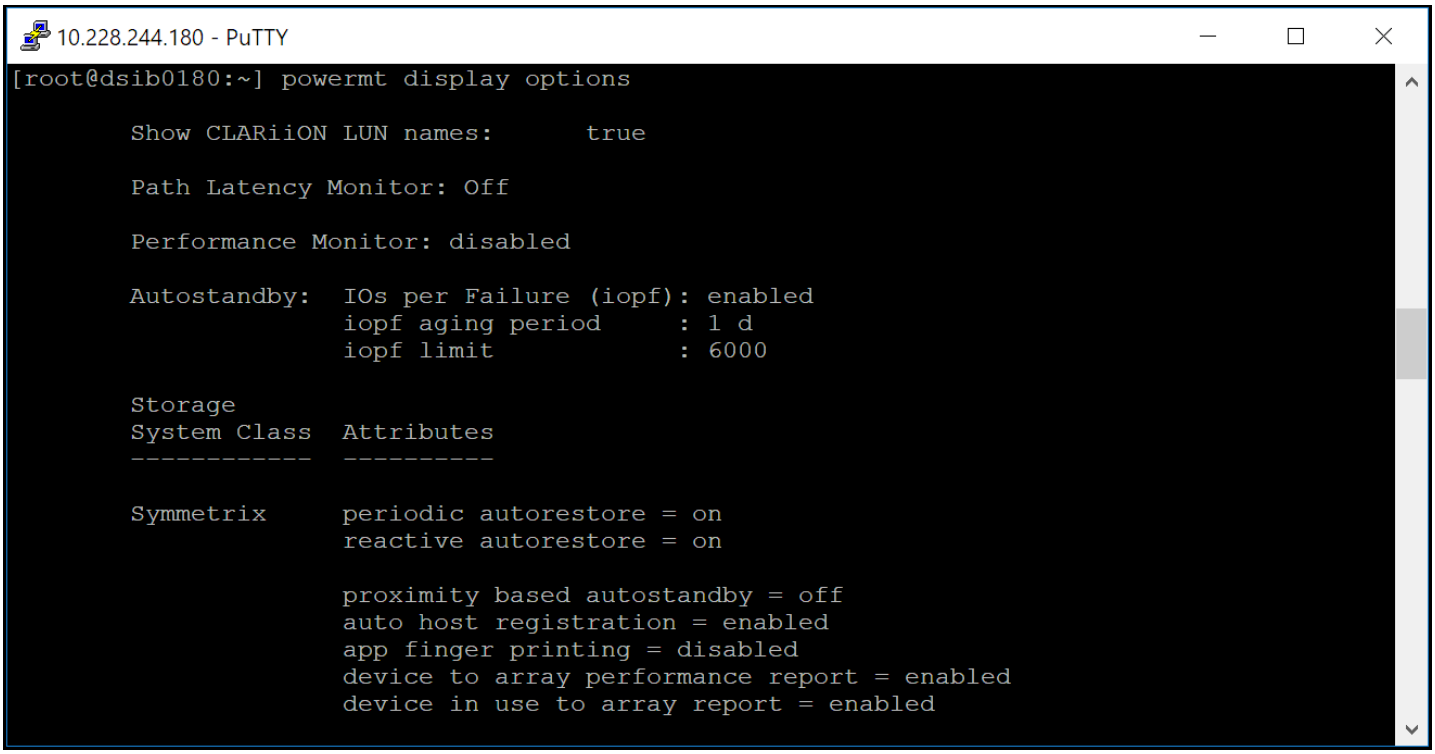



Figure 15. PP/VE defaults

Autostandby has two different modes it can use: proximity and IOs per failure. There is a third mode available known as Autostandby offline which can detect maintenance on ports, but it does require a recent version of PowerMaxOS and as it is not pertinent to the vMSC discussion, we will discuss it. We cover the other two modes, however. Note that the Autostandby feature has changed a number of times since inception. Be sure to review the PowerPath Product Guide for the version used in the environment to understand the Autostandby behavior.

**Note:** Autostandby paths do not persist through reboot. After reboot PowerPath will re-test the paths for latency and assign Autostandby as appropriate.

#### 3.3.2.1 IOs per failure

The first mode is IOs per failure which is enabled by default and labeled as ***asb:iopf***. IOs per failure is most useful in situations where there is the potential for “flaky” paths. When the path has ‘x’ number of failures, the path goes into standby for the aging period (adjustable). Dell does not recommend this mode for vMSC configurations.

#### 3.3.2.2 Proximity

The second is proximity based Autostandby which you can identify by the label ***asb:prox*** on a path. PP/VE disables it by default. This is the mode most pertinent to a uniform vMSC setup as it can determine which paths are remote and which are local. In addition, proximity based Autostandby has a secondary option that you can supply called threshold. The threshold is a latency value which is set and determines when a path should be set to either active or standby. By default, the threshold is zero(0), however it can be set to a value ranging from 0-5000 microseconds. With the threshold set, as long as the latency difference between two paths is above the value, PP/VE will set Autostandby, or asb:prox, on the path(s) with the higher latency. If all paths are below the threshold, then they will be set to active, and none to asb:prox. The idea behind the threshold setting is that if the environment can tolerate up to a particular latency, then it may be desirable to use a threshold value up to that latency to ensure that all paths are set to active. Note that the

default value of zero for the threshold guarantees that if you are running a uniform configuration that one of your array paths will be set to asb:prox. If it is essential that all paths are in an active state, either disable Autostandby (see below), or set the threshold to a high enough value to ensure the latency to the remote array in comparison to the local array will not exceed it.

#### 3.3.2.3 Examples

Since Dell recommends Autostandby with PP/VE and uniform, the following example details how it might work in a real-world environment.

This VMware vSphere Metro Storage Cluster setup is comprised of only two hosts, each attached to its own array in different datacenters. The table in Figure 16 contains the pertinent information for the example. Note each array uses different FA ports, so it is apparent which array is local, and which is remote.

| Host | Device | Site | FA Ports | Array |
|---|---|---|---|---|
| dsib0180.lss.emc.com | 586 | R1 | 1D:4, 2D:4 | 000197600357 |
| dsib0182.lss.emc.com | FE | R2 | 1D:6, 2D:6 | 000197600358 |

Figure 16. Autostandby example environment

As array 357 is the R1, it will supply the external identity to device FE on array 358, the R2. In this example, then, both devices share WWN 6000097000019760035753303035383 6. We presented each device to both hosts on each FA port, resulting in four paths, two to the R1, two to the R2. Figure 17 shows the emcpower1 device on the R1 host prior to enabling the Autostandby feature. Note all paths are active, a true uniform configuration.

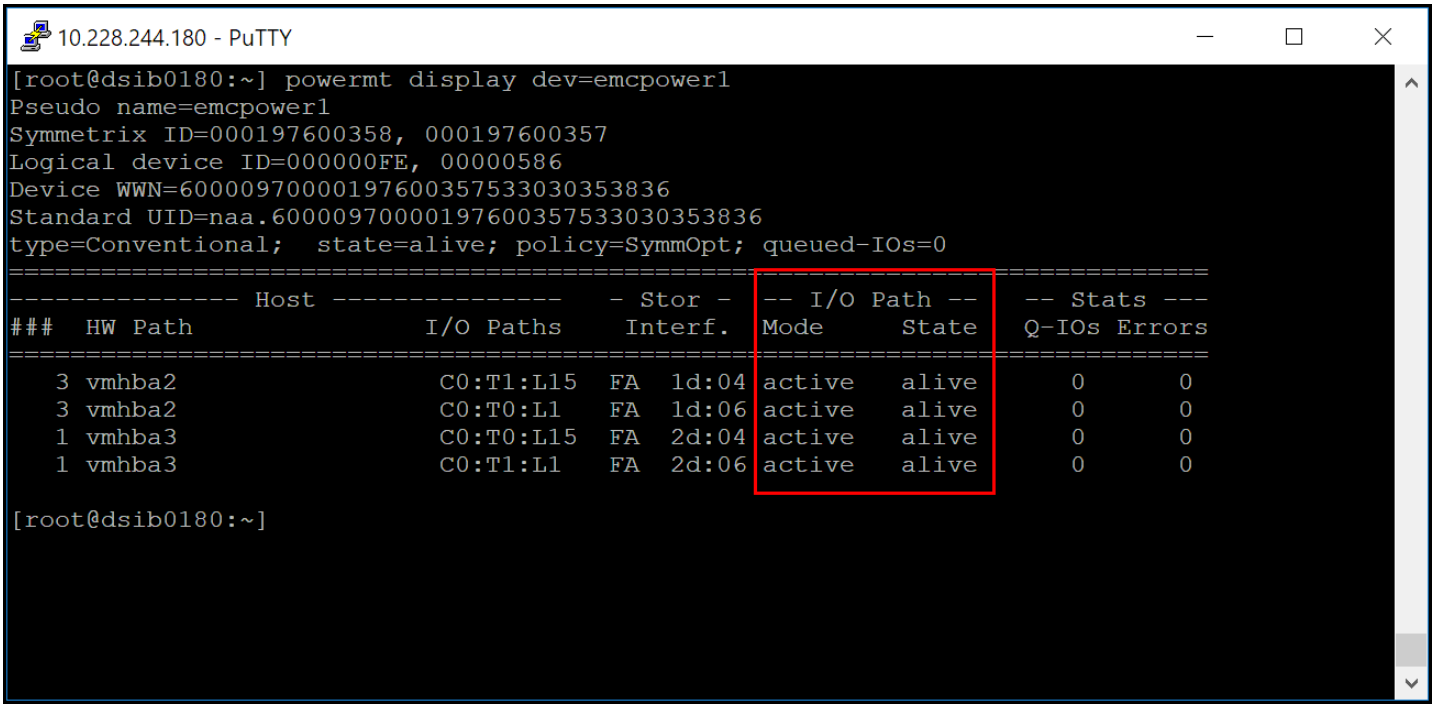



Figure 17. PP/VE path assignment prior to Autostandby activation

Now, enable Autostandby and set the trigger to “prox” or proximity. This will override the default of IOs per failure. PowerPath will immediately determine which paths are local to the host and which

are remote. Figure 18 again shows device emcpower1 on host dsib0180 after enabling Autostandby.

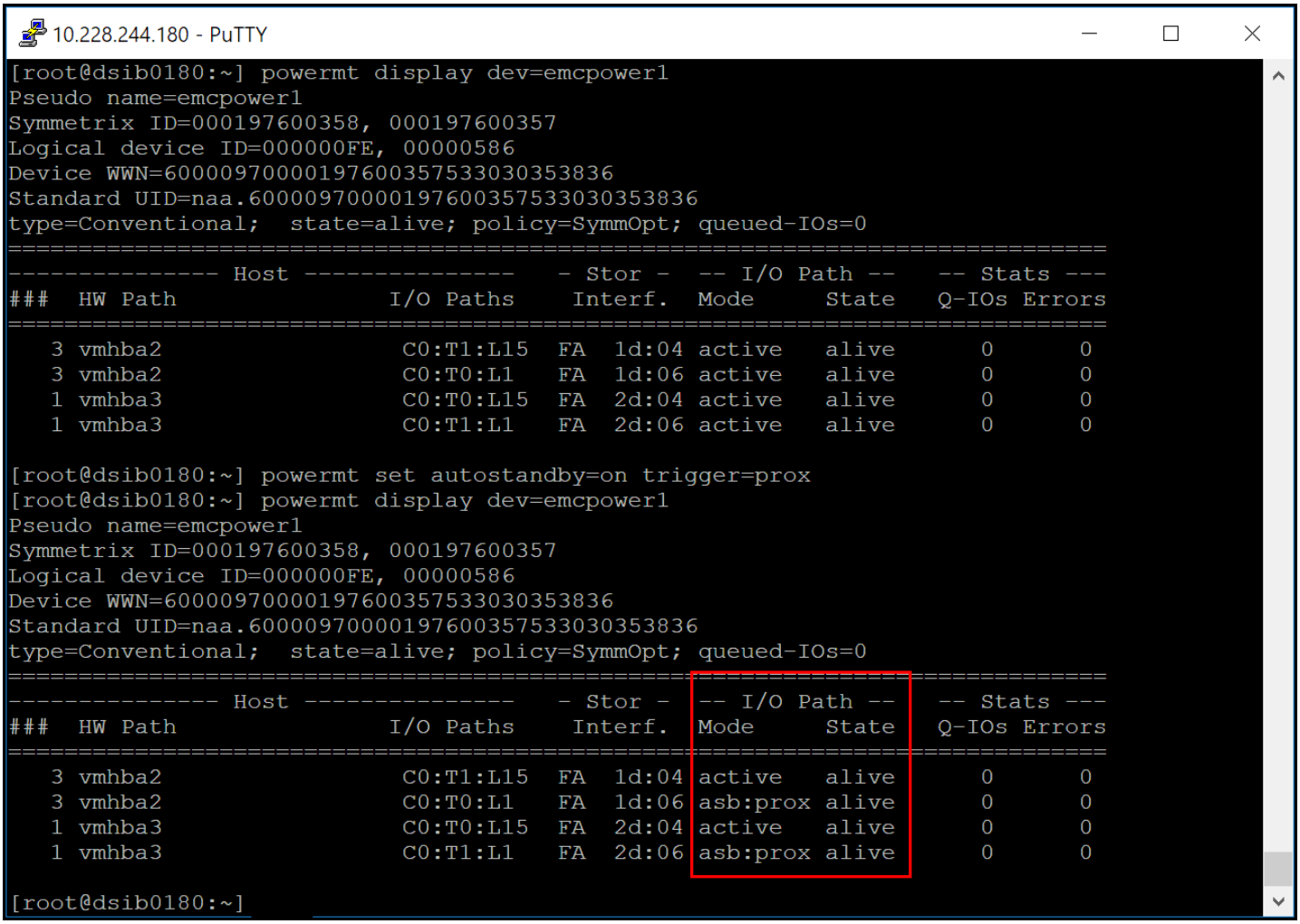



Figure 18. PP/VE path assignment in datacenter A

Notice that the two paths to the R2 device have been set to asb:prox, or Autostandby proximity, while the other two are active. This is the expected result. What this means is that PP/VE evaluated the paths (with default zero threshold) and determined that the paths going to device FE on array 358 had a higher latency than those going to device 586 on array 357 and set them accordingly for dsib0180. And on dsib0182 in Figure 19, PowerPath determines the opposite settings for emcpower2 (which represents the same devices) since array 357 is now the remote one.

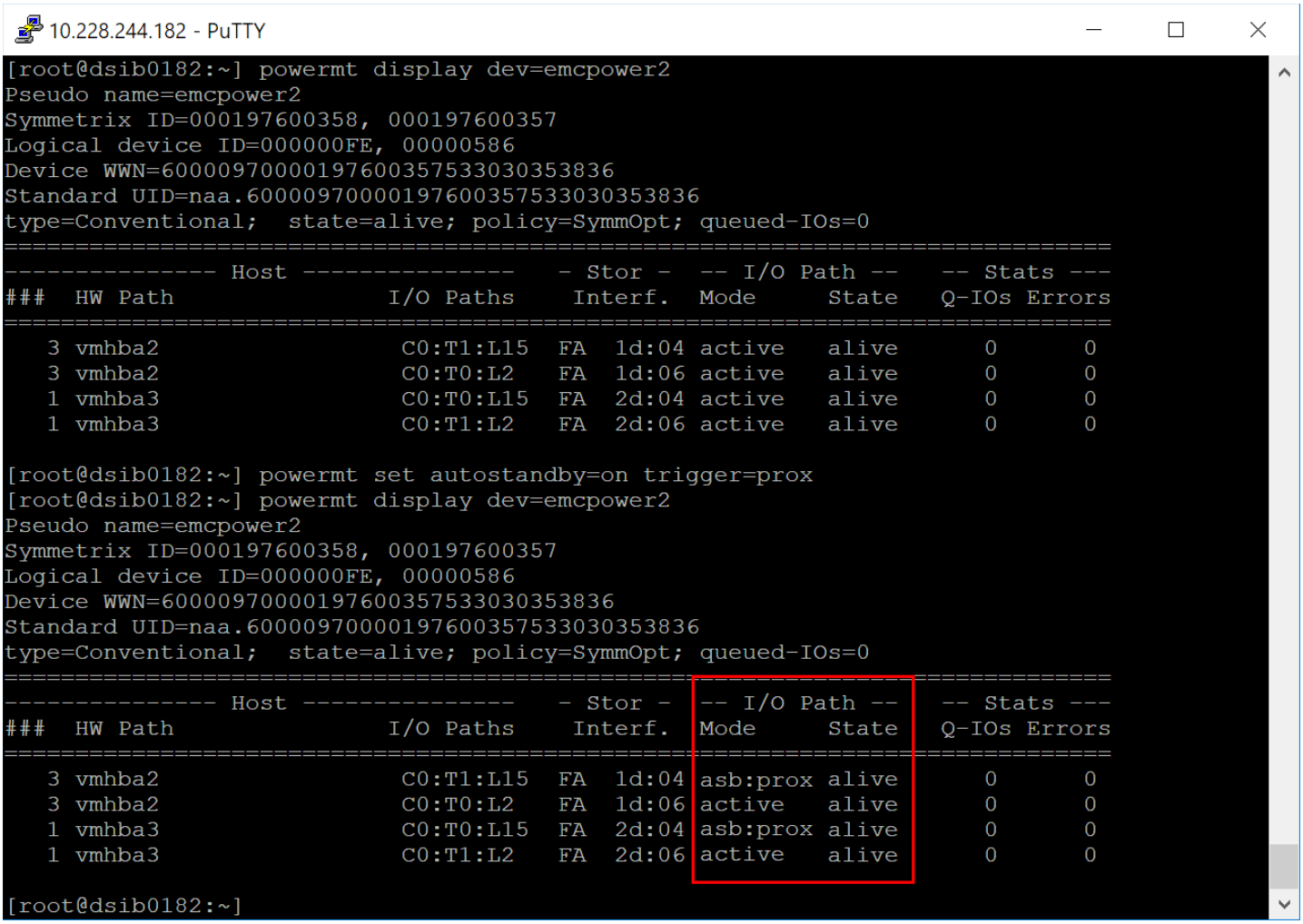



Figure 19. PP/VE path assignment in datacenter B

During normal operation, each ESXi will use only the two active, local paths, but if there is a failure on one of the arrays, one of the ESXi hosts will have IO redirected to the asb:prox paths. Despite the fact the paths are in standby, PowerPath can still use them if it deems it necessary even without a failure. For instance, if one path to the active array becomes extremely slow, PowerPath might choose to send some IO to the standby path.

### 3.3.2.4 Threshold change

Some customers may wish to use a true uniform configuration while latency to the remote array in comparison to the local array remains below a certain threshold. If that is the case, you can adjust the default threshold for Autostandby. Simply follow these steps. Note here that we supply the optional parameter of “class” in case there are other arrays attached to the ESXi host which the customer does not wish to change.

Use the value “reinitialize” to dynamically adjust the threshold:

```
powermt set autostandby=reinitialize trigger=prox class=symm threshold=5000
```

After the change, PP/VE sets the new mode on the device in Figure 20:

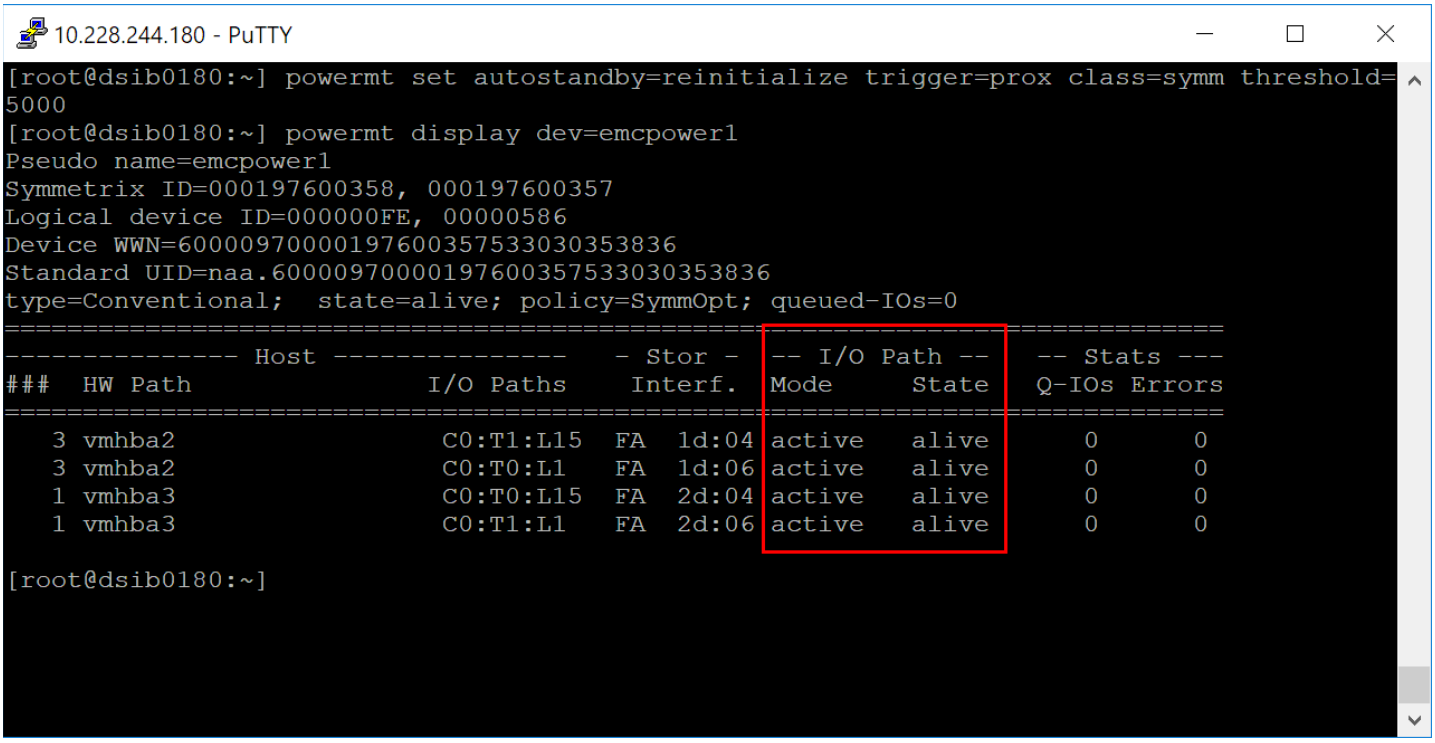



Figure 20. PP/VE settings after change in threshold

After the change, all the paths show as active since the latency difference between the arrays is lower than the threshold. If at some point during normal business operation it appears that one of the arrays is causing latency issues, issuing a reinitialize which will cause PP/VE to re-test the paths and set any paths to asb:prox that exceed the threshold:

```
powermt set autostandby=reinitialize trigger=prox class=symm
```

For customers who have a true stretched cluster where the arrays are co-located and already know they do not wish to use Autostandby, the PowerPath default configuration is appropriate.

### 3.3.2.5 Manual path setting

One final example of using Autostandby is a customer who has a stretched cluster (or co-located arrays) who nonetheless wishes to only use the secondary array as a standby. This environment is a bit tricky for Autostandby because as shown above, PP/VE will set paths to asb:prox in the default configuration and it may not choose the R2 device on the R1 side or the R1 device on the R2 side or if the latency is exactly the same it will set all paths to active. The best way to manage this is to not use Autostandby but rather set standby manually for the paths. The command to do that is:

```
powermt set mode=standby hba=<hba#> dev=all class=all
```

Here are a couple of examples:

```
rpowermt set port_mode=standby dev=vmhba2:C0:T0:L7

rpowermt set mode=standby hba=1 class=symm
```

If it is preferable to keep Autostandby active, changing the paths to active or standby requires the “force” option. Figure 21 demonstrates setting the mode from asb:prox to active.

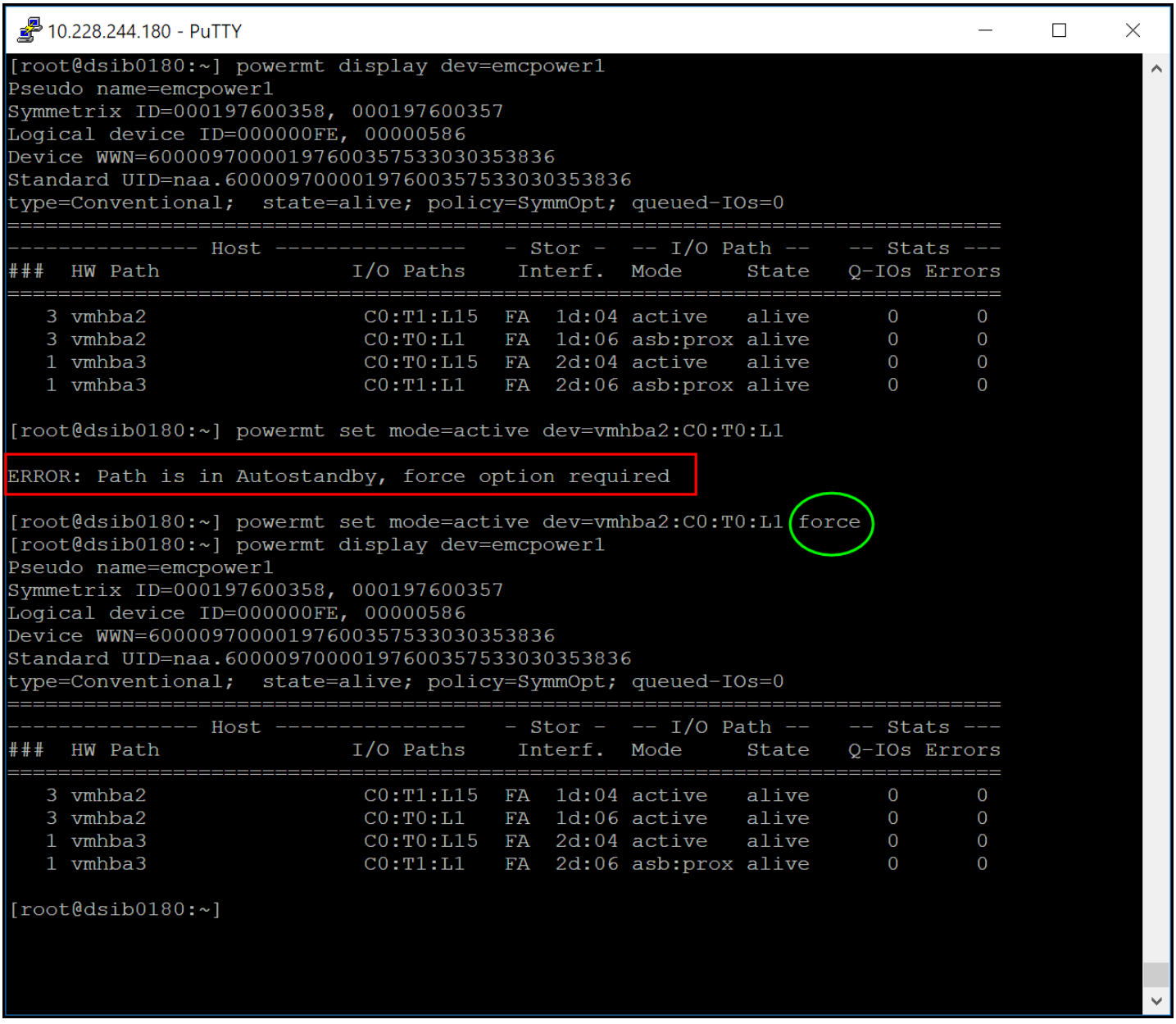



Figure 21. Forcing changes in PP/VE modes

To move a path from active to standby will not require force because the mode is not Autostandby. See this in Figure 22.

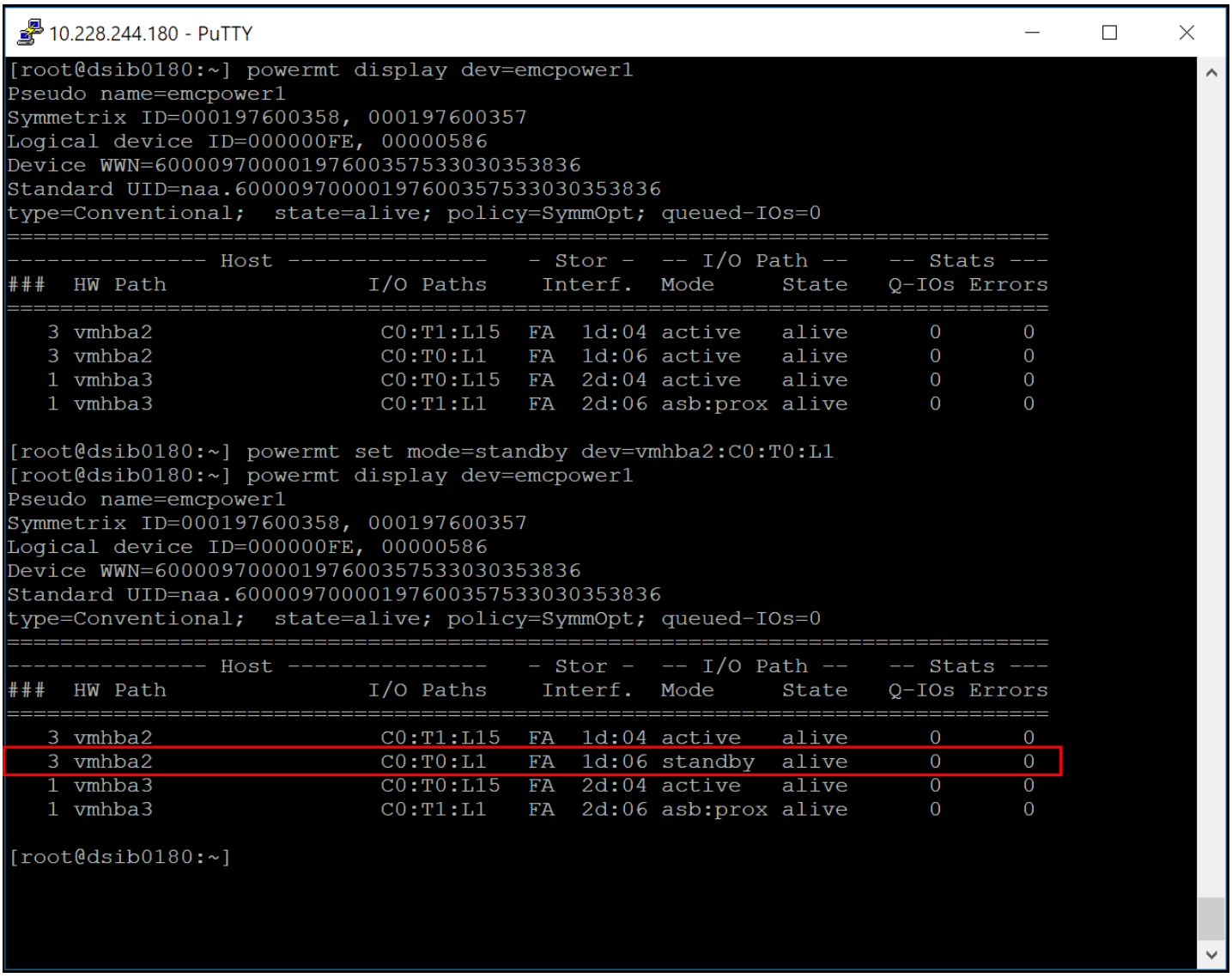



Figure 22. Changing from active to standby path in PP/VE

**Note:** Setting a path to standby does not mean PP/VE will only use it in the event of failure, rather that its use is heavily weighted against the active paths.

One final caveat about Autostandby is if the user improperly assigns the R1 path to Autostandby, any non-Metro devices presented from the R1 array also will be set to Autostandby. Since all paths have the same value, PP/VE will still use them despite being asb:prox; however, the user should manually change them to active. To fix this, simply change the paths to active, remembering to use the force option. Path changes like those demonstrated above will not persist through reboot, unfortunately. It is possible to create a script to assign paths to standby upon reboot.

**Note:** There is no equivalency for Autostandby in NMP. In uniform configurations with NMP, all paths are active. You can only control the path selection policy. Furthermore, it is not possible for NMP to distinguish between the local from the remote PowerMax even when using ALUA.

### 3.3.3 NMP and Latency Round Robin

VMware offers a type of Round Robin NMP called “latency.” The capability enables VMware to evaluate the performance of the paths to a device and route IO appropriately. The feature is known as **Latency Round Robin.**

There are currently only two "types" of pathing available with NMP that are in use with PowerMax – Fixed and Round Robin. Dell no longer supports the third type, Most Recently Used (MRU). Since Fixed is path restrictive, Round Robin is the default policy (VMW_PSP_RR) for PowerMax devices presented to an ESXi host as seen in Figure 23. Note Dell still uses “EMC Symmetrix” as the description and part of the name (SYMM) despite no longer calling the arrays by that name. The reason for this is it provides consistency across ESXi releases to leave it as that name. It is simply a moniker, however, and has no bearing on functionality.

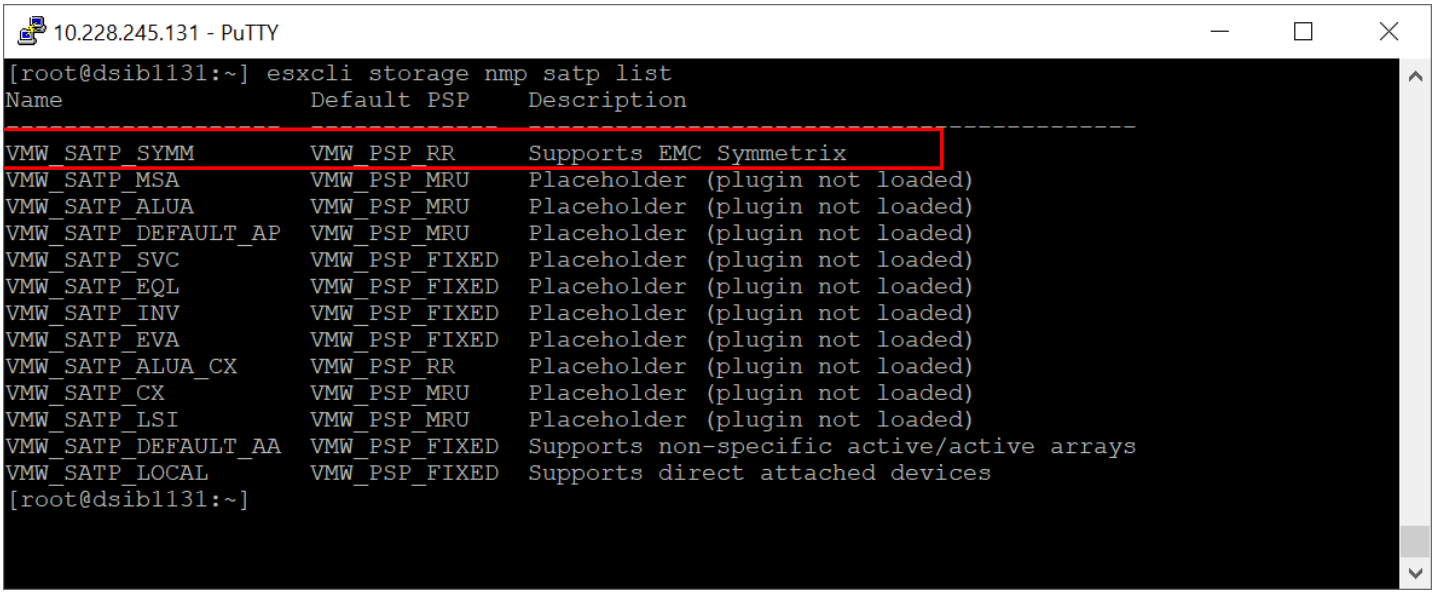



Figure 23. Default SATP for VMAX or PowerMax devices

One thing that has always been lacking in VMware’s NMP implementation is the ability to consider the response time/latency of the path in use. The Round Robin latency setting solves this. When type=latency is set on a device, VMware uses both the latency and pending IOs of a path to determine whether it should send an IO down a particular path.

The parameter that controls latency is known as EnablePSPLatencyPolicy. It is available both in the CLI:

```
[root@dsib0142:~] esxcfg-advcfg -g /Misc/EnablePSPLatencyPolicy
Value of EnablePSPLatencyPolicy is 0
```

and in the UI in Figure 24.

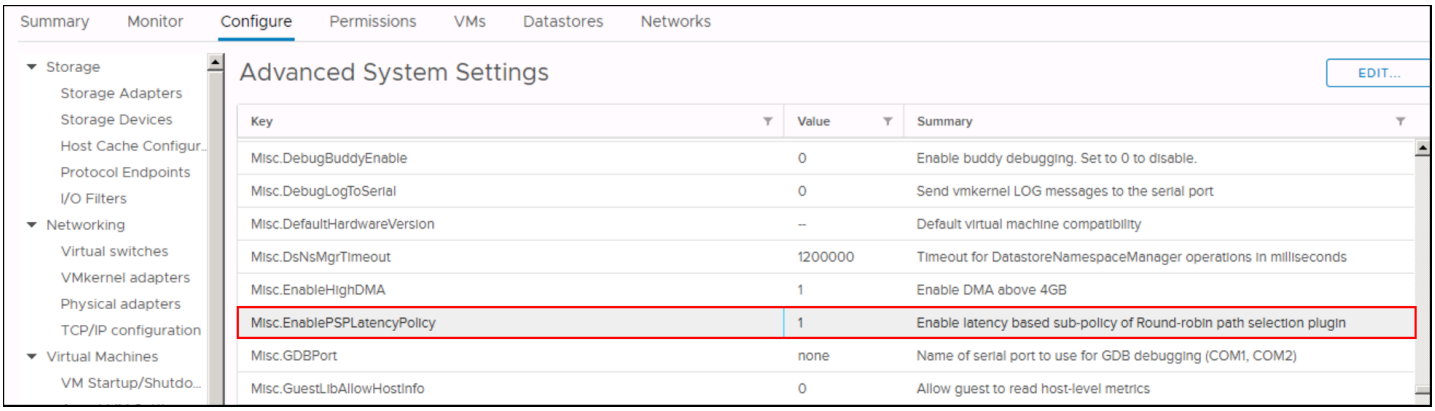



Figure 24. Setting NMP latency in the vSphere Client

If enabled (default), the user can dynamically apply the new type to a device(s):

```
esxcli storage nmp psp roundrobin deviceconfig set -d <Device_ID> --
type=latency
```

Here in Figure 25 is an example of setting latency for a device followed by the command to list the configuration:

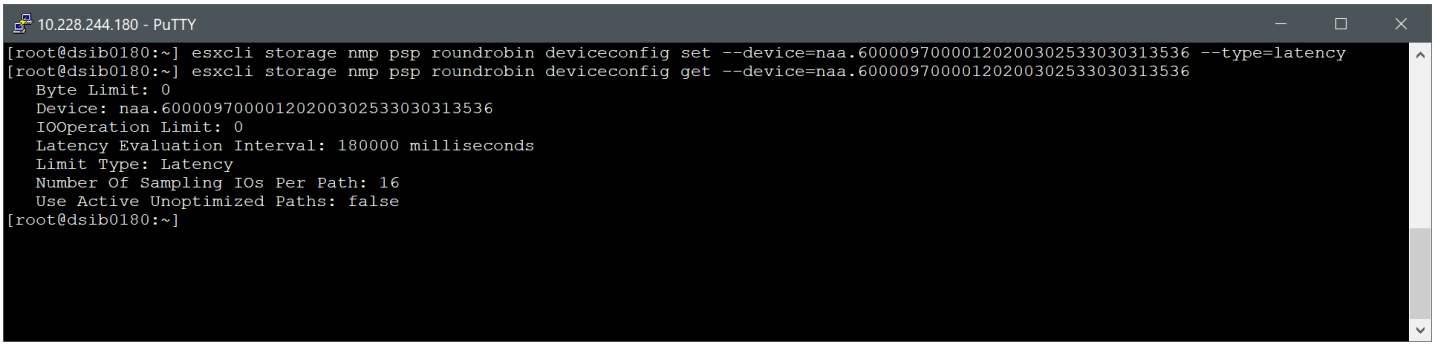



Figure 25. Setting type=latency on a PowerMax device

It is also possible to create an SATP rule that will ensure VMware assigns future devices the type of latency. The command is below:

```
esxcli storage nmp satp rule add -s "VMW_SATP_SYMM" -V "EMC" -M "SYMMETRIX" -
P "VMW_PSP_RR" -O "policy=latency"
```

Be sure that no other user rules exist, or VMware will return an error. For instance, if an iops=1 rule is already present you need to remove it first by running:

```
esxcli storage nmp satp rule remove -s "VMW_SATP_SYMM" -V "EMC" -M
"SYMMETRIX" -P "VMW_PSP_RR" -O "iops=1"
```

In deciding which path to use for a particular IO, VMware monitors all the active paths and calculates the average latency based on either time or number of IOs. Once latency is set on a device, VMware uses the first 16 IOs per active path to calculate the latency. VMware will direct subsequent IOs to the path with the lowest latency. After the initial sampling window, VMware will re-test every 3 minutes by default. The sampling time and amount are present in a device interrogation in the parameters **Latency Evaluation Interval** and **Number of Sampling IOs Per Path** seen in Figure 25.

You can change the time between re-assessing the paths and the number of IOs to use for the sample, though VMware recommends against it as testing shows those values to be effective.

#### 3.3.3.1 Uniform

Uniform vSphere Metro Storage Clusters using SRDF/Metro are a perfect use case for employing the latency type in vSphere 8 or 9. Rather than blindly switching paths every 1000 IOs, VMware will instead evaluate the paths, and if the arrays are any distance apart, VMware will send the majority of the IO down the local paths to the local array. In practice, it turns a uniform configuration into a mostly nonuniform configuration, while still maintaining the remote paths in the case of failure.[4] In environments with significant latency between datacenters, this mechanism can provide better results than the default configuration. We cover the test which demonstrate this next.

#### 3.3.3.2 Nonuniform

The use of the latency type in nonuniform settings has not shown to be more effective than the default recommendation and therefore Dell does not recommend using it in these configurations; however, in environments that nonetheless experience significant congestion on a single array it can be beneficial.

Dell makes these recommendations after extensive testing which in included below.

#### 3.3.3.3 Testing Results

As part of the pathing best practices, Dell evaluated the differences between iops=1000 (default NMP) and the new latency type in a uniform vMSC. The uniform setup utilized a distance simulator between the arrays to mimic a campus cluster with IOMETER generating load.

Figure 26 is a graph of one particular test of the uniform vMSC using IOMETER. It details the following relative results for iops=1000 (blue) and type=latency (orange):

- Average write response
- Average read response
- Average response

The graph also includes the total IOPS in the legend.

[4] When the arrays are co-located, VMware will tend to treat all paths equally, and therefore using Round Robin with iops=1 is an equivalent option in these configurations.

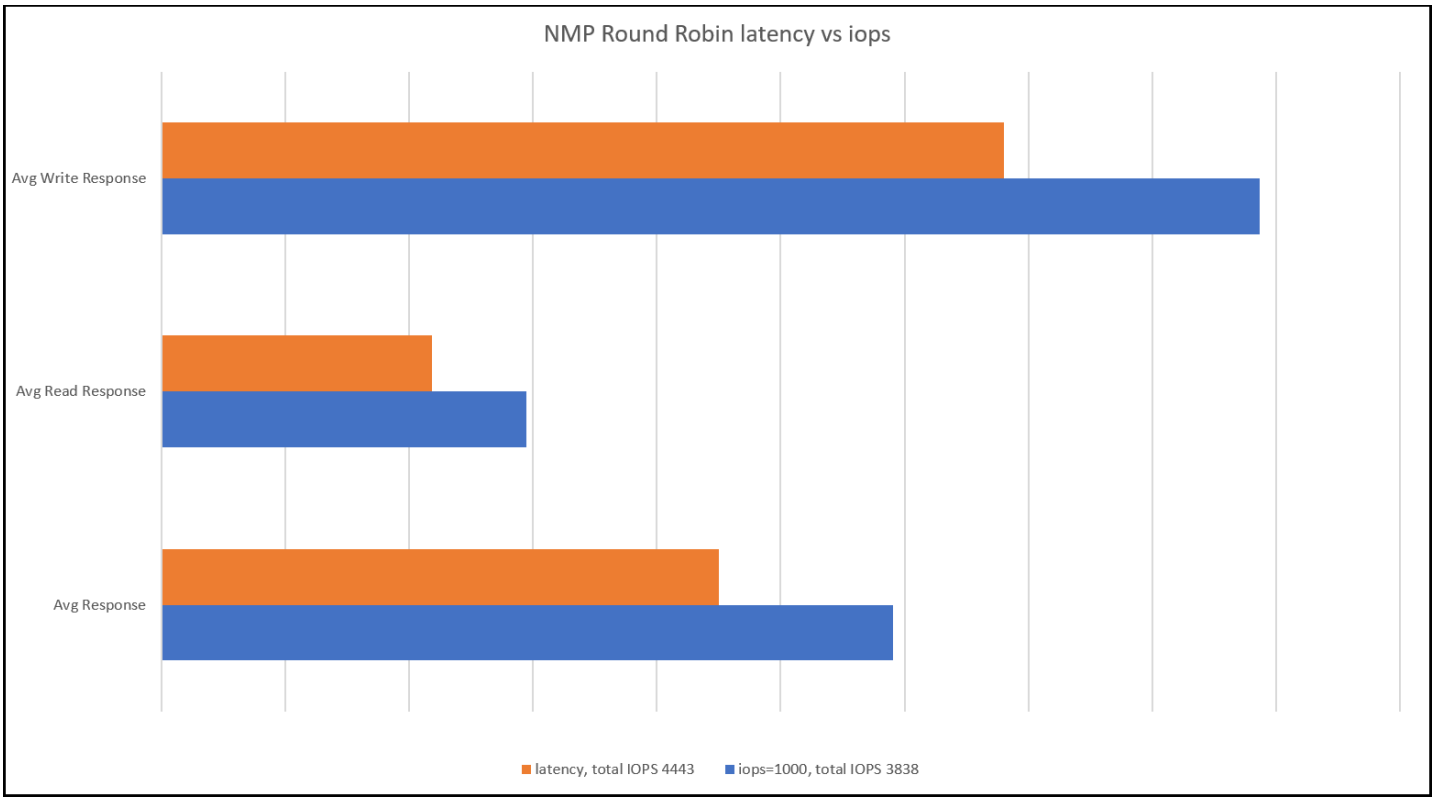



Figure 26. Round Robin type comparison

The results clearly show that the latency type has both superior response times as well as overall IOPS. Detailed analysis determined that while iops=1000 used all paths, regardless of distance, latency used the local paths the majority of the time, producing better performance. These results confirm Dell's recommendation of type=latency when running a uniform vMSC vSphere 8 or 9.

### 3.3.4 Polling time for datastore paths

By default, VMware polls for new datastore paths every 300 seconds. In an SRDF/Metro environment, Dell recommends changing this to 30 seconds to avoid the need to manually rescan after presenting the R2 devices. The value that requires changing is ***Disk.PathEvalTime*** and you must change it on each ESXi host as demonstrated in Figure 27.

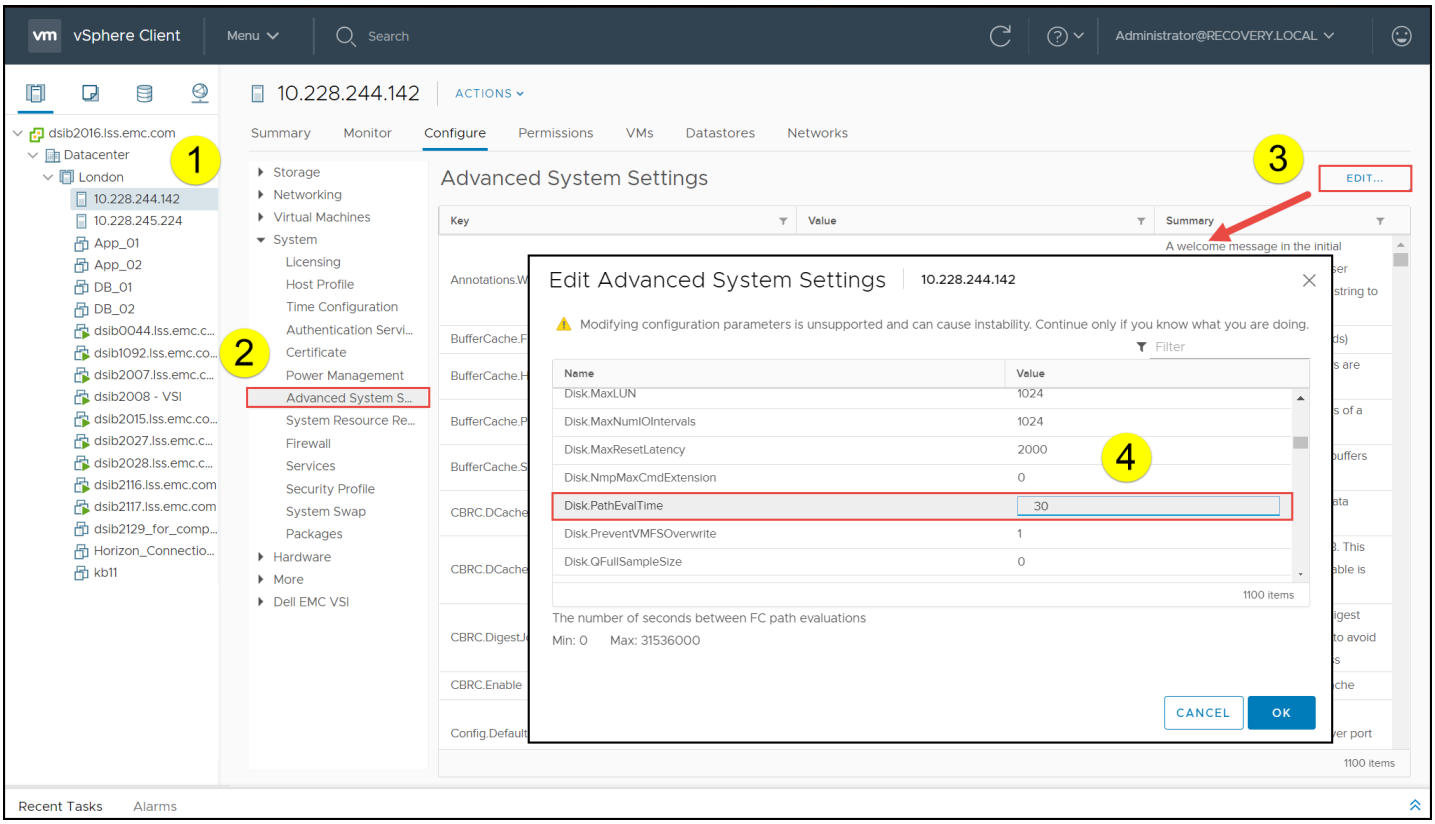



Figure 27. Changing path evaluation time

## 3.3.5 ALUA and Mobility ID

Many array vendors use ALUA with vMSC. On the PowerMax, Dell only supports ALUA with SRDF/Metro when using the Mobility ID (Compatibility ID is default). SRDF/Metro with ALUA considers all paths as active/optimized rather than active/non-optimized which is the more common configuration. Active/optimized treats all paths equally. You cannot change this. Because all paths are the same, active/optimized cannot determine which path is local to the ESXi host. Therefore, it provides no additional benefit to use ALUA on the PowerMax. If your business requirements dictate the use of the Mobility ID, Dell recommends using NMP with type=latency instead of ALUA.

# 4 vSphere Cluster Configuration with SRDF/Metro

This section will specifically focus on the best practices when running vMSC with SRDF/Metro.

## 4.1 vSphere DRS

vSphere Distributed Resource Scheduler or DRS is a VMware utility that balances load across all ESXi hosts in a cluster. Therefore, to properly load balance across the stretched SRDF/Metro cluster, Dell recommends enabling vSphere DRS. **vSphere DRS** is located under the **Cluster -> Configure -> Services** menu in the vSphere Client. Select **vSphere DRS**, then **EDIT**, and move the switch to the right to enable, leaving the defaults in Figure 28.

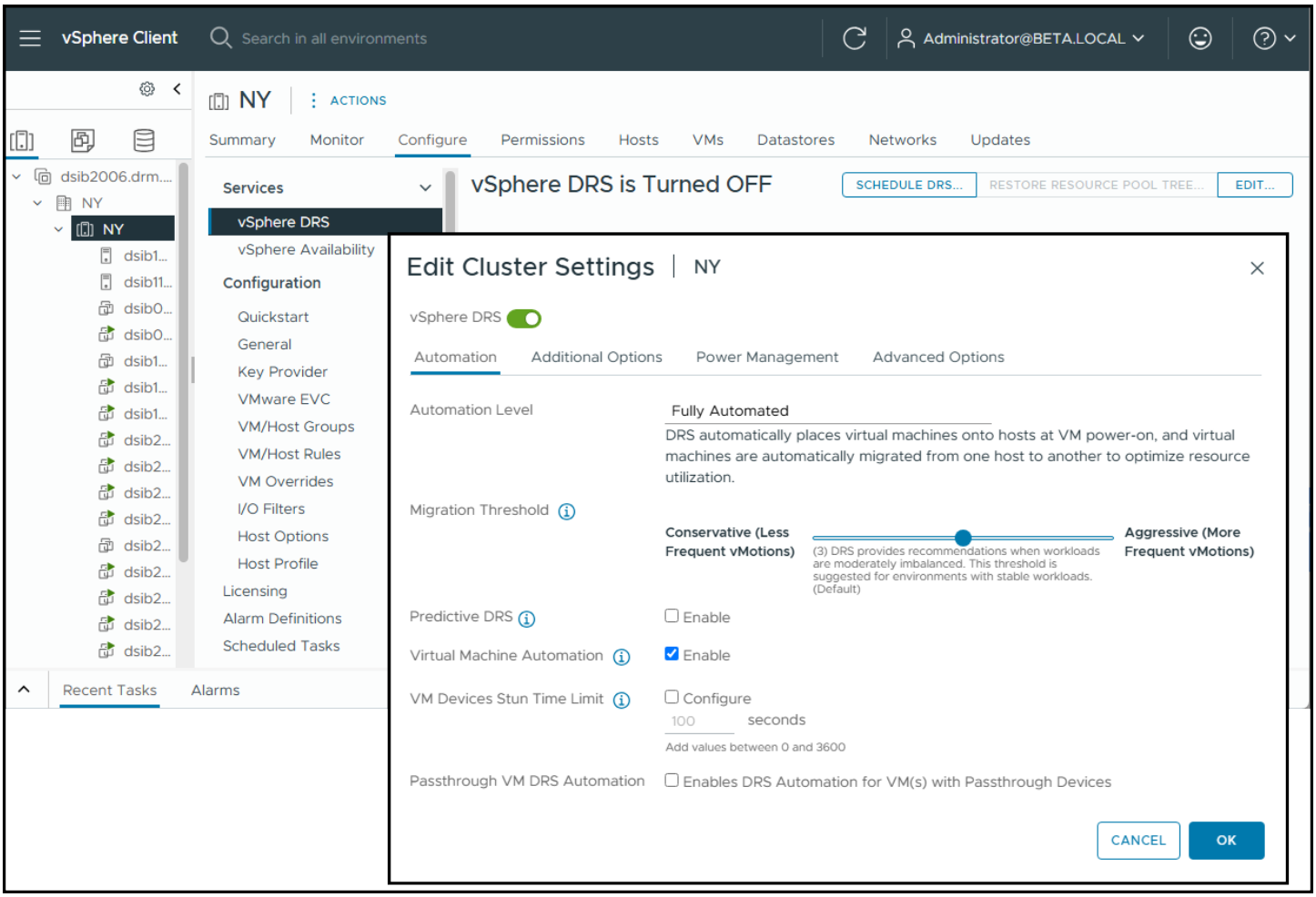



Figure 28. Enabling vSphere DRS in vSphere

vSphere DRS uses host CPU and memory resources to determine how best to load balance across the hosts of a cluster. Depending on the automation level, in order to maintain the CPU/memory equity, VMware will move VMs between ESXi hosts using vMotion. For most environments, however, the default is most appropriate where VMware will make recommendations rather than actually issue the vMotion.

Regardless of load distribution, some customer environments require that particular VMs run at one site (ESXi hosts in one datacenter) except in the event of failure. DRS must comply with these rules. The next section will discuss how to setup these rules.

## 4.2 Site Preference

Many vMSC stretched clusters are an active/passive solution, wherein only one device of a pair is read/write at any time. This means only the hosts from one datacenter access the datastore residing on that device at any one time. In such configurations, therefore, it is essential that the VMs on that datastore "prefer" to run at the read/write site unless there is a failure. This is site affinity. VMware offers the opportunity to create rules that govern this site affinity. Fortunately, the Dell solution of SRDF/Metro is an active/active solution permitting read/write on both devices. This means that a VM can run on any host in the cluster, at any datacenter and does not require these site affinity rules for that purpose; however, that does not mean the rules cannot be useful in an SRDF/Metro vMSC. For instance, take a common multi-tier application like Oracle ERP which requires two application (e.g., Oracle Apps) and two database (e.g., Oracle RAC) VMs. In this environment to provide the best availability, ensuring there is an application and database VM running on each datacenter is preferable. Since DRS only makes decisions based on CPU and memory, it is unlikely to separate VMs properly based on function. Fortunately, through VM/Group and VM/Host rules, users can pre-empt DRS by telling VMware that certain VMs should be associated with certain hosts if possible. DRS will work to enforce these rules when necessary.

### 4.2.1 VMware VM/Host Groups and VM/Host Rules

To ensure that VMs run on the desired hosts and datacenter while the cluster is healthy, it is necessary to create VM/Host Groups and VM/Host Rules. Setting these up is a fairly simple procedure in the vSphere Client. These rules will set forth which VMs "belong" to which host, and thus which datacenter.

To access the wizards to create Host Groups and Rules, navigate to **Cluster -> Configure -> Configuration**, expanding the group. Here in Figure 29 the two categories are: **VM/Host Groups** and **VM/Host Rules**. Note that the DRS automation level must be set to **Fully automated** (default) to permit VMware to move the virtual machines as necessary as previously seen in Figure 28.

**Note:** The screenshots herein are from an early version of vSphere 7. Newer releases have subtle differences in the icons.

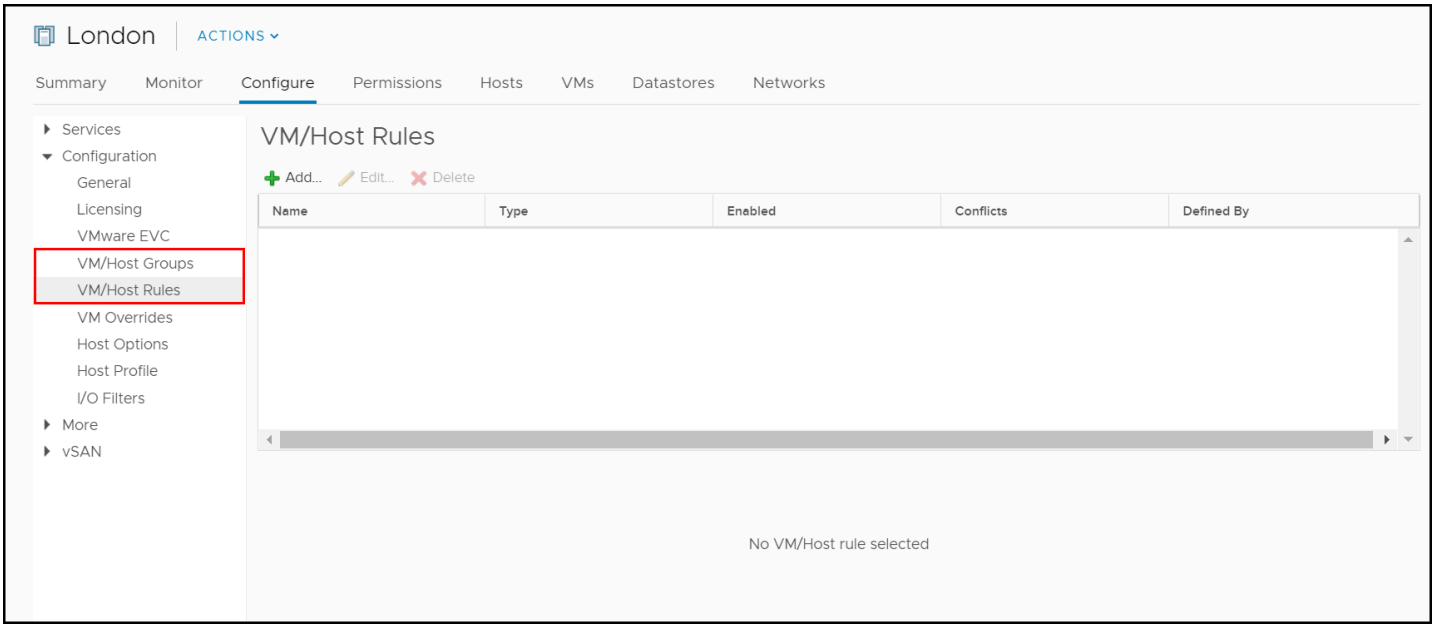



Figure 29. Locating DRS Groups and Rules

For VM/Host Groups, create two VM Groups and two Host Groups, representing the two datacenters. In this example, cluster London consists of an East side and West side datacenter. The East side is the **East_Side_Host_Group**, containing one ESXi host[5], and the West side is the **West_Side_Host_Group** containing the other ESXi host in the cluster. Similarly, each datacenter has a VM Group containing the VMs that should be associated with their respective datacenters. Figure 30 shows the detail of the four groups while Figure 31 is the completed summary.

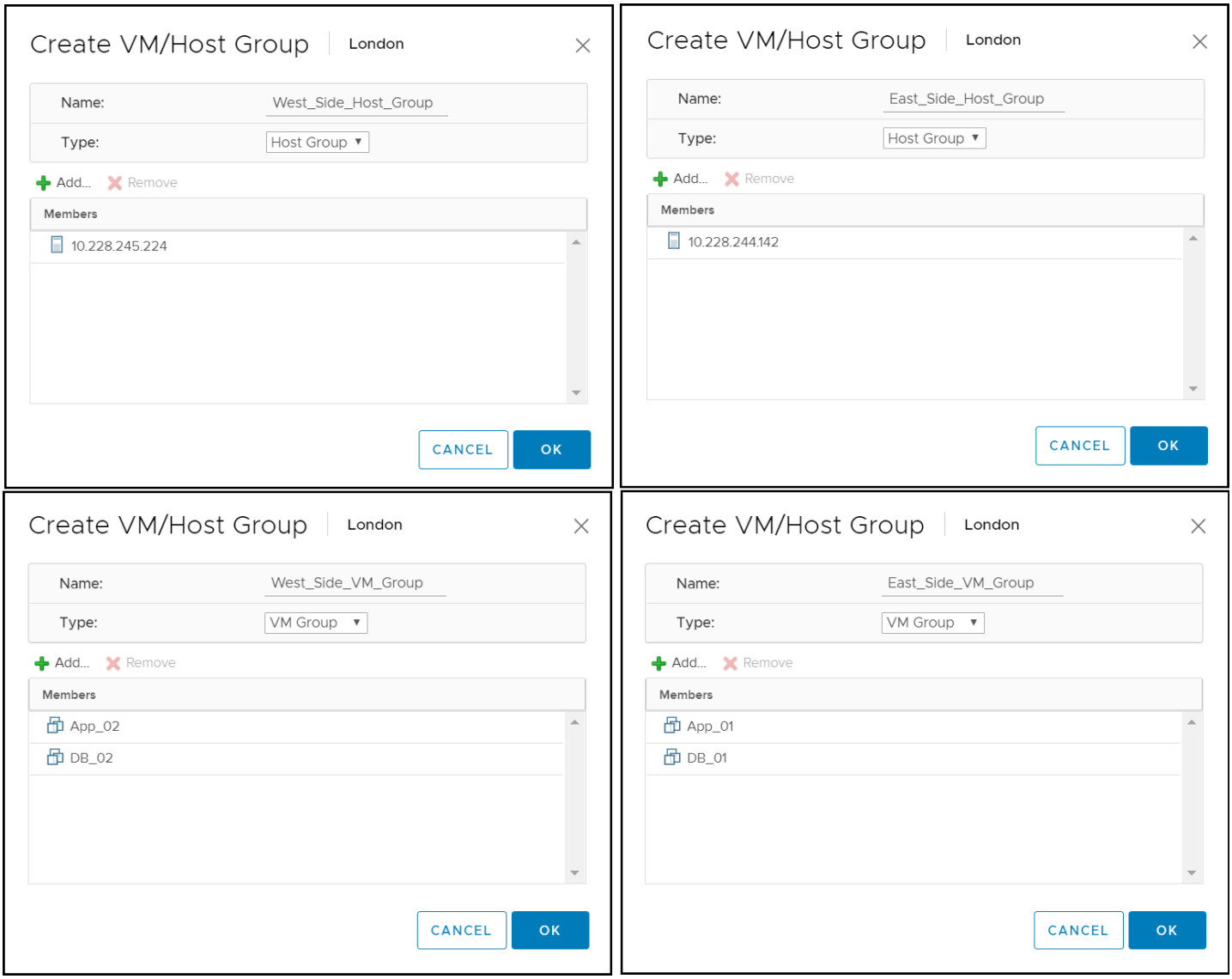



Figure 30. Creation of groups with their hosts and virtual machines

[5] A production environment would contain many more ESXi hosts than this lab example. Add all ESXi hosts present in each respective datacenter to the Host Group. Upon power on, vSphere DRS will place the VM on one of the ESXi servers automatically.

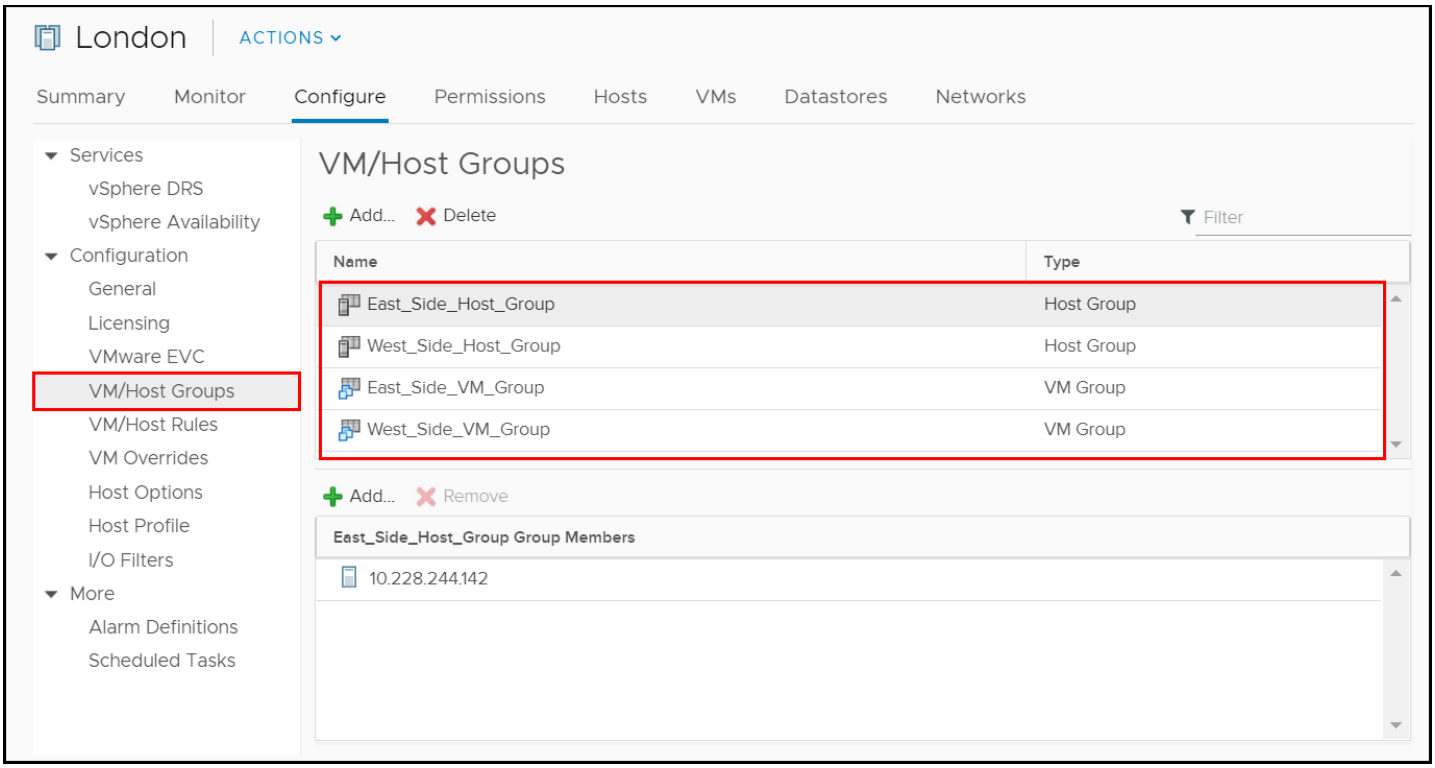



Figure 31. Summary of groups

With groups in place, you need to create rules to govern the site affinity (or host affinity) of the VMs. There are two rules, one that applies to the East side and one that applies to the West side. When setting up the rules, there are the following options in Figure 32 for how the VM Group can operate with Host Group:

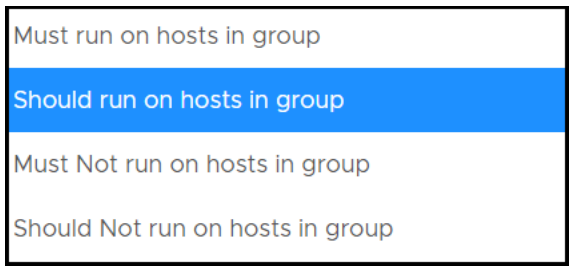



Figure 32. Rule options

It is essential you select the option **Should run on hosts in group** when configuring the rules and not **Must run on hosts in group**. There must be flexibility for the VMs to start-up on the other host(s) that are not part of the Host Group if needed. If you select **Must run on hosts in group**, this will mean in the event of a datacenter failure, VMware cannot restart the failed VMs on ESXi hosts running in the other datacenter. The rules configured here, however, permit VMware to restart the VMs. In addition, if you recover the failed hosts, DRS will automatically migrate the VMs back to their original hosts due to the rule.

Figure 33 has the new rules **East_Side_Rule** and **West_Side_Rule**. Each rule directs that the VMs in the VM Group **Should run on hosts in group** in the Host Group. In practice it means the VMs have affinity to the hosts in one or the other datacenter.

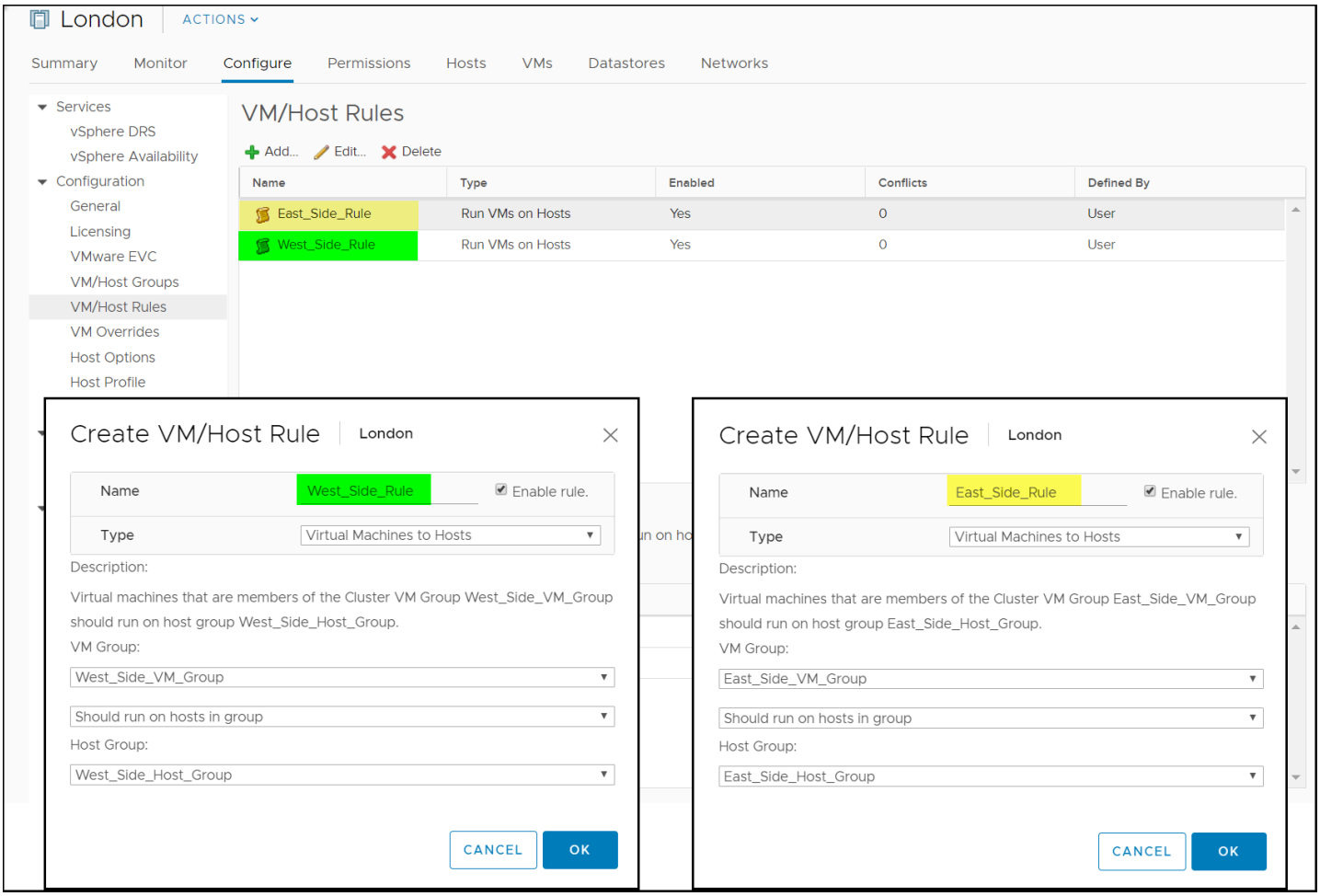



Figure 33. Creation of VM/Host Rules for groups

## 4.3 vSphere HA

In a vMSC environment utilizing SRDF/Metro, vSphere HA is critical to provide the needed availability by guarding against not just individual host, network, and storage failures, but also complete site failures. SRDF/Metro allows both sides to provide coherent read/write access to the same virtual volume. That means that on the remote site, the paths are up, and the storage is available even before any failover happens. When you combine this with host failover clustering technologies such as VMware HA, it creates a fully automatic application restart for any site-level disaster. The system rides through component failures within a site, including the failure of an entire array.

In this configuration, a virtual machine can write to the same virtual device from either cluster. In other words, if you are using vSphere DRS, which allows the automatic load distribution on virtual machines across multiple ESXi servers, you can move a virtual machine from an ESXi server attached to the R1 array to an ESXi server attached to R2 array without losing access to the underlying storage. This configuration allows virtual machines to move between two geographically disparate locations with up to 150ms of latency, the limit to which VMware supports vMotion across hosts.

### 4.3.1 Site failure example

In the event of a complete site failure, shown in Figure 34, SRDF/Metro Witness automatically assigns the R2 array as the winner, rather than following the R1 site bias. vSphere HA detects the failure of the virtual machines and restarts the virtual machines automatically at the surviving site with no user intervention. It is therefore critical to properly configure vSphere HA to aid in this automated recovery.

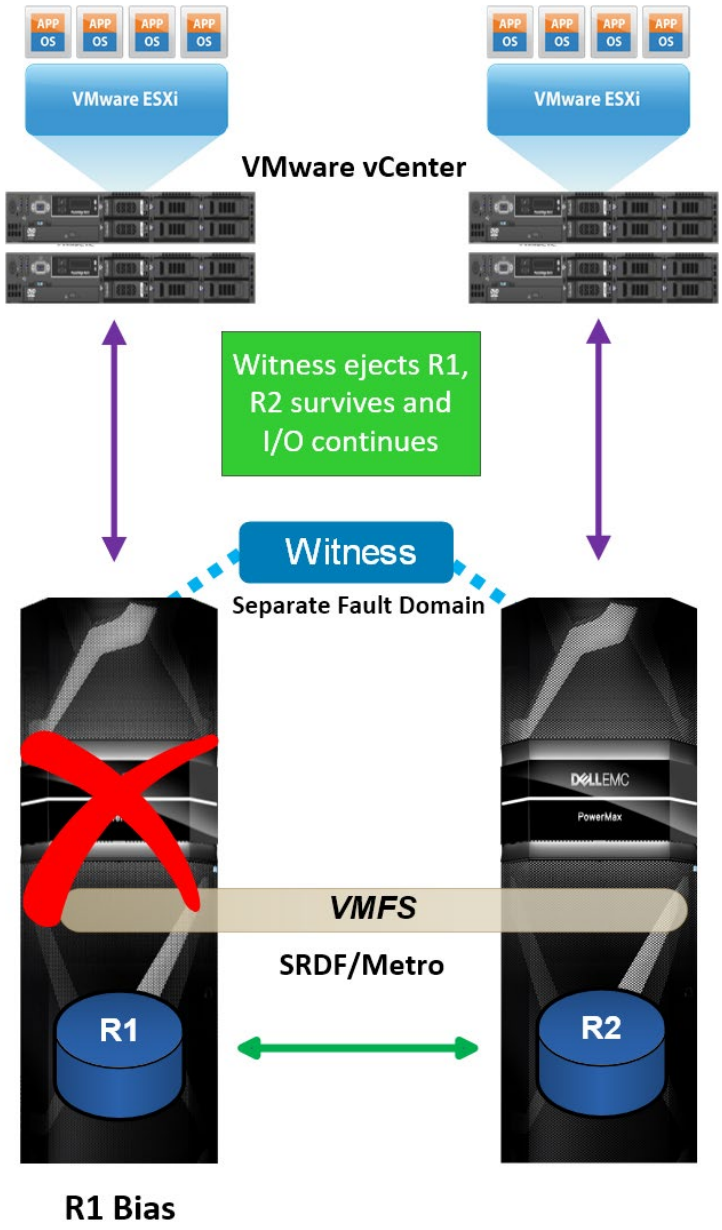



Figure 34. vMSC with SRDF/Metro Witness

**Note:** As this paper will not detail all failure scenarios of an SRDF/Metro configuration, please see the Technical Note *SRDF/Metro Overview and Best Practices* in the References section if more information is desired.

## 4.4 Enabling vSphere HA

To enable vSphere HA, navigate to **Cluster -> Configure -> vSphere Availability** and then **EDIT vSphere HA**. To activate, slide the bar to the right and green in the steps outlined in Figure 35.

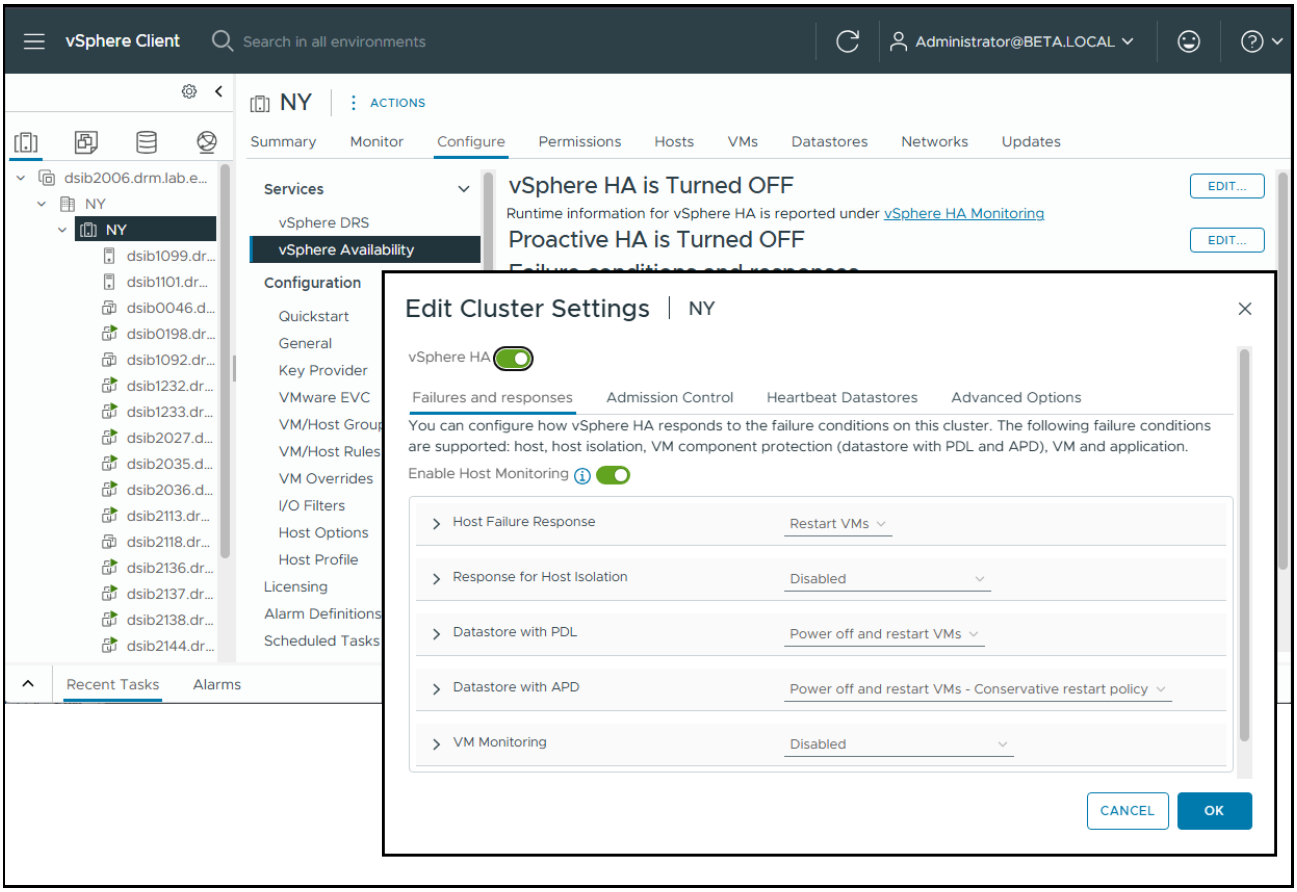



Figure 35. Setting vSphere HA services

The subsequent sections detail the configuration of vSphere HA. You modify the settings while in the edit screen in Figure 35.

### 4.4.1 Admission Control

As the core business driver of SRDF/Metro and vSphere Metro Storage Cluster is high availability, it is important to ensure that server resources exist to failover to a single site. Both VMware and Dell recommend using a percentage-based policy for Admission Control with HA as it is flexible and does not require changes when you add additional hosts to the cluster. Therefore, you should configure the Admission Control policy of vSphere HA for 50% CPU and 50% memory. An example is in Figure 36.

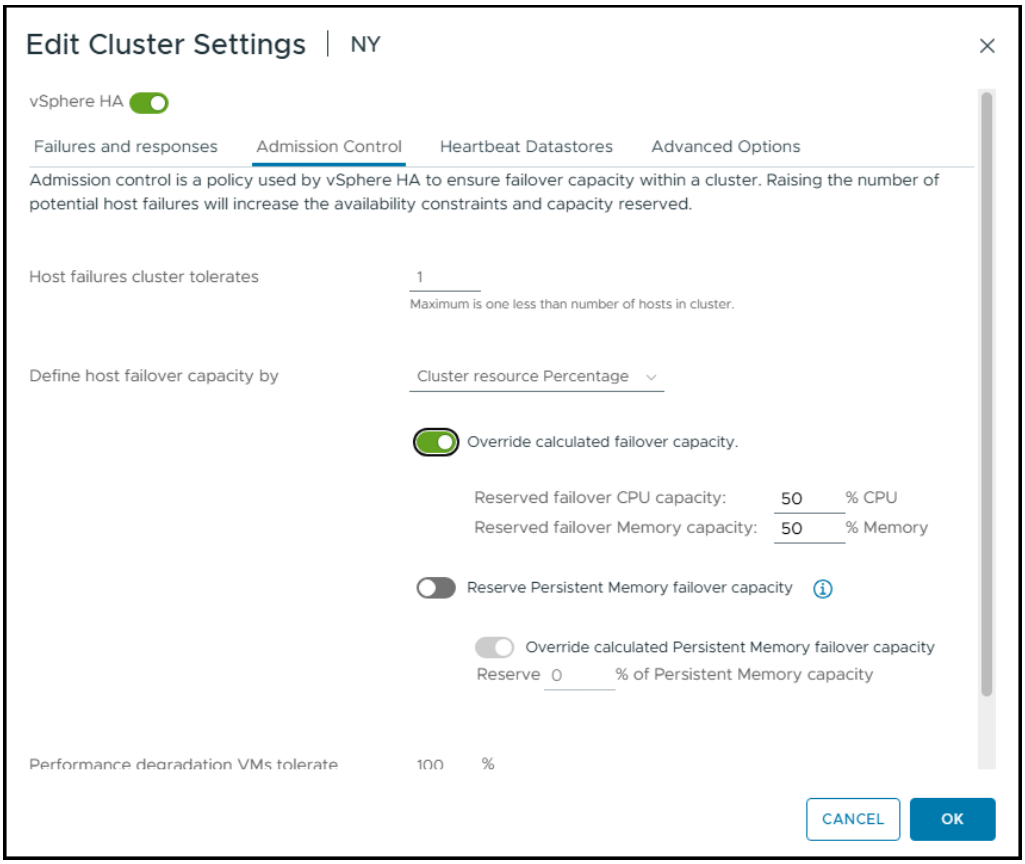



Figure 36. vSphere HA Admission Control

## 4.4.2 Heartbeating

vSphere HA uses heartbeat mechanisms to validate the state of a host. There are two types of heartbeating:

- network (primary)
- datastore (secondary)

If vSphere HA fails to determine the state of the host with network heartbeating, it will then use datastore heartbeating. If a host is not receiving any heartbeats, it uses a fail-safe mechanism to detect if it is merely isolated from its master node or completely isolated from the network. It does this by pinging the default gateway. It is possible to configure additional isolation addresses in case the gateway is down. VMware recommends specifying a minimum of two additional isolation addresses. Each address should be local to one datacenter. You configure this in the Advanced Options of vSphere HA using the option name ***das.isolationAddress.x*** (increment 'x' for each address starting with zero) as shown in Figure 37.

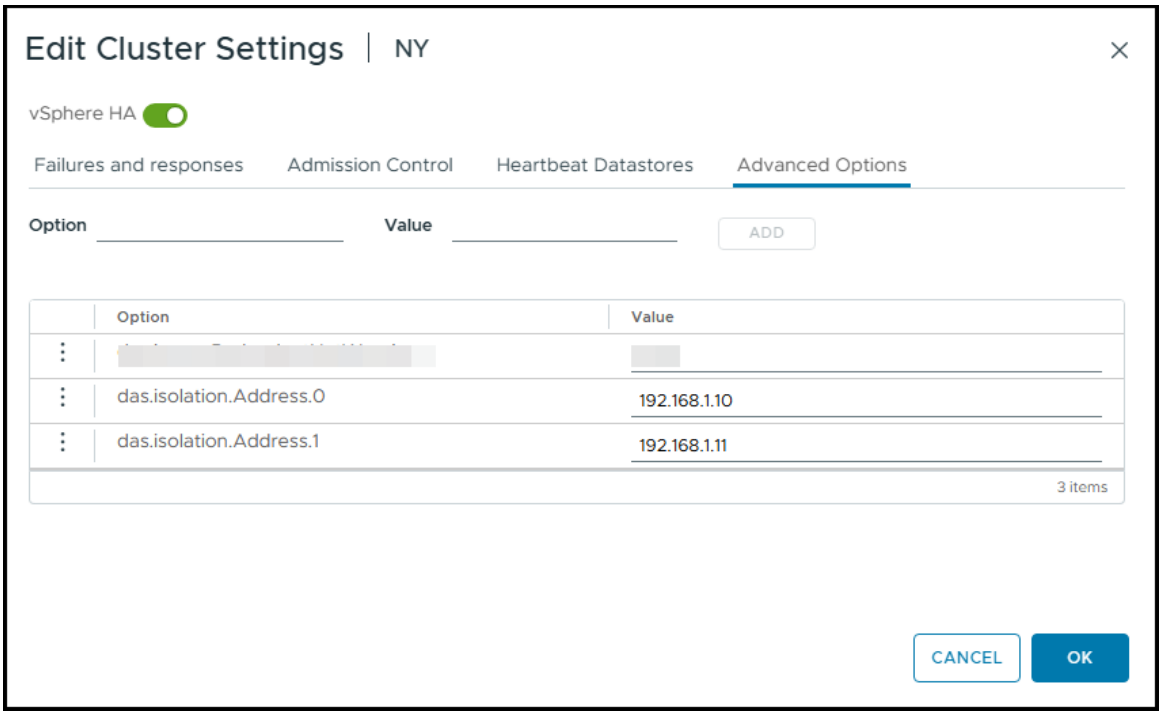



Figure 37. Adding network isolation addresses for vSphere HA

For the heartbeat mechanism, the minimum number of heartbeat datastores is two and the maximum is five. VMware recommends increasing the number of heartbeat datastores from two to four in a stretched cluster environment. This provides full redundancy for both datacenter locations. Dell recommends defining four specific datastores as preferred heartbeat datastores. To increase the minimum heartbeat datastores, in the Advanced Options of vSphere HA add a new option named ***das.heartbeatDsPerHost*** and set the value to 4, depicted in Figure 38.

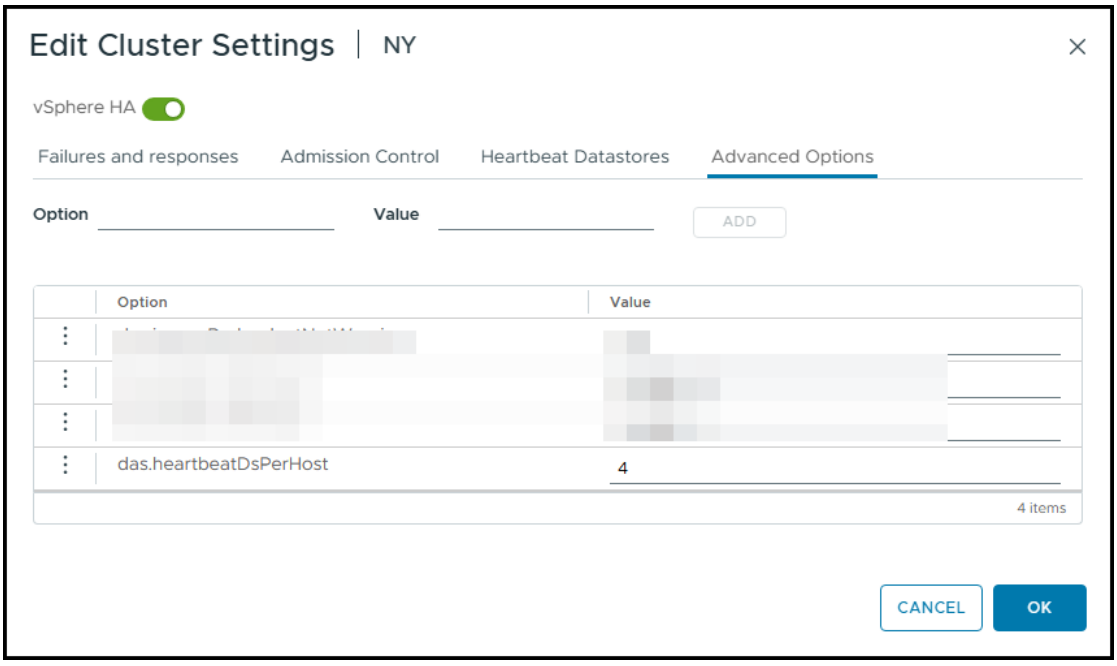



Figure 38. Change minimum heartbeating datastores

Once you configure the minimum heartbeat datastores, under the tab **Heartbeat Datastores** select the radio button **Use datastores from the specified list and complement automatically if needed**. Then select the four specific datastores to use in the configuration. Since SRDF/Metro is an active/active solution, it is perfectly acceptable to select datastores that are visible on all hosts. Figure 39 shows the recommended setup.

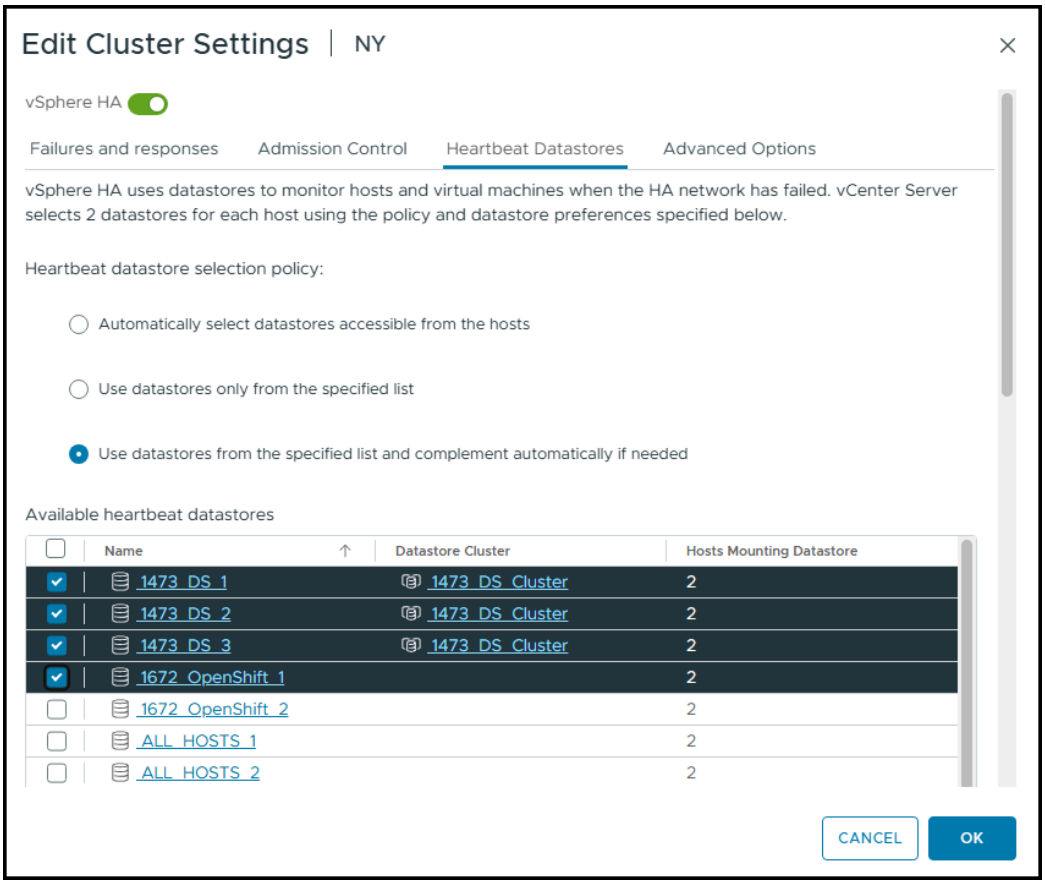



Figure 39. Modify heartbeat datastore selection policy

## 4.4.3 All Paths Down (APD) and Permanent Data Loss (PDL)

An additional, important feature around vSphere HA is Host Hardware Monitoring or VM Component Protection. This relates to the conditions All Paths Down and Permanent Data Loss.

### 4.4.3.1 APD

All paths down or APD occurs on an ESXi host when you remove a storage device in an uncontrolled manner from the host (or the device fails), and the VMkernel core storage stack does not know how long the loss of device access will last. VMware, however, assumes the condition is temporary. If you remove zoning it is a typical way of getting into APD.

### 4.4.3.2 PDL

Permanent device loss or PDL, is similar to APD (and hence why initially VMware could not distinguish between the two) except it represents an unrecoverable loss of access to the storage. VMware assumes the storage is never coming back. Removing the device from the storage group that underlies the datastore from would produce the error.

**Note:** VMware relies on SCSI sense codes to detect PDL. If a device fails in a manner that does not return the proper sense codes, VMware will default to APD behavior.

#### 4.4.3.3 PDL AutoRemove

PDL AutoRemove is a feature that automatically removes a device from a host when it enters a PDL state. PDL AutoRemove occurs only if there are no open handles left on the device. The auto-remove takes place when the last handle on the device closes. If the device recovers, or the user re-adds it after accidentally removal, VMware treats it as a new device. In such cases VMware does not guarantee consistency for VMs on that datastore.

In a vMSC environment, such as with SRDF/Metro, VMware recommends leaving AutoRemove in the default state, enabled. In an SRDF/Metro environment this is particularly important because if you disable it, and then suspend an SRDF/Metro pair, the non-biased side will experience a loss of communication to the devices. Once this occurs you must reboot the host.

For more detail on AutoRemove refer to VMware KB 310084.

### 4.4.4 VMCP

vSphere offers some capabilities around APD and PDL for the vSphere HA cluster which allow automated recovery of VMs. The capabilities are enabled through a feature in vSphere called VM Component Protection or VMCP. When VMCP is enabled, vSphere can detect datastore accessibility failures, APD or PDL, and then recover affected virtual machines. VMCP allows the user to determine the response that vSphere HA will make, ranging from the creation of event alarms to virtual machine restarts on other hosts.

VMCP is part of vSphere HA and enabled by default in vSphere HA. VMware labels it as **Enable Host Monitoring** in Figure 40. Note how the switch is already enabled even before vSphere HA is enabled.

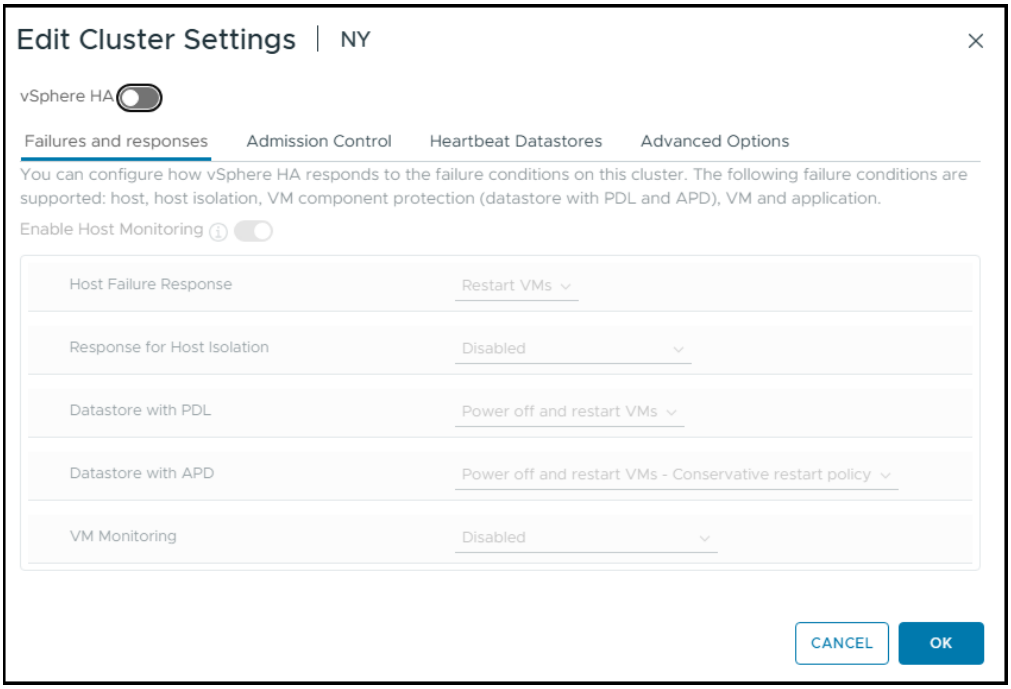



Figure 40. VMCP

Once vSphere HA is enabled, therefore, you can choose storage protection levels and virtual machine remediation for APD and PDL conditions. Figure 41 demonstrates the recommended

settings. By default, VMware sets the most common options for each category. For example, **Host Failure Response** is set to **Restart VMs**, as otherwise HA would leave the VMs in a failed state. We address each category in the next sections.

**Note:** To enable some of the VMCP actions, install VMware Tools on the VMs participating in the vSphere HA cluster.

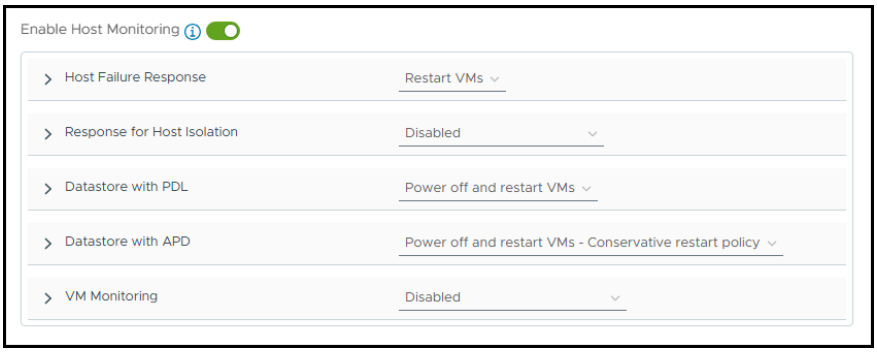



Figure 41. Storage and VM settings for VMCP

### 4.4.4.1 Response for Host Isolation

Host isolation is when a particular host in the cluster cannot communicate with other hosts in the cluster. At that point, the cluster isolates the host, so it is no longer part of the cluster. The options for managing this event are in Figure 42, with **Disabled** being the default.

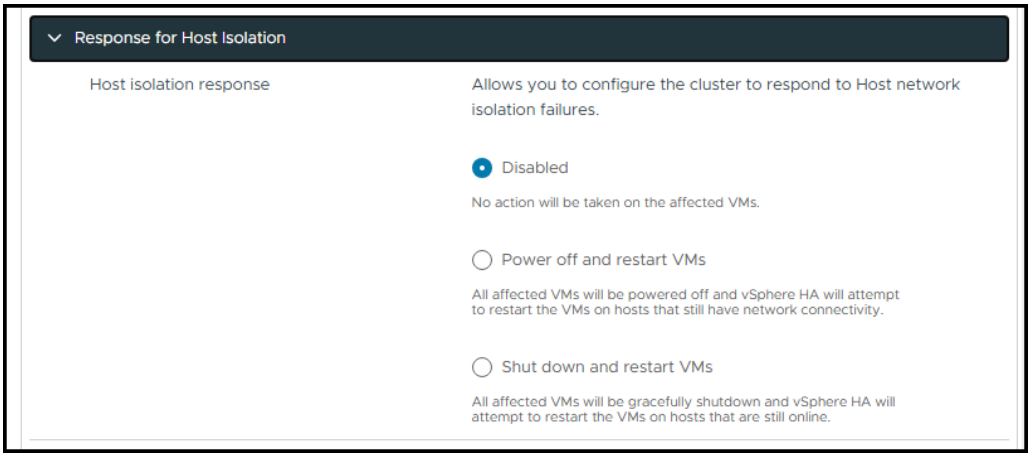



Figure 42. VMCP - Response for Host Isolation

Host isolation may be a transient condition due to an inability for the host to receive a heartbeat, or it may be a more serious condition. It is perfectly acceptable, therefore, to leave the setting as Disabled, i.e., VMware will do nothing, or to set it to shut down and restart the VMs (preferable to forced power off). Previously Dell and VMware recommended leaving this setting as default, but more recent documentation appears suggests setting it to the options **Shut down and restart VMs**, though without detailed explanation. Therefore, Dell makes no definitive recommendation as either option is acceptable.

### 4.4.4.2 PDL VMCP settings

The PDL settings are the simpler of the two failure conditions between APD and PDL to configure. This is because there are only two choices other than leaving it disabled:

- Issue events

- Power off and restart VMs

As the purpose of HA is to keep the VMs running, the recommendation is to **Power off and restart the VMs**. As this is the default, leave the radio button next to this option set as in Figure 43.

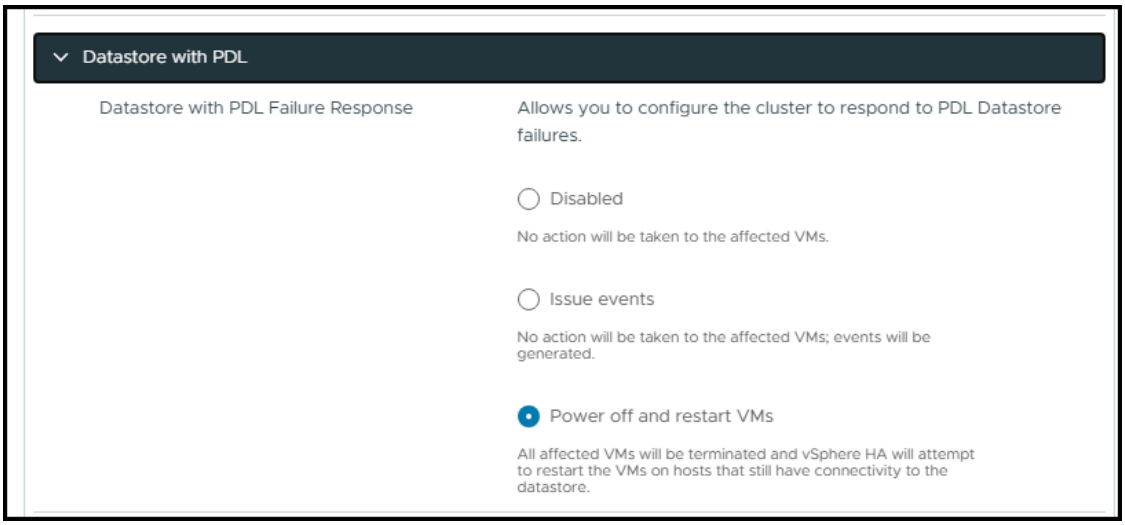



Figure 43. VMCP - PDL recommended setting

#### 4.4.4.3 APD VMCP settings

As APD events are by nature transient, and not a permanent condition like PDL, VMware provides a more nuanced ability to control the behavior within VMCP. However, there are still two options to choose from:

- vSphere can issue events.
- vSphere can initiate power off of the VMs and restart them on the surviving host(s) (aggressively or conservatively).

In general, when vSphere recognizes an APD condition it starts a 140 second countdown timer. When the timer ends, VMware marks the device as an APD timeout. At this point, vSphere HA starts a 180 second countdown timer before it acts, however using VMCP you can enhance this behavior by setting vSphere to power off and restart the VMs either conservatively or aggressively instead of waiting. The difference between how vSphere behaves in a conservative or aggressive configuration has to do with the other hosts in the cluster. If you select conservative, VMware will attempt to power off and restart the VM only if it is able to determine if there is another host in the cluster with access to the datastore where the VM resides. In the aggressive setting, VMware powers off the VM and then determines if a host is available for restart. This may mean the VM will not be able to restart. Note that if the cluster does not have sufficient resources, neither approach will terminate the VM. Dell recommends a conservative approach which is the default. This is the setting seen in Figure 44.

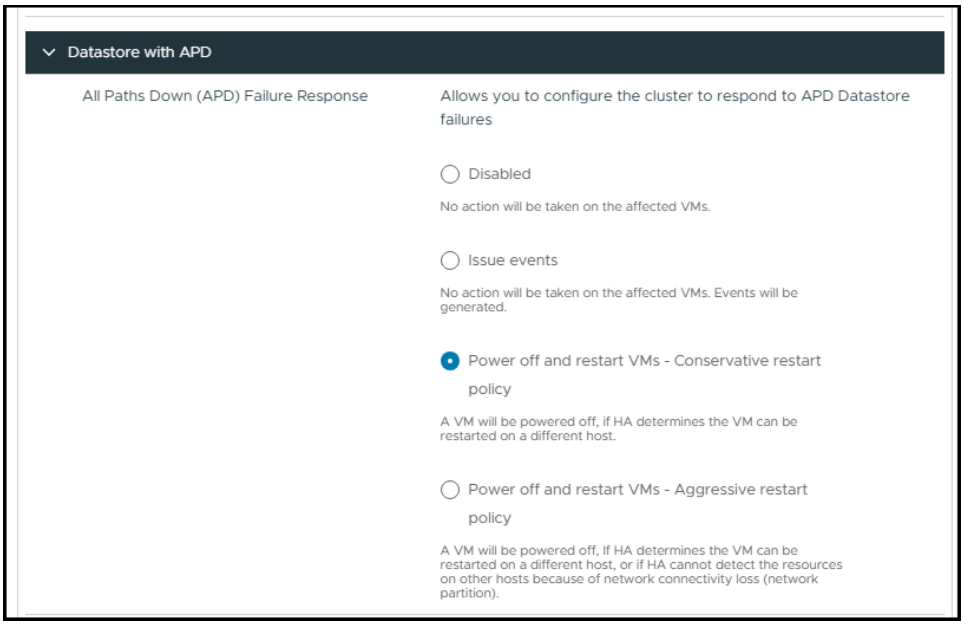



Figure 44. VMCP - APD recommended setting

If you select **Issue events**, vSphere will do nothing more than notify the user through events when an APD event occurs. As such, no further configuration is necessary. If, however, you choose either aggressive or conservative restart of the VMs, you may select an additional option to further define how vSphere is to behave.

#### 4.4.4.4 Response recovery

In Figure 44 the bottom box contains a policy setting called **Response recovery**. In addition to setting the delay for the restart of the VMs, the user can choose whether vSphere should act if the APD condition resolves before reaching the user-configured delay period. If the setting **Response recovery** is set to **Reset VMs**, and APD recovers before the delay is complete, VMware resets the affected VMs which will recover the applications impacted by the IO failures. This setting does not have any impact if vSphere is set to only issue events in the case of APD. VMware and Dell recommend leaving this set to **Disabled** (the default setting) to avoid disrupting the VMs.

**Note:** If you disable either the Host Monitoring or VM Restart Priority settings, VMCP cannot perform virtual machine restarts. VMware, however, will continue to monitor storage health and issue events.

#### 4.4.4.5 VM Monitoring

The purpose of VM Monitoring is to restart VMs if the VMware Tools heartbeat is not responding and there is no storage or network IO over a specified period – 120 seconds in the recommended preset configuration. Dell recommends activating **VM Monitoring Only** as seen in Figure 45, accepting the default values.

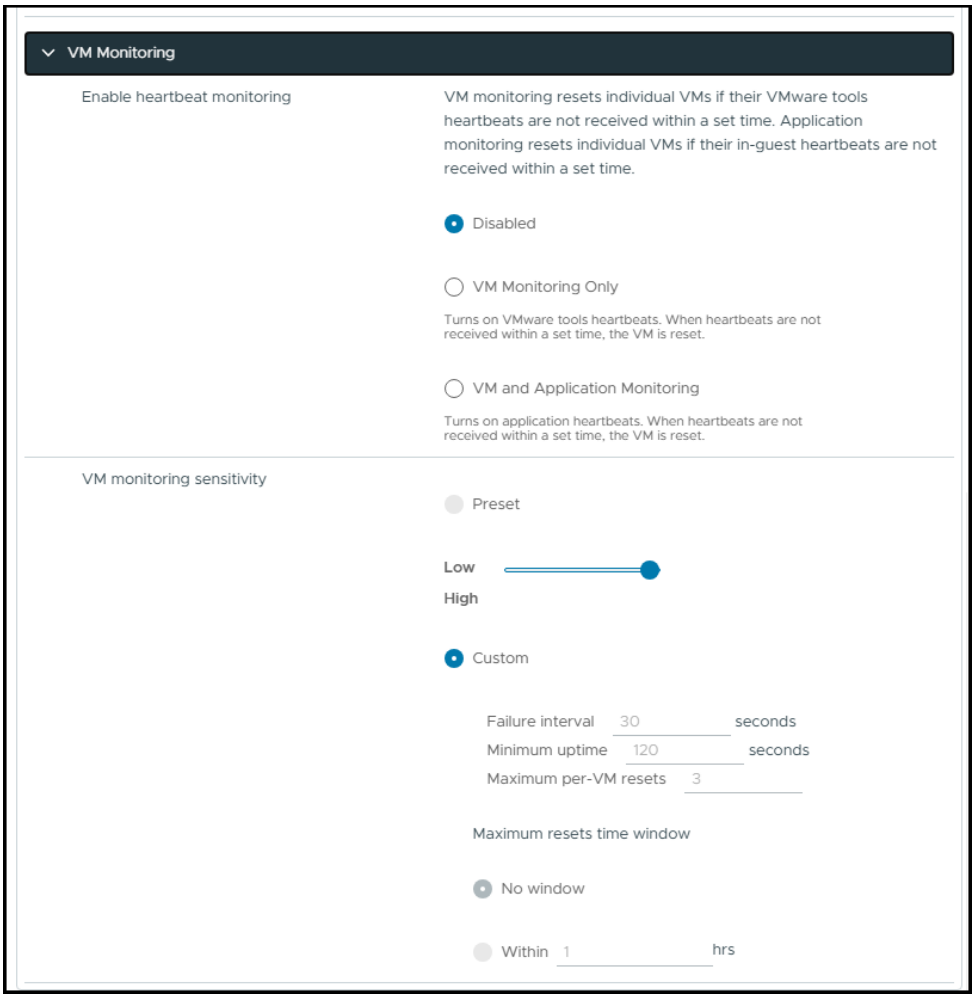



Figure 45. VMCP - VM Monitoring

The completed recommendations for all VMCP are in Figure 46.

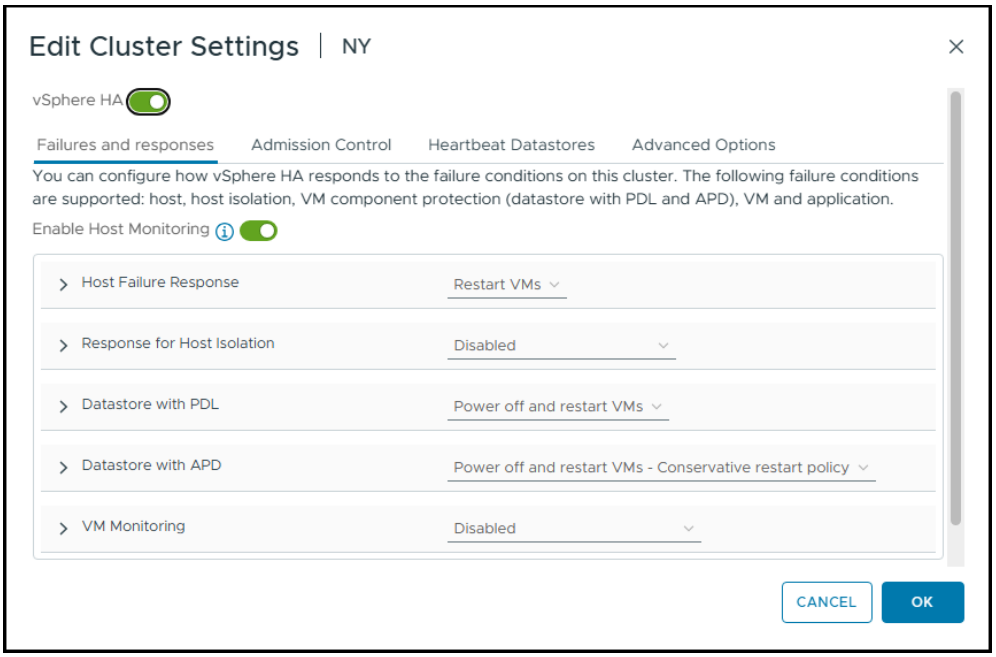



Figure 46. VMCP summary

## 4.5 VM restart order/priority

When you need to restart VMs after a failure, the order in which they come up can be critical. Fortunately, VMware provides a mechanism to set both the order and priority. The VM restart priority consists of the following categories:

- Lowest
- Low
- Medium
- High
- Highest

By default, all VMs have a priority of Medium.

Order priority can be essential for VMs that provide the infrastructure of an environment such as DNS or single sign-on. In addition, multi-tier applications like Oracle Apps that consist of a web application and database tier, require that the database be running before the application can start, and in turn the application must be running before the web tier can start. VMware offer VM Overrides for these situations.

### 4.5.1 VM Overrides

The **VM Overrides** is accessible from the **Cluster -> Configure -> Configuration -> VM Overrides** menu here in Figure 47.

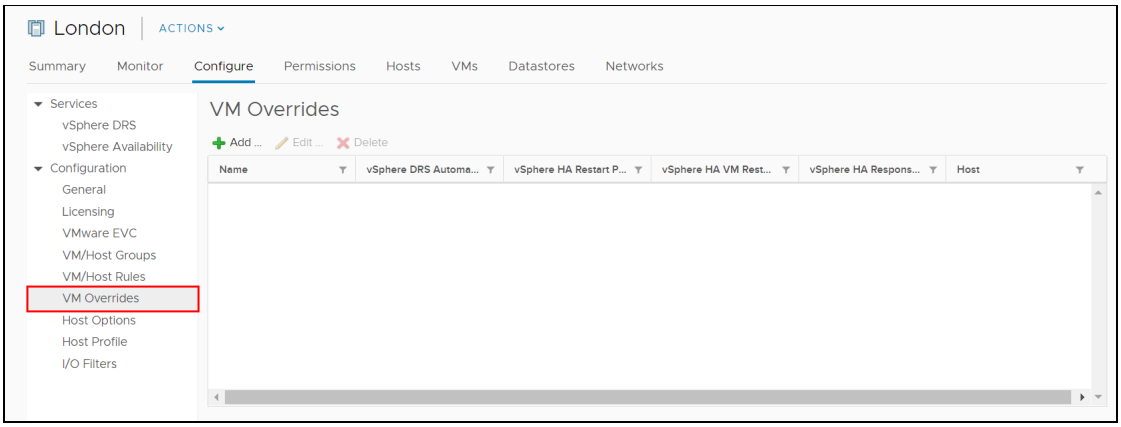



Figure 47. VM Overrides

The wizard has two steps. In the first step in Figure 48, select the VMs that require special priority.

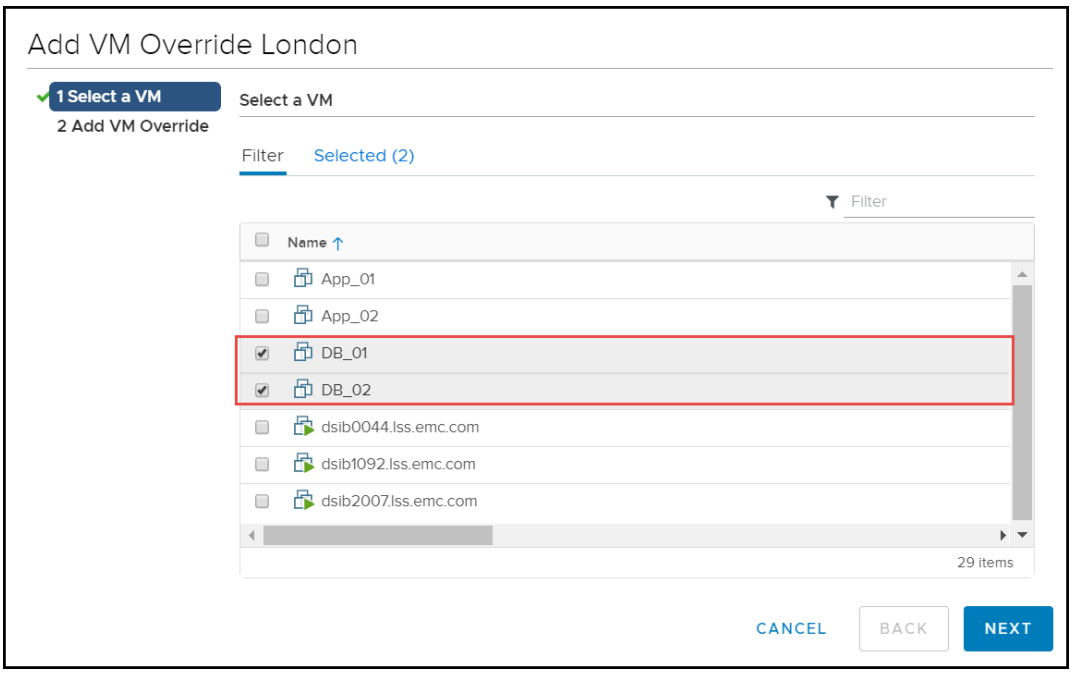



Figure 48. VM Overrides - Step 1

In the second step, override the restart priority and set it to the highest, to ensure those VMs, in this case database VMs, start first. Also, select the next box to override when to start the next VMs. In this example in Figure 49, it will use the guest heartbeat to determine when to start the next highest priority VMs.

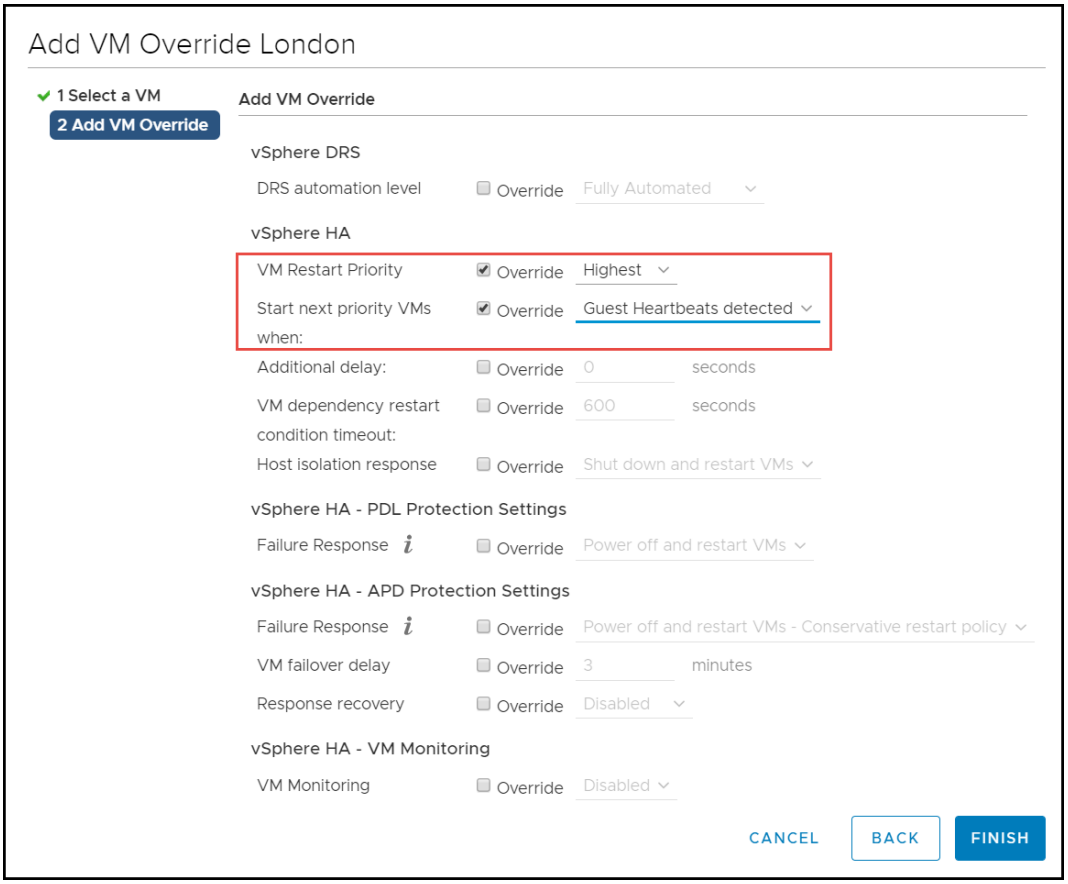



Figure 49. VM Overrides - Step 2

By default, vSphere will start the next VMs after 600 seconds even if it cannot detect a heartbeat. VMware does not recommend adjusting this value. In this example, you would create a second VM Override for the application tier as High, then one for the web tier as Medium.

#### 4.5.1.1 VM startup dependency

A second method can be employed to determine startup order. This involves setting up the previously discussed VMware VM/Host Groups and VM/Host Rules. Rather than configuring site affinity, VM Groups can be setup with dependencies upon one another. For example, using the database and application tiers, setup a VM Group for each as in Figure 50.

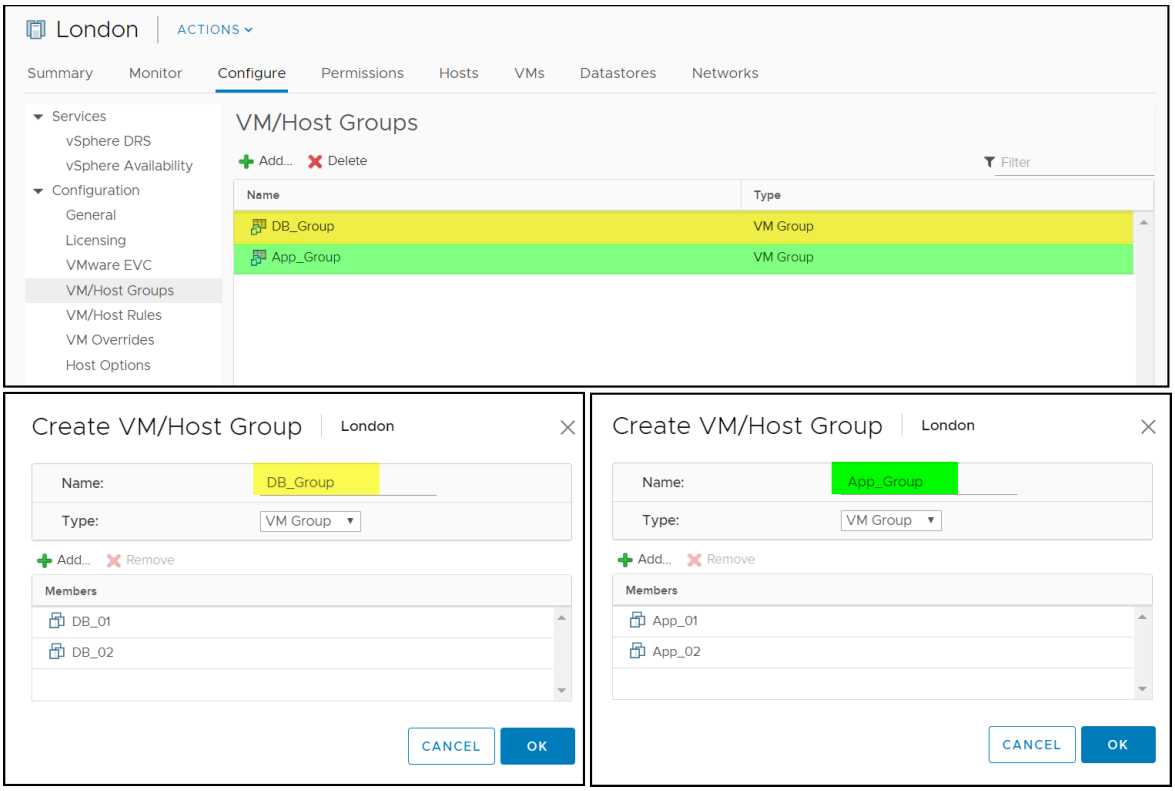



Figure 50. VM Groups for startup dependency

Now, setup a VM Rule which governs how the two VM Groups should interact. In Figure 51 the **Oracle_Startup** rule says that before the **App_Group**, containing the application VMs, can startup, the **DB_Group**, containing the database VMs, must first start. One significant difference between this startup method and the previous one using VM Overrides is that you cannot break VM Rules. In other words, if the **DB_Group** fails to start, vSphere will not start the **App_Group**. Recall that after 600 seconds the startup priority moves forward, regardless of whether the higher priority VM startups have succeeded.

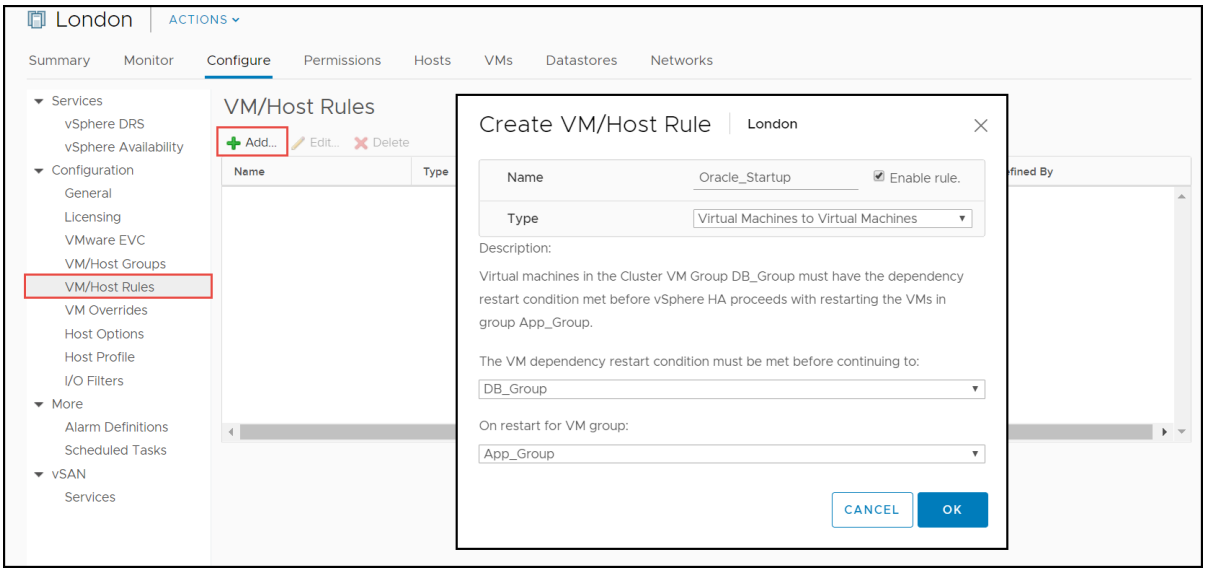



Figure 51. VM Rule for startup dependency

# 5 Conclusion

Dell SRDF/Metro is an enterprise-class technology that dissolves distance by providing active/active access to dispersed PowerMax arrays enhancing availability and mobility. Using SRDF/Metro in conjunction with a VMware vSphere Metro Storage Cluster (vMSC), including vSphere DRS and HA, provides new levels of availability suitable for the most mission critical environments without compromise.

These technologies provide the basis by which a customer can ensure high availability at both the hardware and software level – through the nature of SRDF/Metro and vMSC.

# 6 References

## 6.1 Dell

- The Dell PowerMax and VMware vSphere Configuration Guide

  https://infohub.delltechnologies.com/t/dell-powermax-and-vmware-vsphere-configuration-guide-1/

- Dell SRDF/Metro vWitness Configuration Guide

  https://dl.dell.com/content/manual60079529-dell-srdf-metro-vwitness-configuration-guide.pdf?language=en-us

- Using VMware Storage APIs for Array Integration with Dell PowerMax

  https://www.delltechnologies.com/asset/en-us/solutions/infrastructure-solutions/technical-support/h8115-powermax-vmware-vaai-wp.pdf

- Unisphere for PowerMax Product Guide

  https://www.dell.com/support/product-details/en-us/product/unisphere-powermax/overview

## 6.2 VMware

- vSphere General Documentation

  https://techdocs.broadcom.com/

- SRDF/Metro vMSC support VMware KB article
  https://knowledge.broadcom.com/external/article?articleNumber=323113

- VMware vSphere Metro Storage Cluster Recommended Practices
  https://core.vmware.com/resource/vmware-vsphere-metro-storage-cluster-vmsc

# Appendix

This appendix covers the steps required to setup SRDF/Metro for use in a vMSC environment. The example uses Unisphere for PowerMax 10.0.1.0, though other versions of Unisphere have similar interfaces.

## 1.1 Setting up SRDF/Metro

You can configure SRDF/Metro with Solutions Enabler (CLI) or with Unisphere for PowerMax. Dell recommends using Unisphere to configure SRDF/Metro to reduce complexity. This section details the setup and includes those tasks related to SRDF/Metro setup. In this example we created the following objects as their creation is independent of SRDF/Metro:

- Initiator groups for each site.
- Port groups for each site.
- Devices on each site created and placed in a single storage group on each array.
- A masking view exists for the R1 array, but NOT for the R2.

We utilized a vWitness as that is the most common implementation, but for completeness we include the process to create an array witness.

### 1.1.1 Array witness group creation

To create an array witness, within Unisphere for PowerMax navigate to the **Data Protection -> SRDF Groups** and select **Create SRDF Group** as in Figure 52.

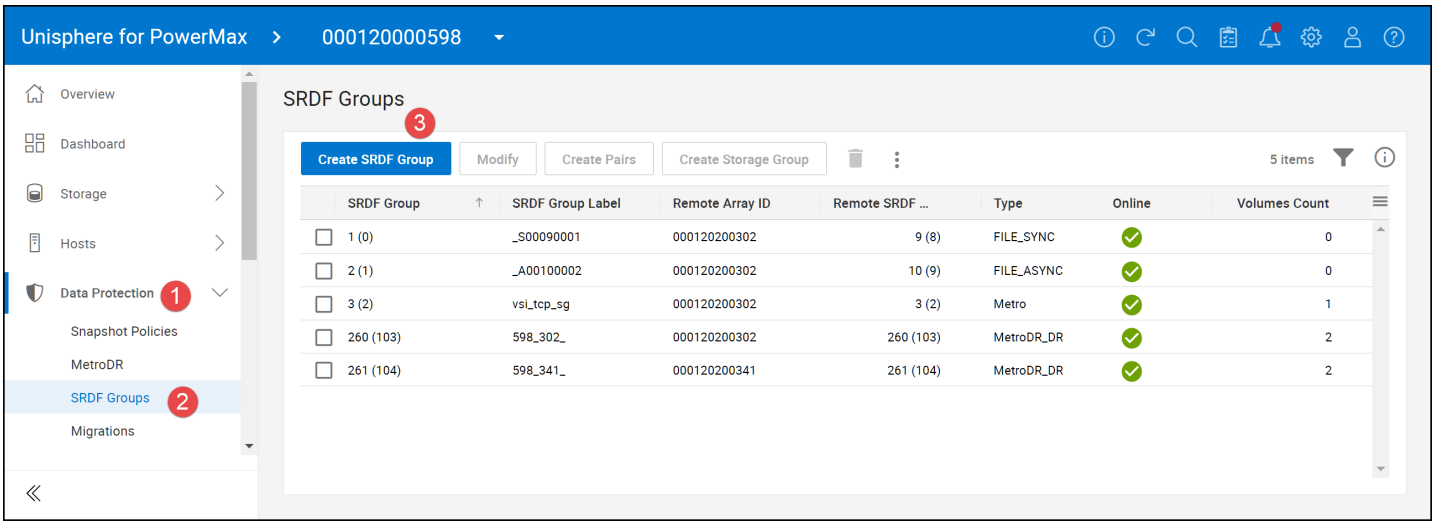



Figure 52. SRDF group creation in Unisphere for PowerMax

Provide an SRDF Group Label to designate the function of the group. In this example in Figure 53, the label “aw-302598” indicates this is a witness group between the R1 302 array and the witness array, 598. Note we create the group here from the witness array, but you could create it from the R1 as well. Next, check the box **SRDF/Metro Witness Group**.

If an array witness group already exists to the remote array, Unisphere will gray out the setting for SRDF/Metro Witness Group.

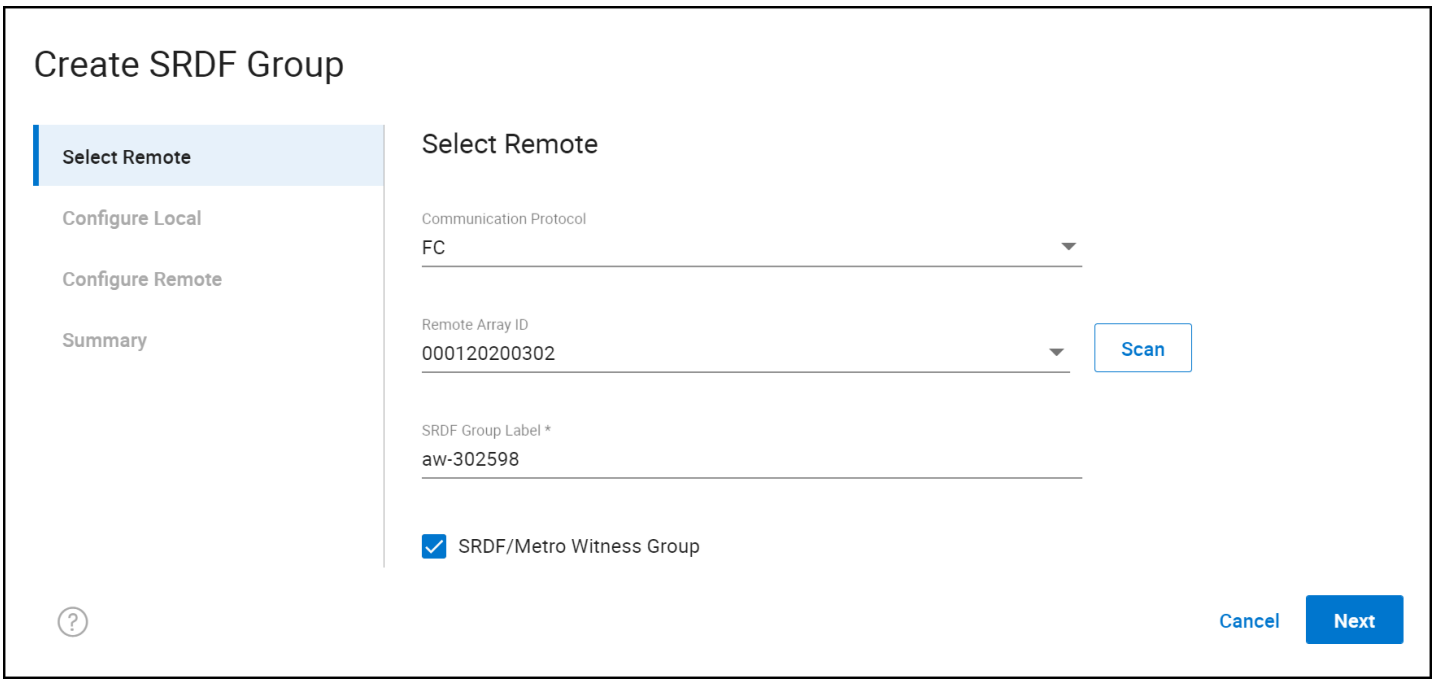



Figure 53. Creating the SRDF/Metro Witness Group from R1 to Witness array -

In steps 2 and 3, select the RDF ports and assign an RDF group. It is best to use the same number on each array for consistency. We use group 65 in this example,. Figure 54 is the summary in step 4 before creation.

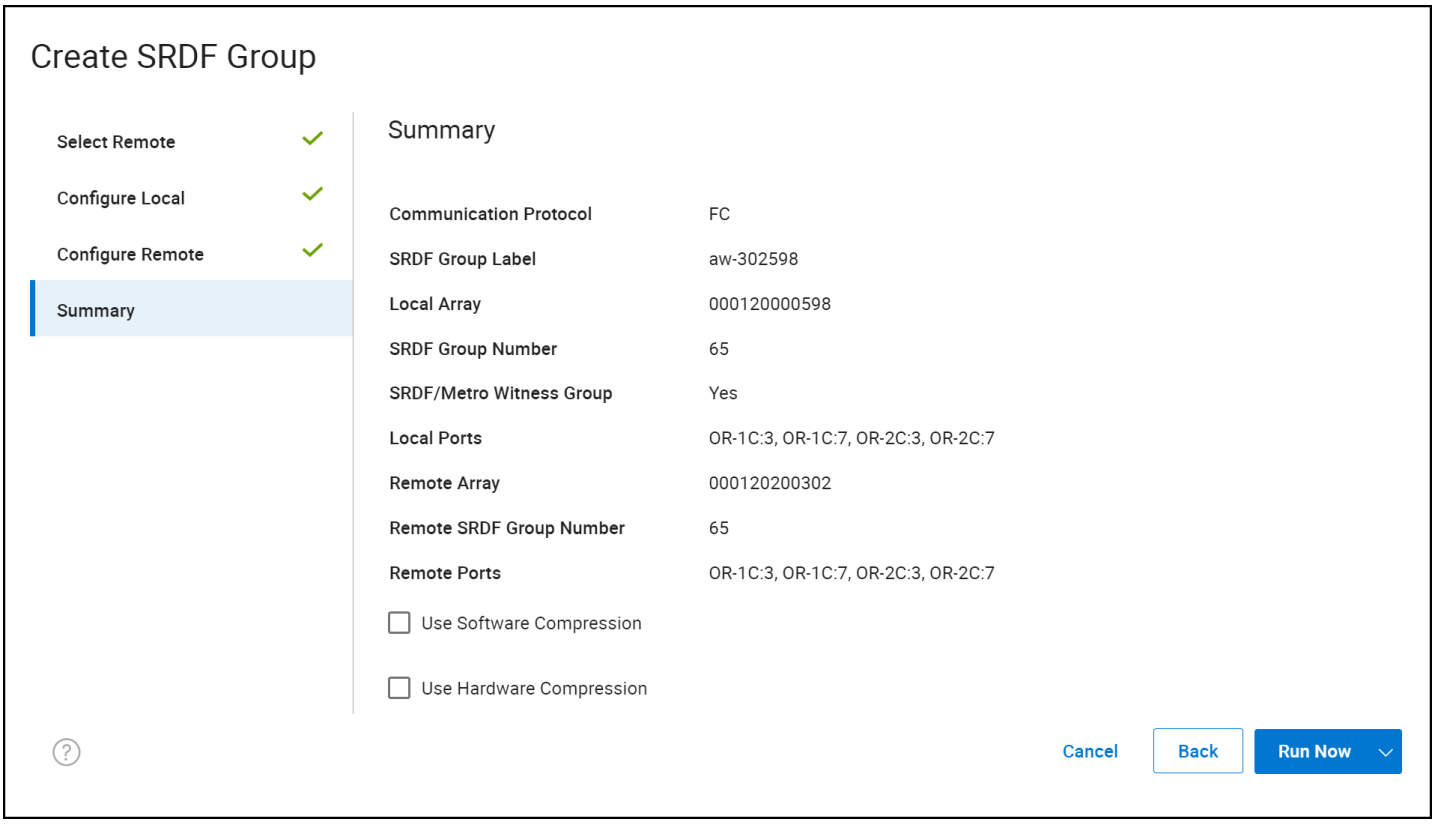



Figure 54. Creating the SRDF/Metro Witness Group from R1 to Witness array - Summary

You need to run the same steps between the R2 array, 341, and the witness array 598. You will need a new SRDF group number. When complete, compare the two groups in Figure 55 and note that the **Witness SRDF Group Type**.

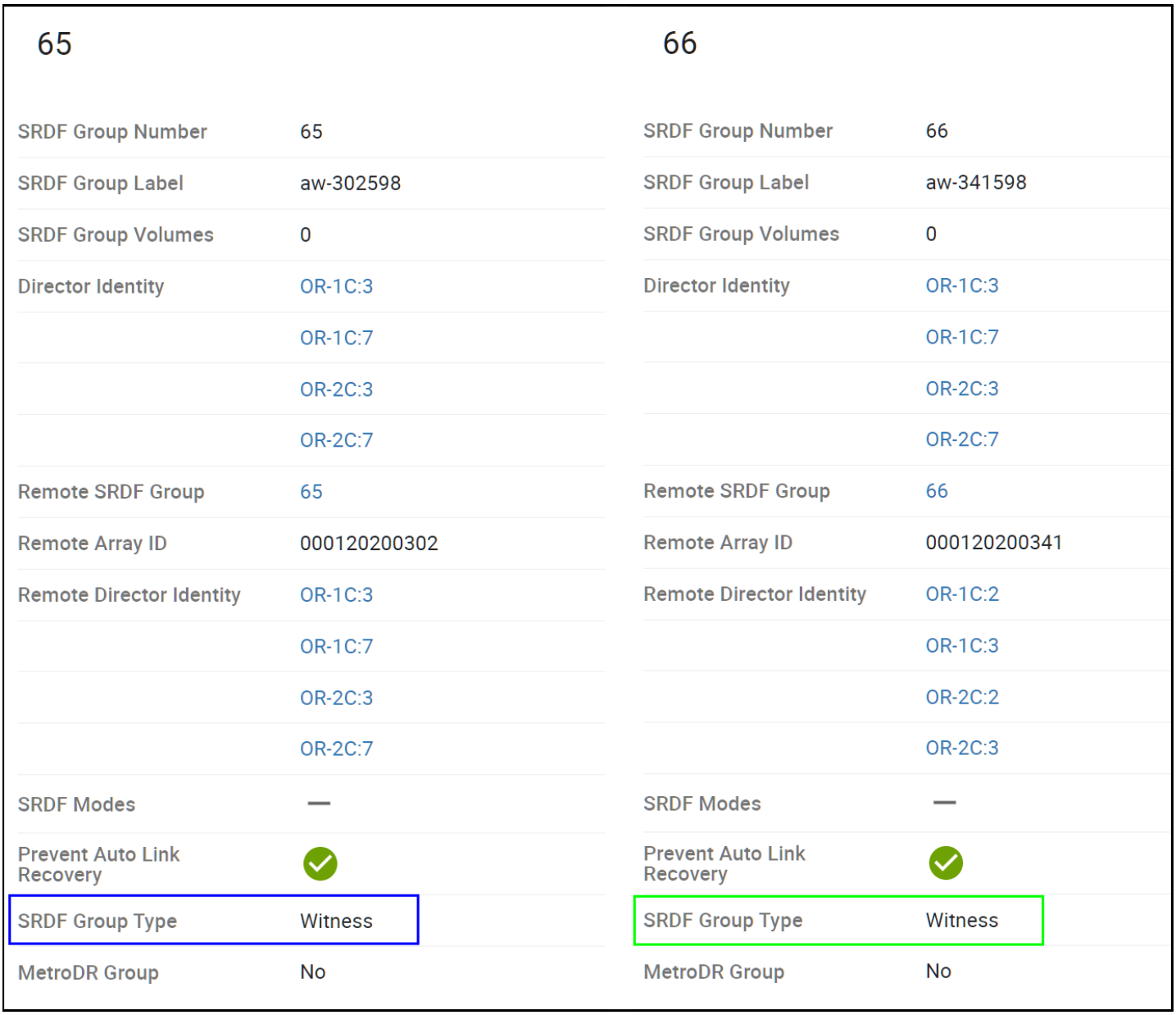



Figure 55. Witness group summary

To add an SRDF group for the witness using Solutions Enabler, specify the **`-witness`** flag to a standard SRDF group creation statement shown in Figure 56. We use different arrays here in the CLI than the GUI.

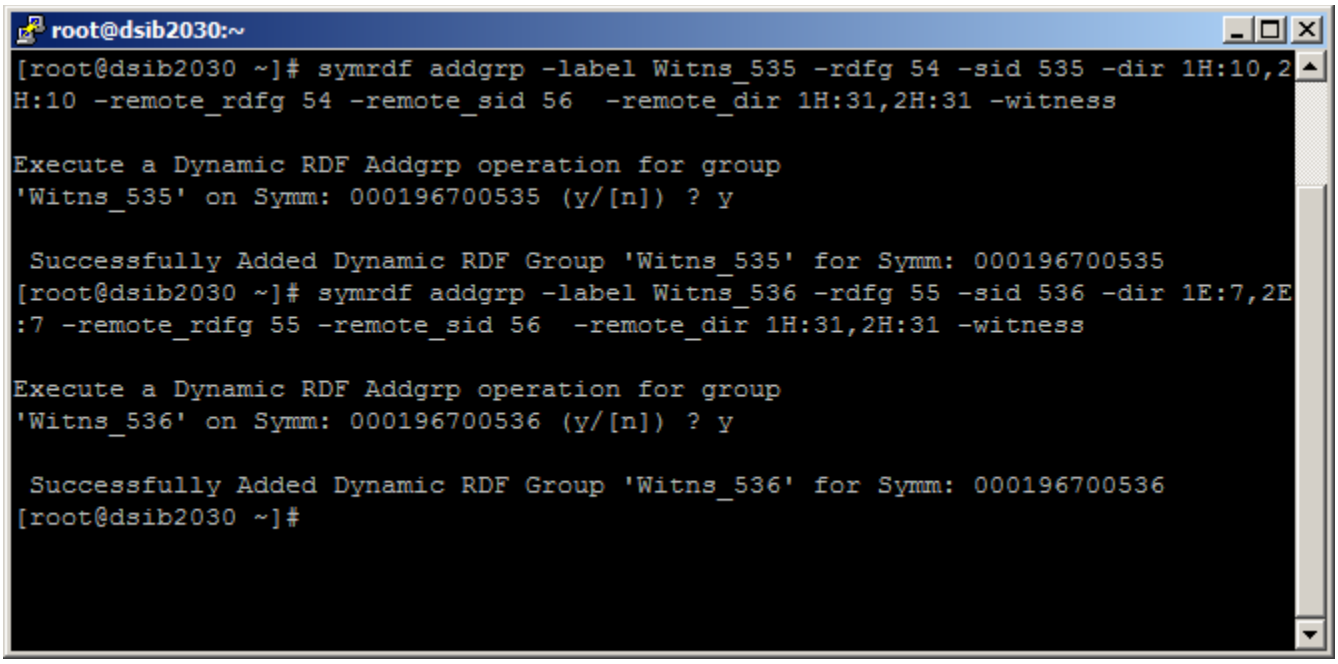



Figure 56. Creating SRDF groups for the Witness through CLI

If a Witness group already exists, whether or not it is in use, Solutions Enabler will return the following error: *A Witness RDF group already exists between the two Symmetrix arrays*.

## 1.1.2 vWitness creation

Beginning with PowerMaxOS 10, you must deploy the virtual witness on a supported OS. The software package is independent of Solutions Enabler, but you deploy it similarly. After deployment, the Lock Service Daemon should be running as in Figure 57.

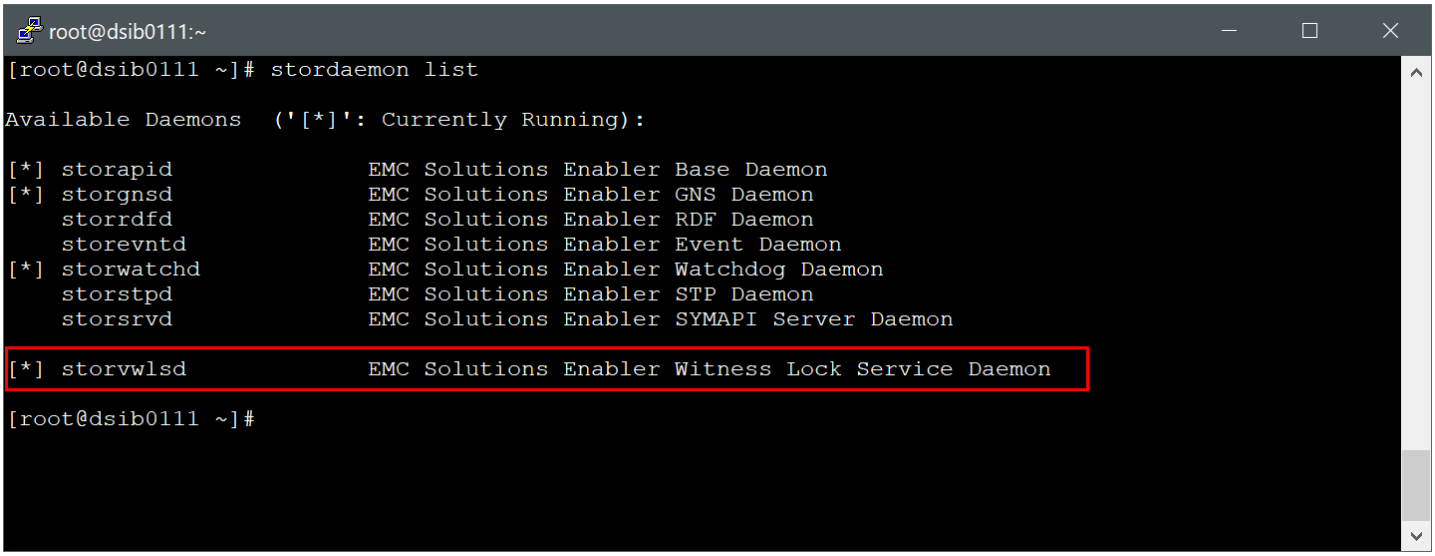



Figure 57. vWitness deployment

### 1.1.3 Witness use

Though the status of an SRDF/Metro pair will designate whether the witness is in use, it is not possible to tell what type, array or virtual, within Unisphere. Only in Solutions Enabler CLI is this information found. You can use the `symcfg` command to reveal the name and type of witness. Figure 58 shows the command to issue to see witness detail. Note the column **Witness Identifier**. If this field has a value for a row, it indicates there is a witness associated with the group. There are two types of values in this row. If Dell is using the array witness, the value will be the witness array SID as is present for **RA-Grp** 11. If a virtual witness is in use, the value will be the name given to the virtual witness as in **RA-Grp** 40 and 41.

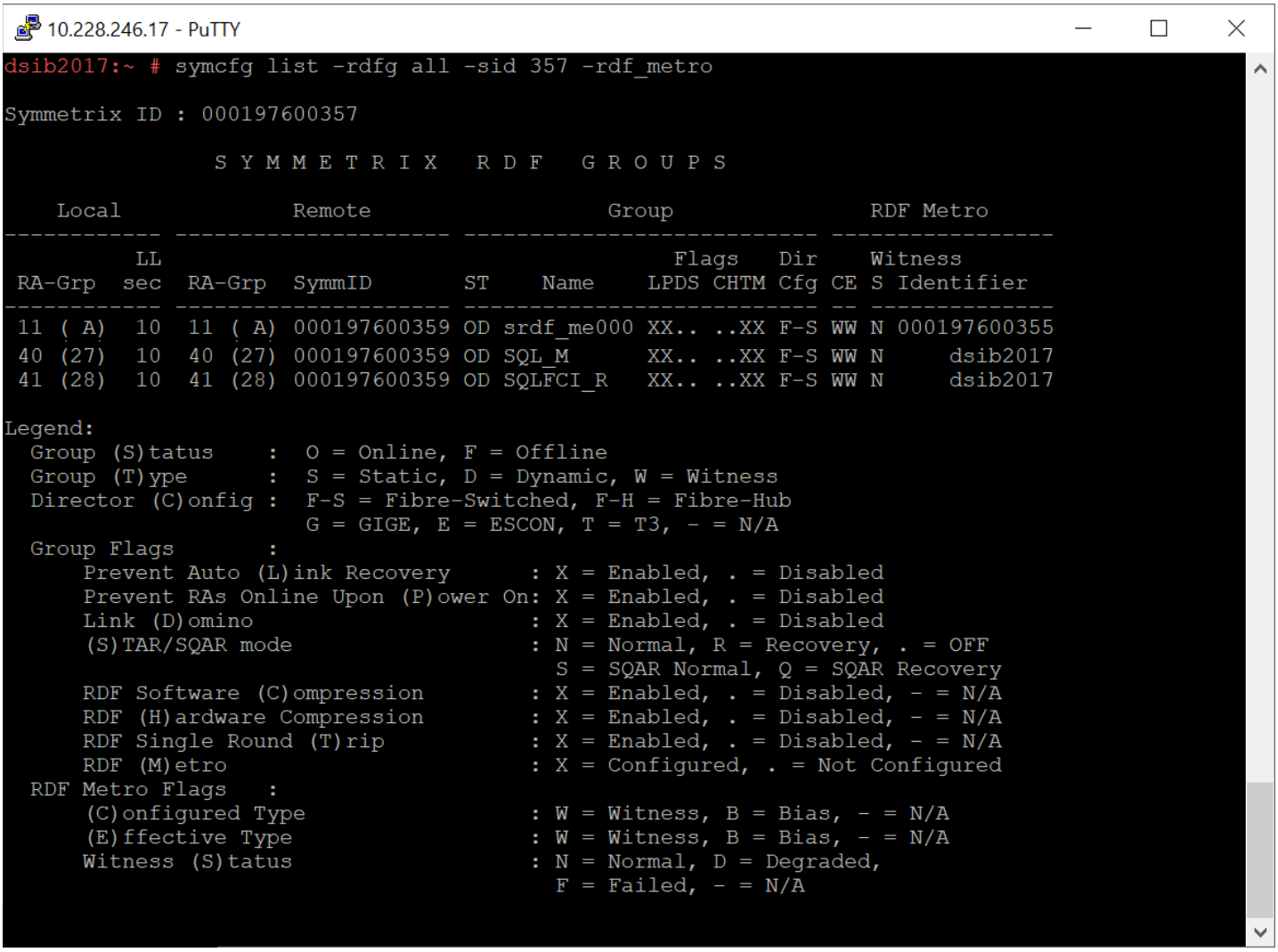



Figure 58. Witness detail

### 1.1.4 SRDF/Metro pair creation

The Storage Groups screen in Unisphere provides an easy-to-use wizard to enable SRDF/Metro on a storage group. Before starting the wizard, it is important to remember that though this is an active-active device configuration, do not present the R2 to the hosts until Dell completes device synchronization.

Start by navigating to the **Storage -> Storage Groups** seen in Figure 59. Highlight the desired storage group for protection and select the button **Protect** at the top menu.

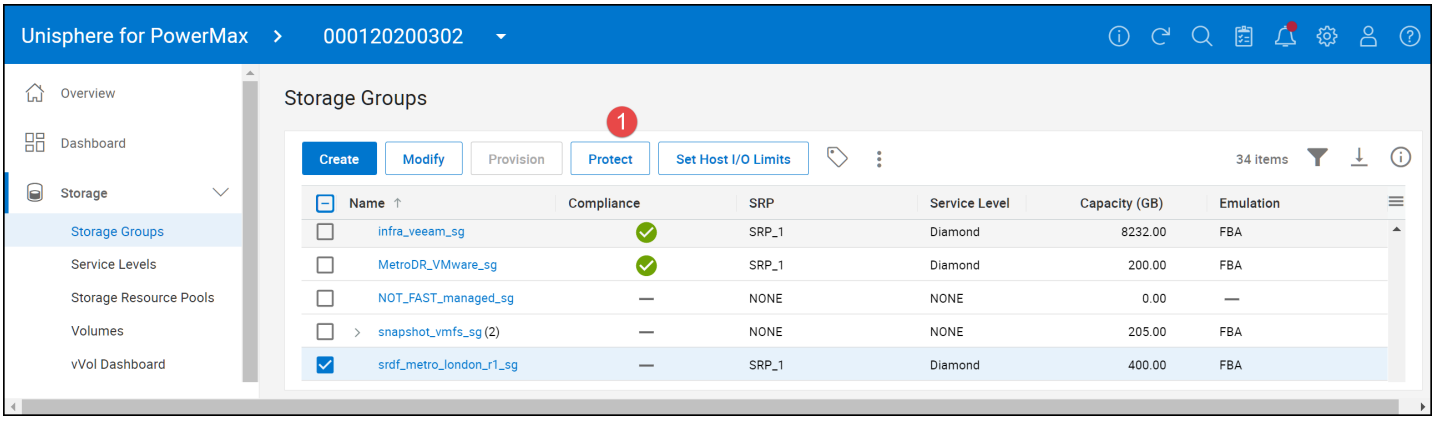



Figure 59. Storage group selection for SRDF/Metro setup in Unisphere

With the storage group selected, Unisphere provides the available protection types on the array. Select **Setup High Availability using SRDF/Metro** radio button and then **Next**. This step is in Figure 60.

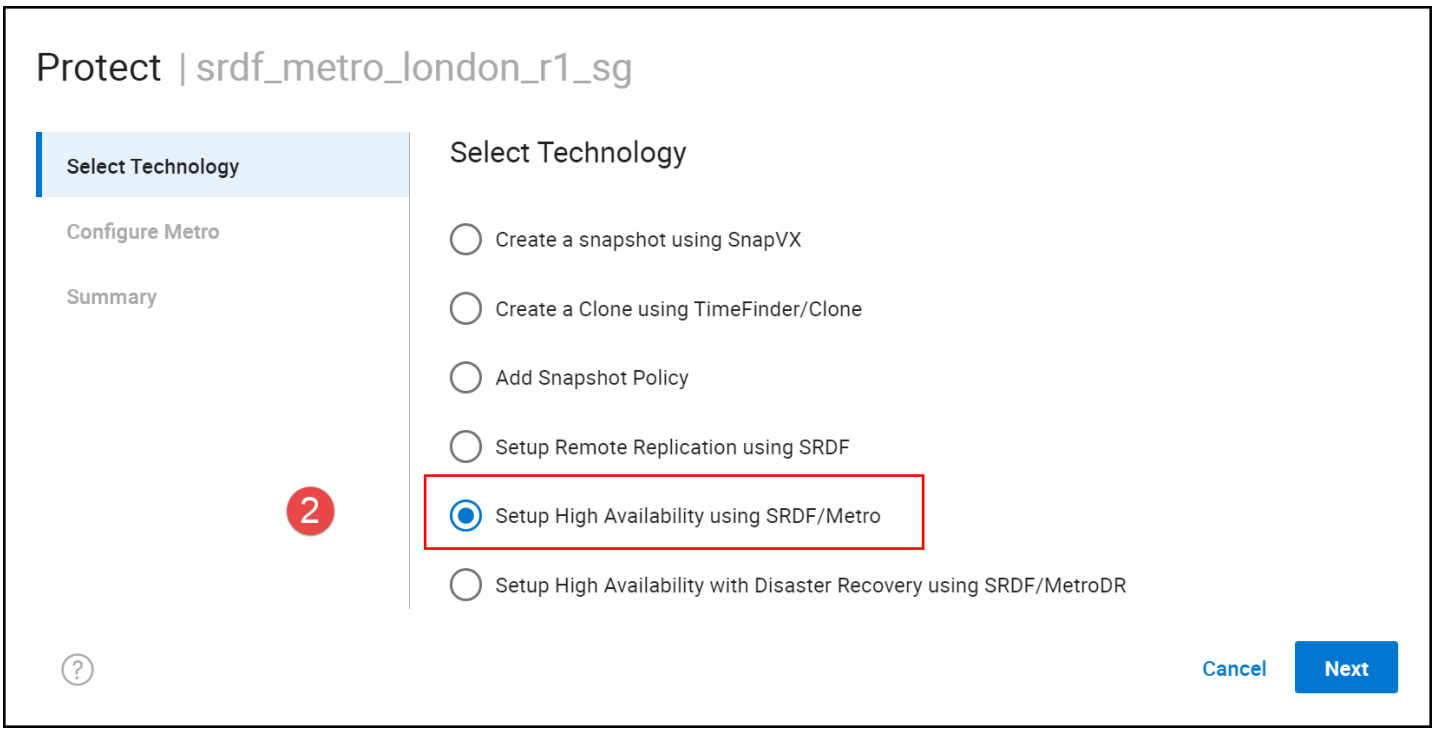



Figure 60. Protection Type in protection wizard

In step 3, Figure 61, Unisphere will default the following:

- Automatic for the **SRDF Group**
- Box checked for Establish SRDF Pairs
- Bias for **Protected By** – chang this to Witness as best practice. Note that Unisphere will gray out the witness if there are none.

- The remote storage group name will be set to the same name as the protected group, though you can change it.
- The same service level as the R1 group.
- Box checked for **Enabled Data Reduction** if it is enabled on the R1.

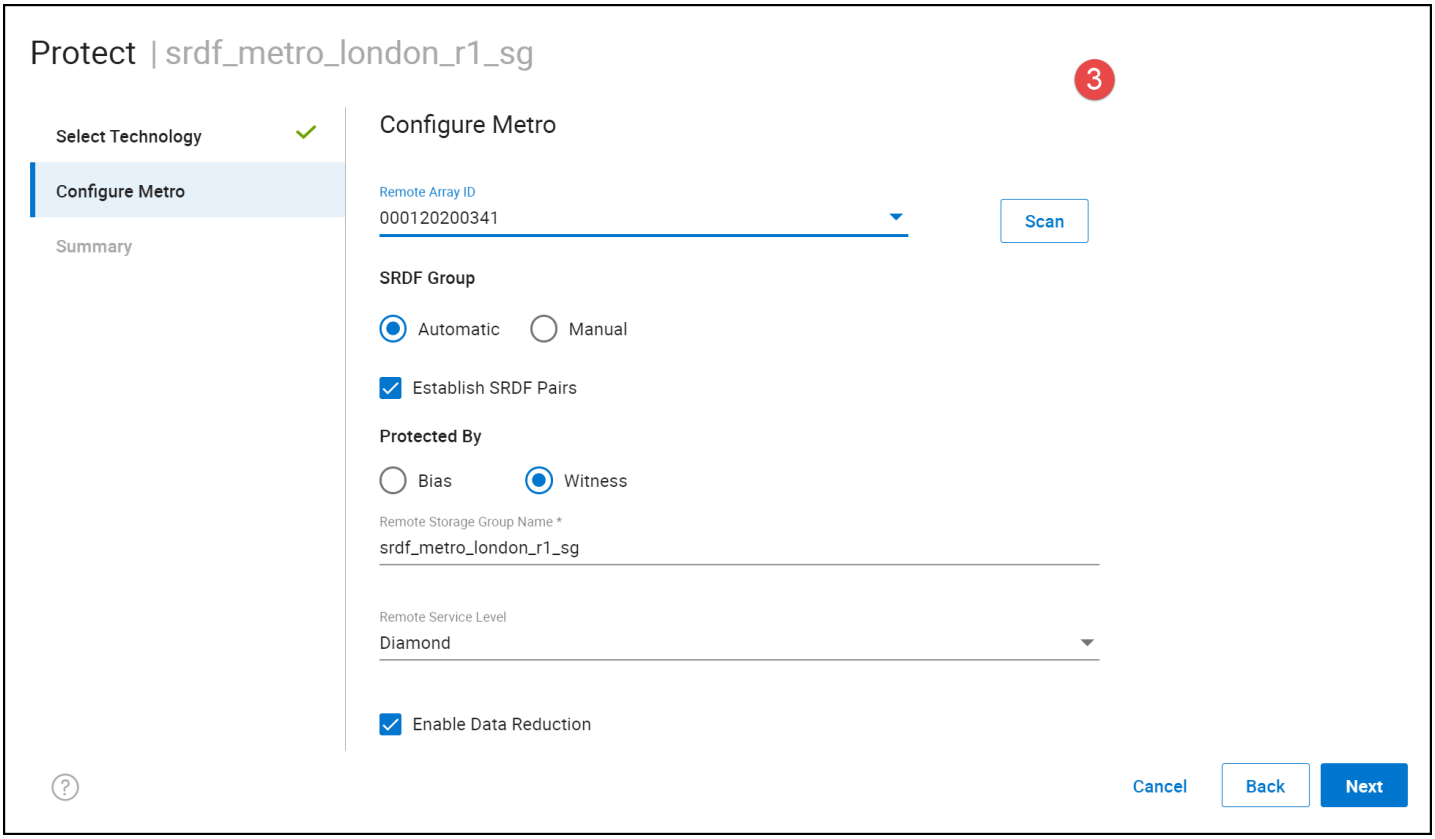



Figure 61. SRDF/Metro Connectivity in in protection wizard

Finally review the proposed changes in step 4 in Figure 62 and run the task.

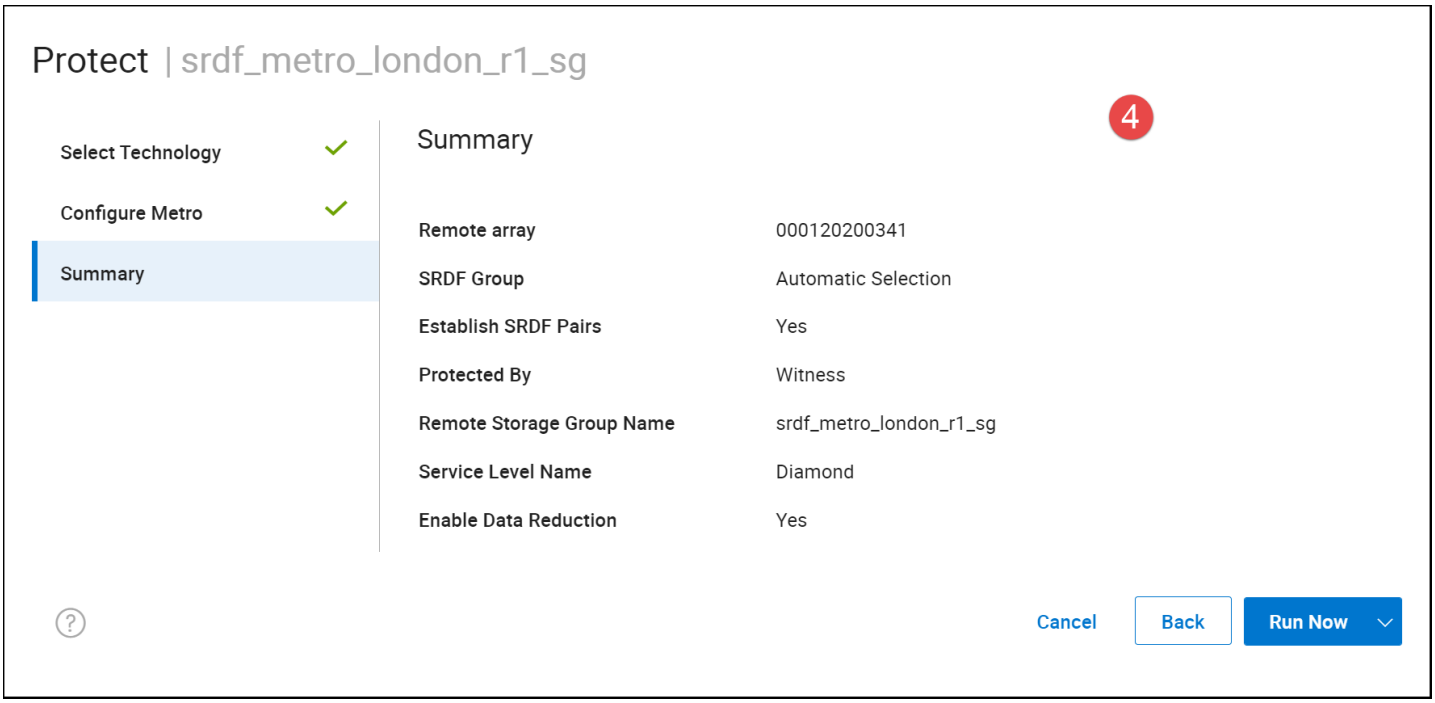



Figure 62. Complete protection wizard

Once completed, the SRDF/Metro dashboard in Figure 63 will show the group syncing.

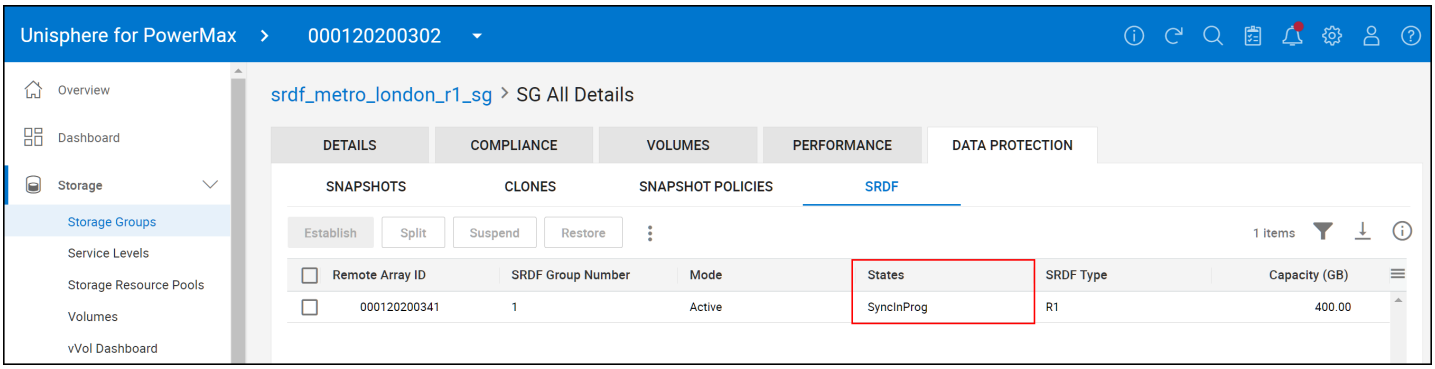



Figure 63. SRDF/Metro dashboard

The user should wait until the State in Figure 63 reads **ActiveActive** before creating a masking view for the R2 devices. That will mean the synchronization is complete. Alternatively, if using bias, the synchronized state is **ActiveBias**. You can use Solutions Enabler to verify the state of the RDF group as in Figure 64.

```
root@dsib2005:~
[root@dsib2005 ~]# symrdf list -sid 302 -rdfg 1

Symmetrix ID: 000120200302

                              Local Device View
-------------------------------------------------------------------------------
                     STATUS    FLAGS                        RDF  S T A T E S
Sym   Sym    RDF     --------- -----  R1 Inv   R2 Inv  ----------------------
Dev   RDev   Typ:G    SA RA LNK MTES   Tracks   Tracks  Dev RDev Pair
----- ----- --------- --------- ----- -------- -------- --- ---- -------------

0013F 0010B   R1:1    RW RW RW  T1.E         0        0 RW  RW   ActiveActive
00158 0010C   R1:1    RW RW RW  T1.E         0        0 RW  RW   ActiveActive

Total                                -------- --------
  Track(s)                                  0        0
  MB(s)                                   0.0      0.0

Legend for FLAGS:

 (M)ode of Operation : A = Async, S = Sync, D = Adaptive Copy Disk Mode
                     : T = Active
 Mirror (T)ype       : 1 = R1, 2 = R2
 (E)xempt            : X = Enabled, . = Disabled, M = Mixed, - = N/A
 R1/R2 Device (S)ize : E = R1 EQ R2, 1 = R1 GT R2, 2 = R2 GT R1, - = N/A

[root@dsib2005 ~]#
```

Figure 64. Verify SRDF/Metro state in CLI

Once the state is in the proper active state, you can present the R2 devices to the R2 host(s) using the masking view wizard in Unisphere, displayed in Figure 65. In this example the masking view name changes the r1 to r2. This is useful initially, but recall if there is a failure, or a re-establish after a change in architecture (e.g., a cascaded leg added), the R2 may become an R1.




Figure 65. Create R2 masking view

As shown above, when setting up SRDF/Metro pairs, Unisphere for PowerMax wizards work at the storage group level. If the devices are not already in a storage group, you can still use Unisphere in more of a manual fashion. For instance, you can create pairs in the **SRDF Groups** screen in the Data Protection area of Unisphere. Here, using the checkboxes, check one of the groups and then select **Create Pairs** as in Figure 66.

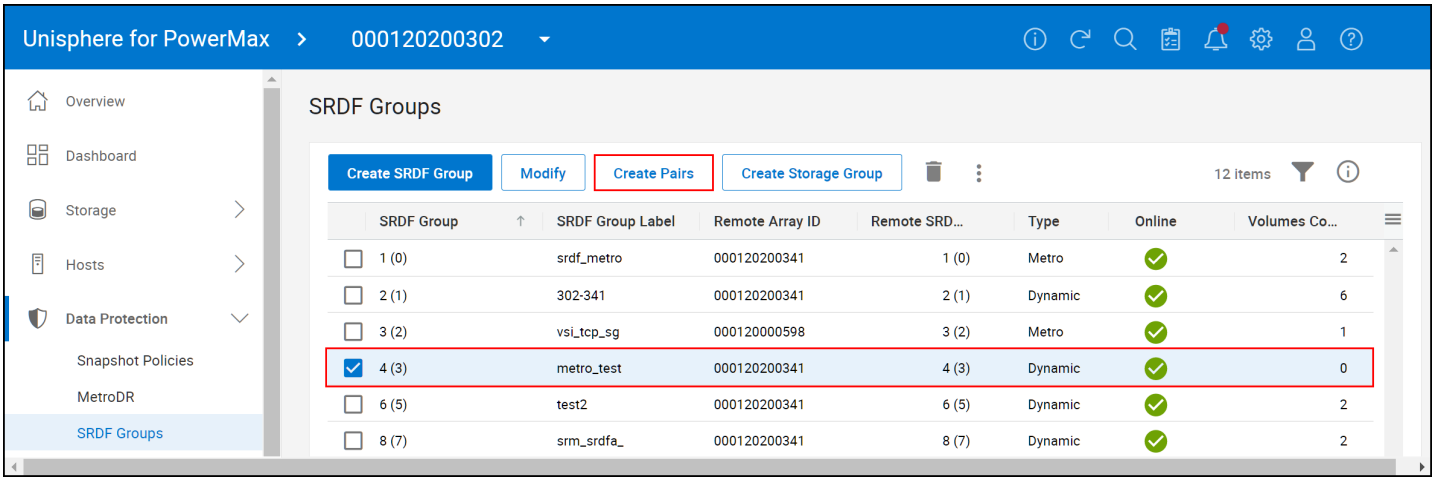



Figure 66. SRDF/Metro create pairs in an empty RDF group

A dialog box appears where the user can select the SRDF Mode **Active** from the drop-down list. At that point there is the ability to set **Establish** or **Restore** and **Bias** or **Witness** (if available). Choosing between Establish and Restore sets which device is the source for the data. Establish means SRDF will copy the data from the R1 to the R2, Restore means it will copy the data from the R2 to the R1. Note that choosing Restore does not mean the R2 becomes an R1. Unisphere then provides the ability to manually select existing local and remote devices, or Unisphere can look for available devices based on a designated size. If Unisphere cannot find available devices, it will create them. You may place the devices in an existing storage group on one or both sides. All four steps are in Figure 67.

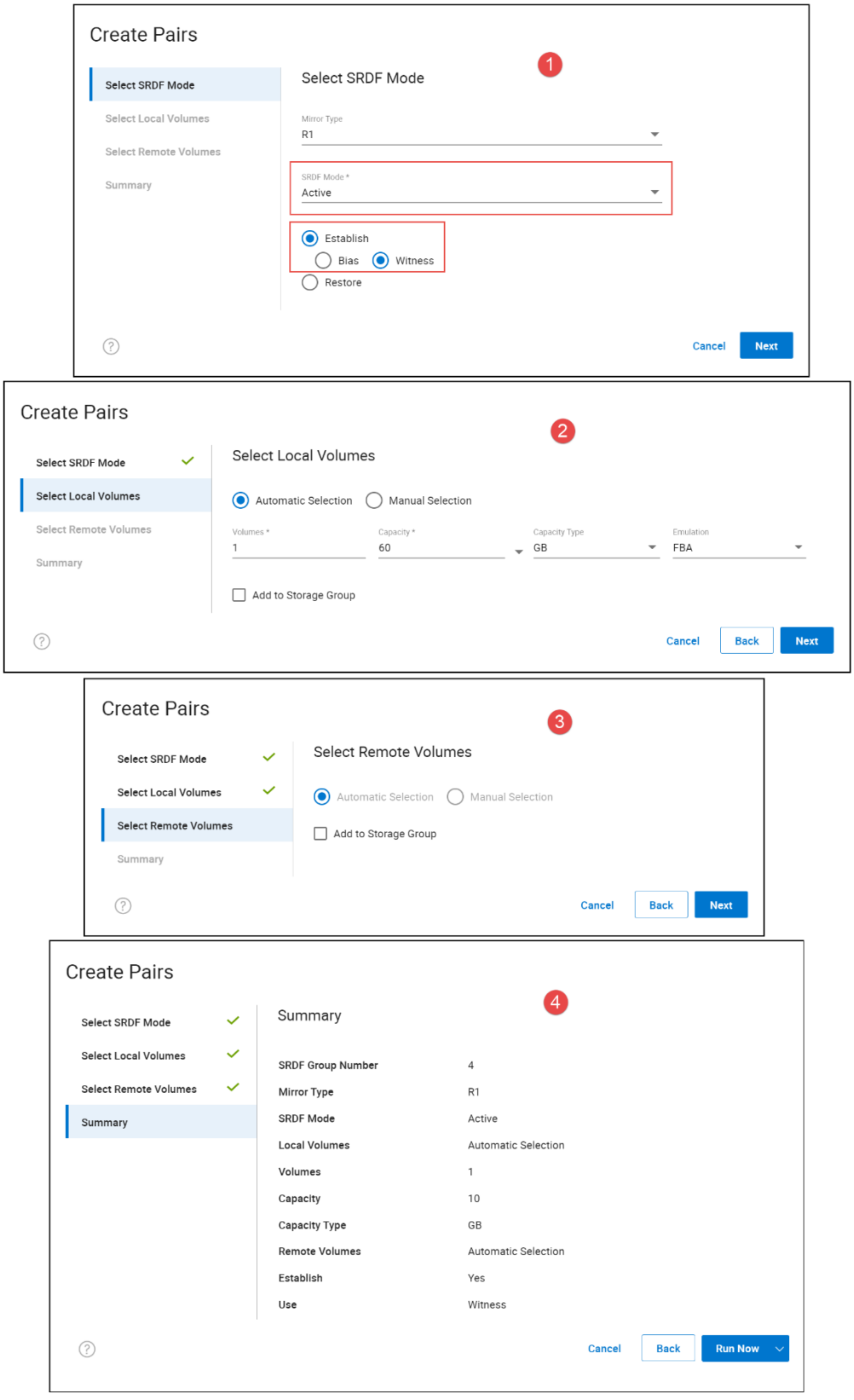



Figure 67. Creating SRDF/Metro pairs in Unisphere without a storage group

**Note:** If the user selects an existing SRDF/Metro RDF group, the SRDF mode will automatically be set to **Active,** and you cannot adjust it. An RDF group with SRDF/Metro pairs cannot have a second SRDF mode present. Though Dell recommends using Unisphere, it is possible to setup SRDF/Metro using Solutions Enabler. In general, setting up SRDF/Metro is akin to any SRDF configuration, save for during the ***createpair*** command when you specify the switch ***–metro*** for an SRDF/Metro pair.

### 1.1.5 Adding new pairs to an existing RDF group online

Beginning with PowerMaxOS 5978.144.144, you can dynamically add existing devices to a SRDF/Metro RDF group without any loss of data. Prior to this release, you could only add new or formatted devices to an existing group with the `-format` option, potentially causing data loss.